AI-empowered UAV for Intelligent Transportation Systems

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Department of Mechanical and Automation Engineering



Personal Introduction

🎓 Educational Background

2022-2026	The Chinese University of Hong Kong, Shenzhen	Ph.D
2018-2021	Beijing University of Posts and Telecommunications	Master
2014-2018	Beijing University of Posts and Telecommunications	Bachelor

💼 Internship Experience

2022-2026	Shanghai AI Lab	Joint Program
2024 -2025	Nokia Bell Lab	Intern Researcher
2020-2021	SenseTime Co., Ltd	Intern Researcher
2018	Didi Chuxing	Department of Products and Technology

Bonus & Course Project

- 🎁 **Bonus Points**
  - **Participate in class and earn bonus points! (Answer questions)**
- 🚀 **Course Project Preview**
  - ***Design an unmanned-system solution for a real-world application***
  - **Why?** Why are unmanned systems needed?
  - **What?** What type of unmanned system will you use?
  - **How?** How will you design the system?
  - **Compare.** How does your solution differ from existing approaches?



Prof. Chen Xi

Research Project under CUHK-TJU Joint Research Laboratory for Smart Transportation



Prof. Fang Shou'en



Prof. Ben M. Chen

Aerial Photography and AI Empowered Simulator for Responsive Traffic Management

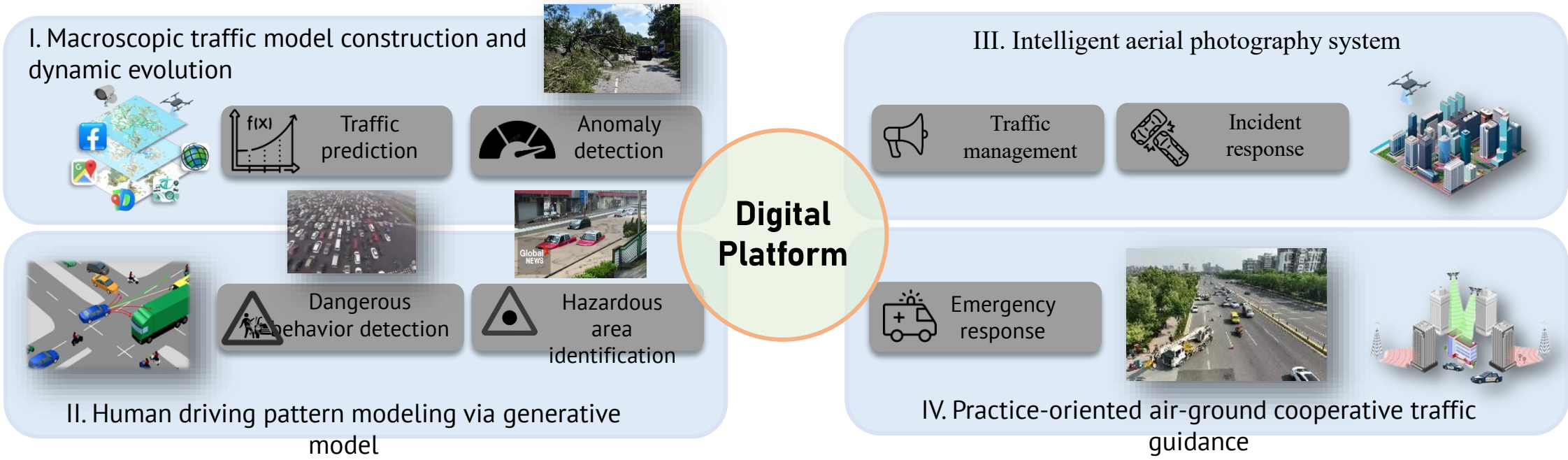


Prof. Anthony Chen

Project background & key innovation

With the rise of smart cities, aerial photography has emerged as a new remote sensing platform increasingly applied in urban traffic management. They enable real-time monitoring, digital analysis, and emergency response. However, most current aerial photography applications remain limited to "**camera**" functions, lacking **intelligent analysis**, **decision-making support**, and **integration across platforms and departments**.

Project targets:



Supporting organizations:

香港特別行政區政府 香港警務處 | 機電工程署 EMSD | AECOM | DJI 大疆车载 | OntheGo | 中国移动 China Mobile

R&D tasks and challenges

Intelligent aerial photography system

Key limitations: low autonomy, flexibility & endurance for traditional drones



Long-term traffic monitoring

Autonomous takeoff & return

Easy to deploy



24-hour on call

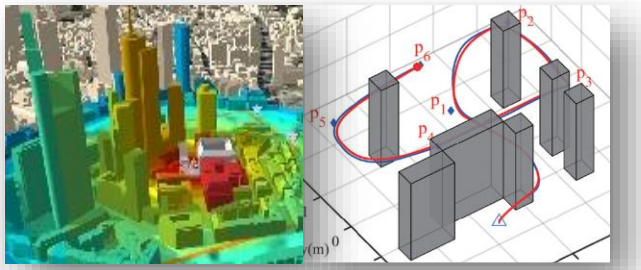
Autonomous takeoff & return

Cloud-based task assignment

Autonomous UAV duty and emergency response platform



Drone-based RGB-Infrared multi-object detection and tracking



Intelligent path planning & motion planning in 3D urban environment

Practice-oriented air-ground cooperative traffic guidance -- Emergency Response Demo

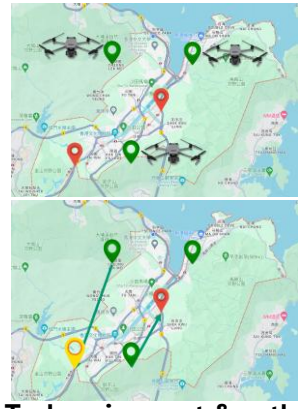
Key challenge: Government-supported, practice-driven drone applications



Accident happened

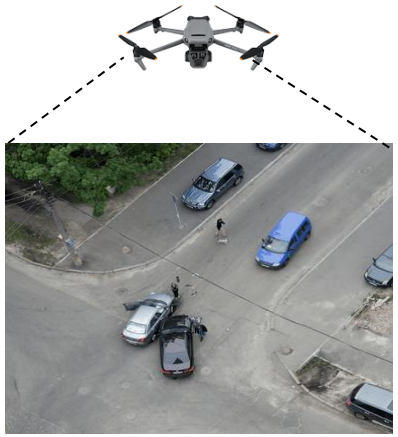


Rapid response



Task assignment & path planning for drones

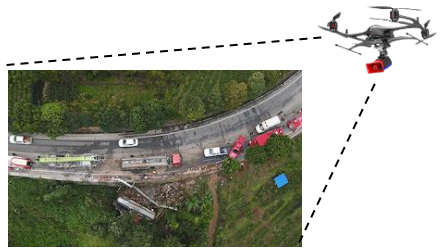
Aerial response



Drone survey of accident scene

- Evidence collection
- Accident analysis
- Casualty assessment
- Influence on traffic flow

- License plate record
- Injury detection
- Accident severity assessment
- Fire detection



Traffic flow guidance





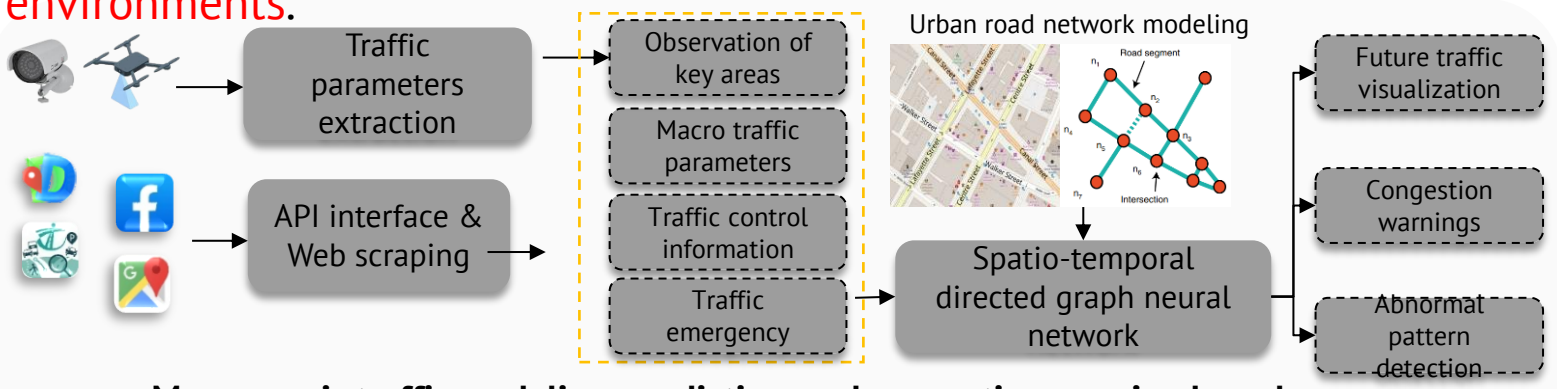
Optimal task scheduling

Post-accident handling

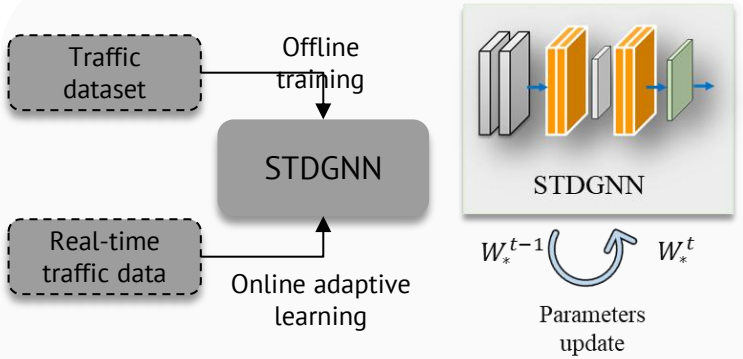
R&D tasks and challenges

Macroscopic traffic model construction and dynamic evolution

Key challenge: 1. Heterogeneous data fusion; 2. Model evolution in dynamic environments.



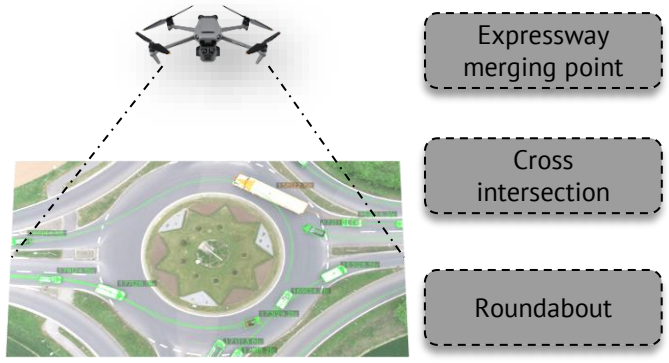
Macroscopic traffic modeling, prediction, and congestion warning based on heterogeneous traffic information fusion



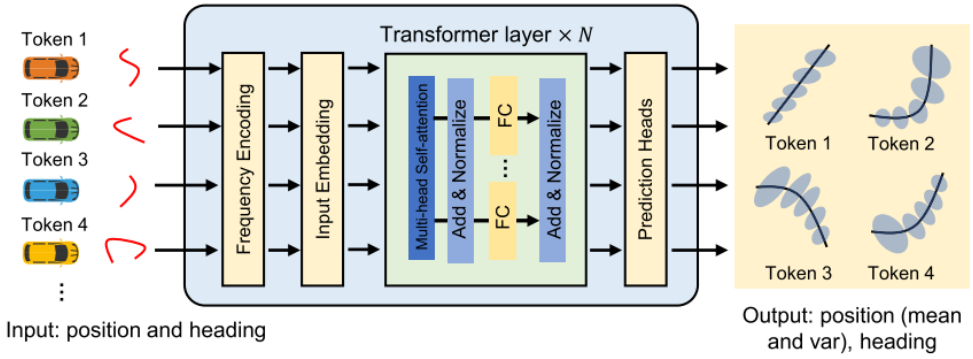
Dynamic evolution for deep neural network

Human driving pattern modeling via generative model

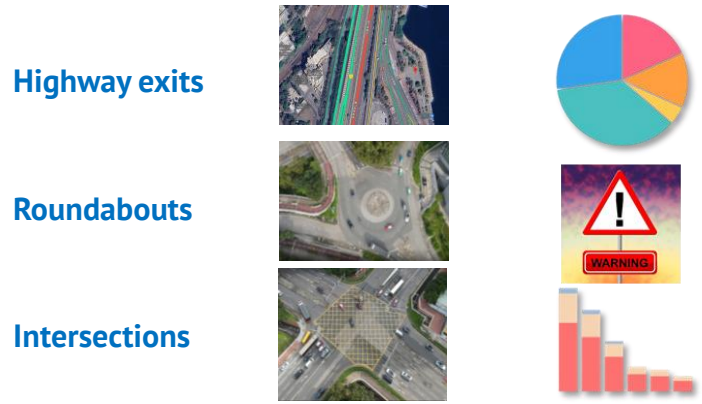
Key challenge: 1. Scarcity of high-quality driving trajectory data; 2. Uncertainty in human driving pattern.



Drone-collected driving trajectory data



Human driving pattern modelling



Aerial traffic dataset construction

UAV Traffic Detection



1. Drone Data Capture

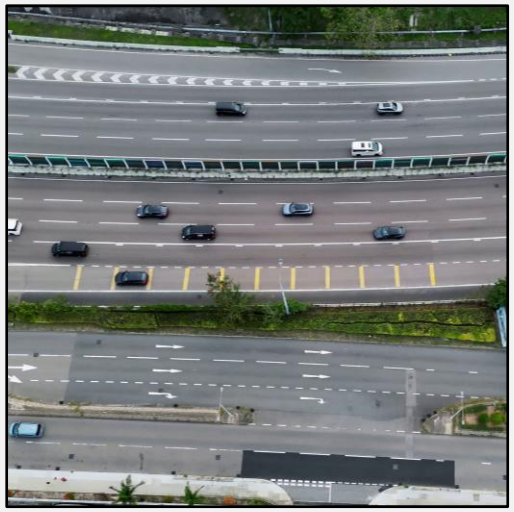


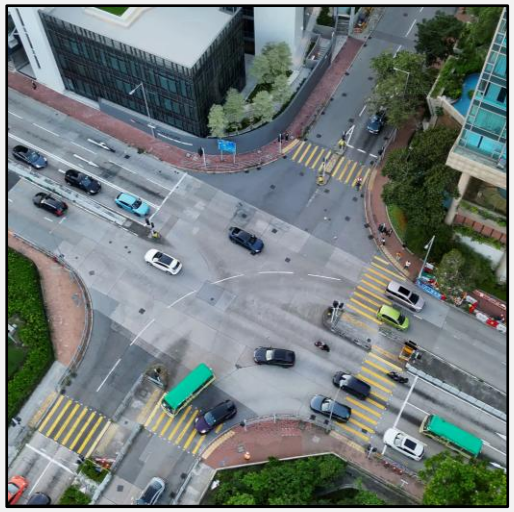
2. Image Extraction

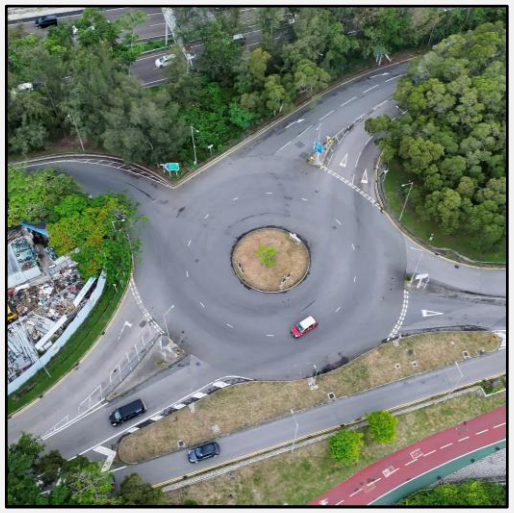


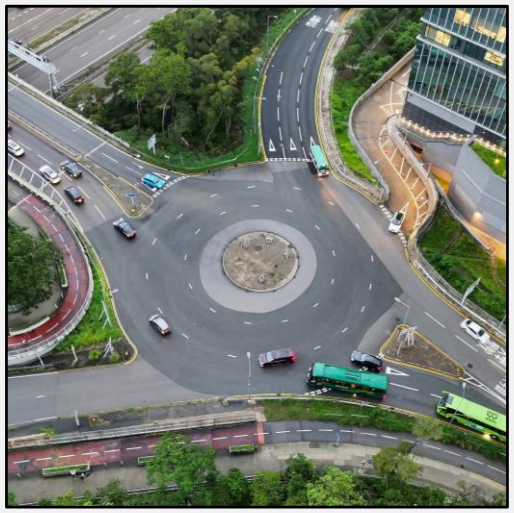
3. Annotation & Labeling

UAV Traffic Detection





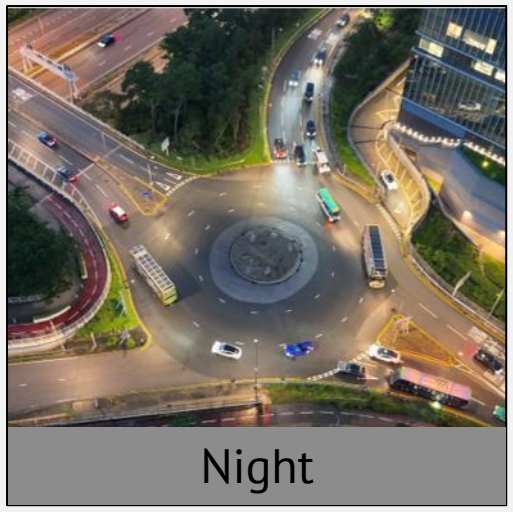




Four Locations



Day



Night

Day/Night



Sunny

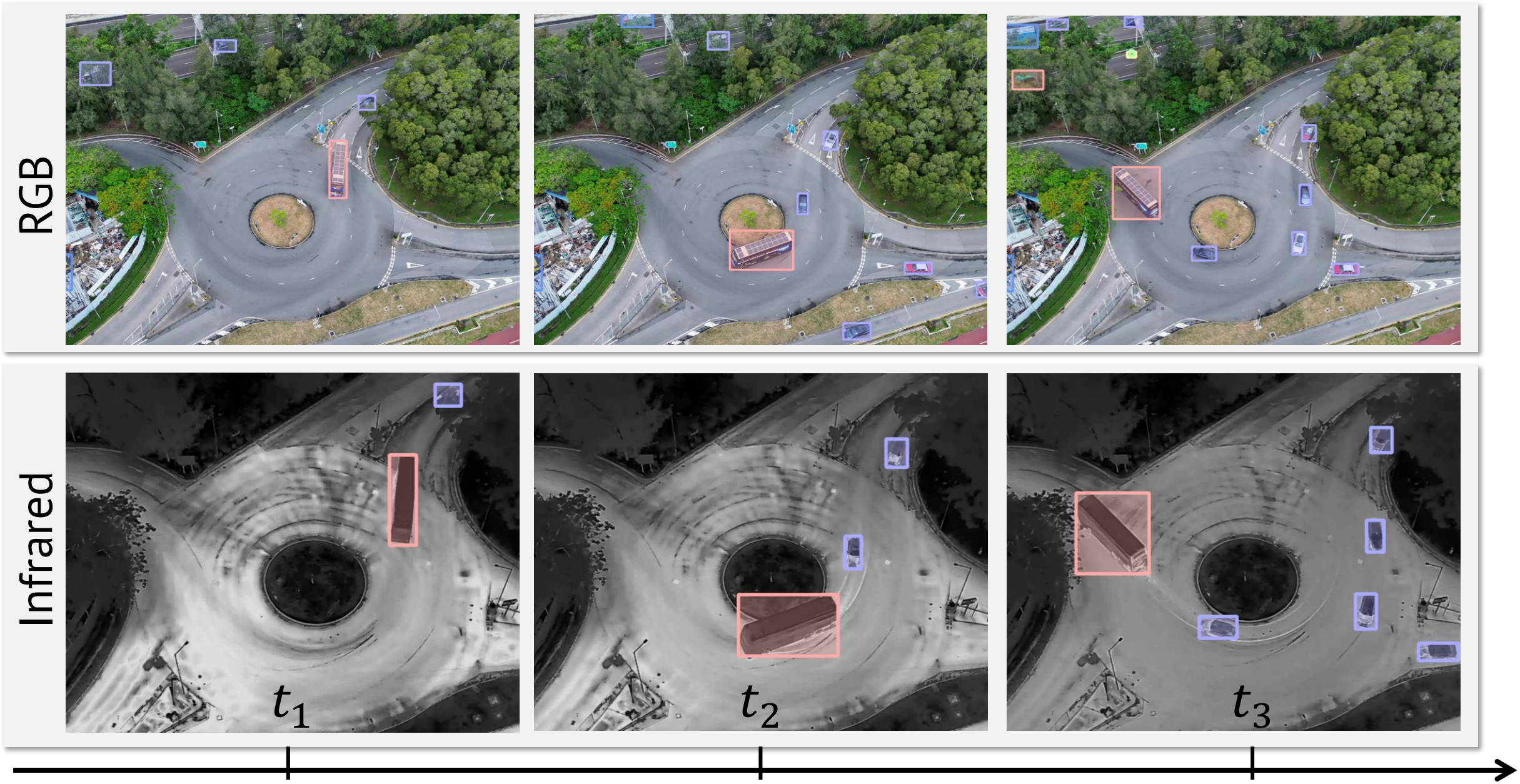


Light Rain

Weather

Time & Weather Coverage

UAV Traffic Detection



UAV Traffic Detection

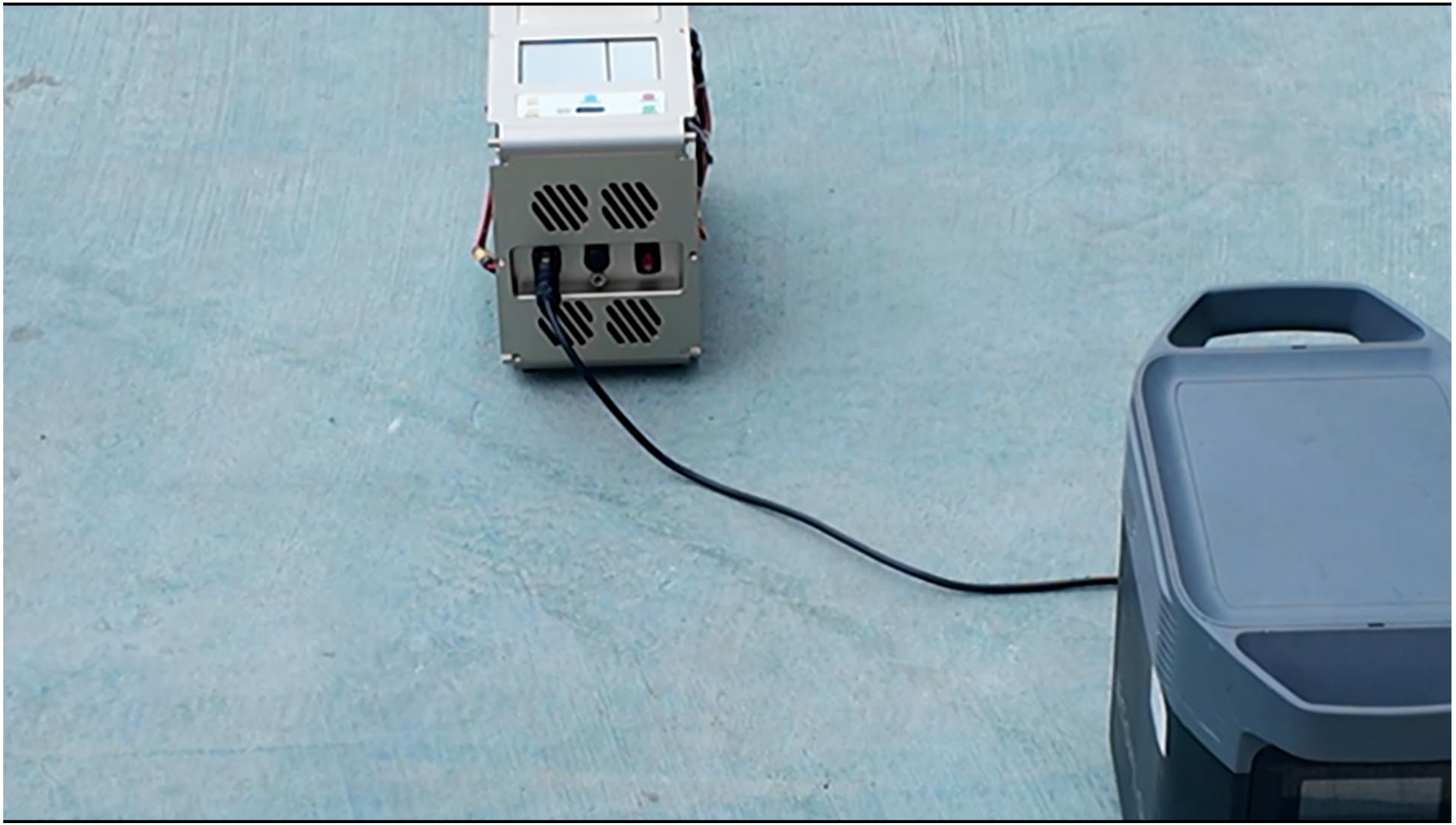


Tethered Ground Station

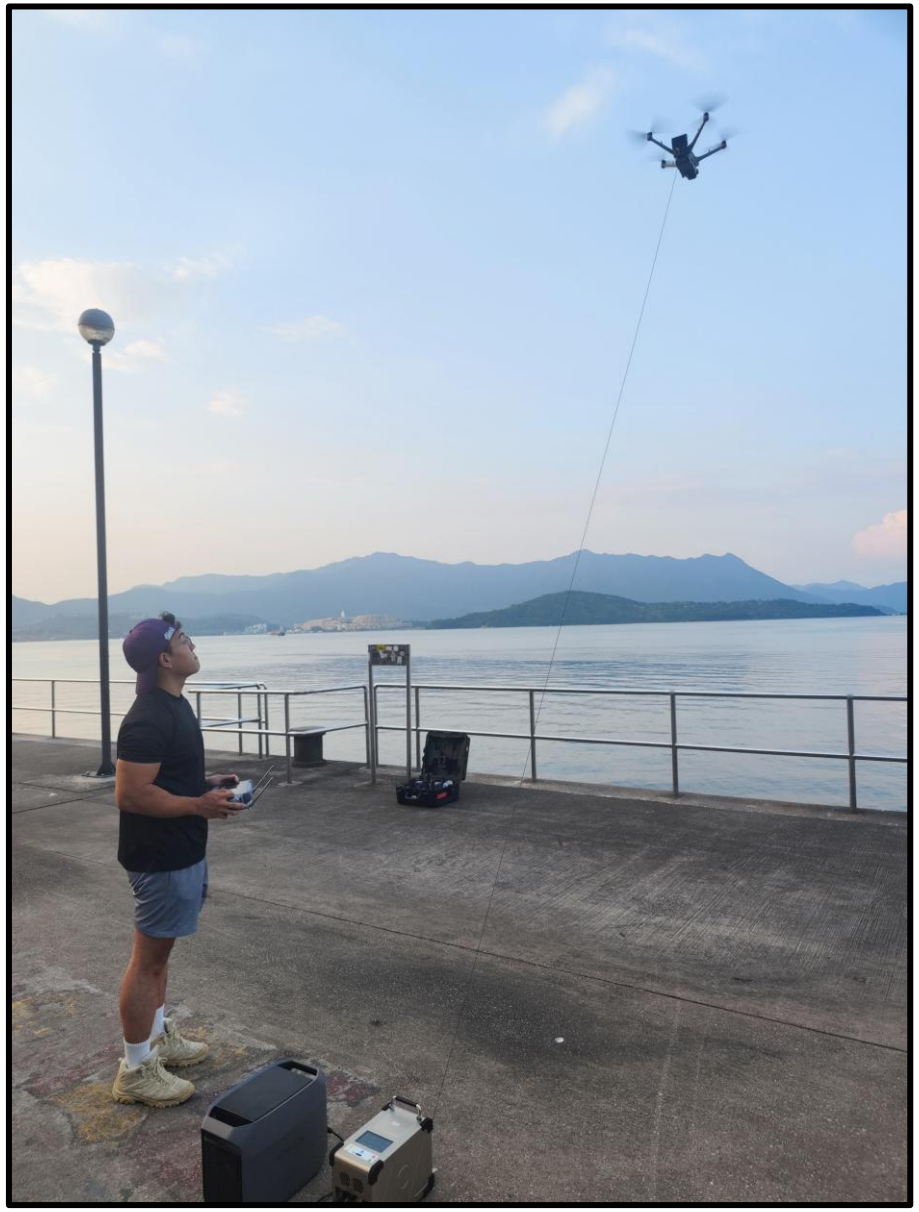


Onboard Power Adapter

UAV Traffic Detection



UAV Traffic Detection

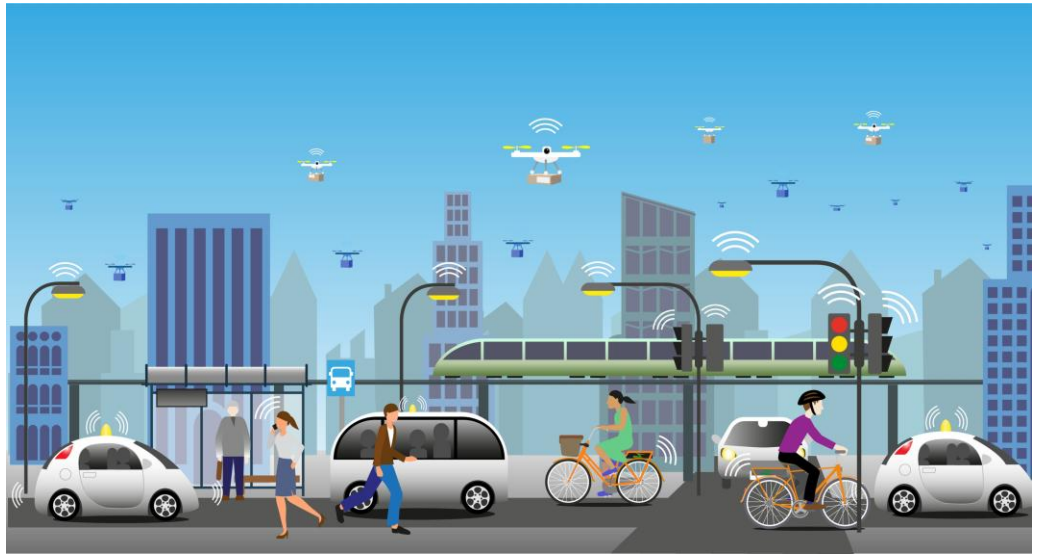






Understanding Urban Mobility

- **Defining Urban Mobility**
  - The essence of city movement
  - Vital for daily life functionality
- **The Impact of Traffic Congestion**
  - **Economic impacts:** productivity loss, increased costs
  - **Environmental effects:** pollution, carbon footprint
  - **Social implications:** stress, health issues





Economic Impacts



Environmental Effects



Social Implications

Why Use Drones for Traffic?

Traffic detection capabilities are limited to the main roads in each district. Meanwhile, the deployment of traditional traffic sensors faces challenges including **high costs**, **limited field of view**, and **complex installation processes**.







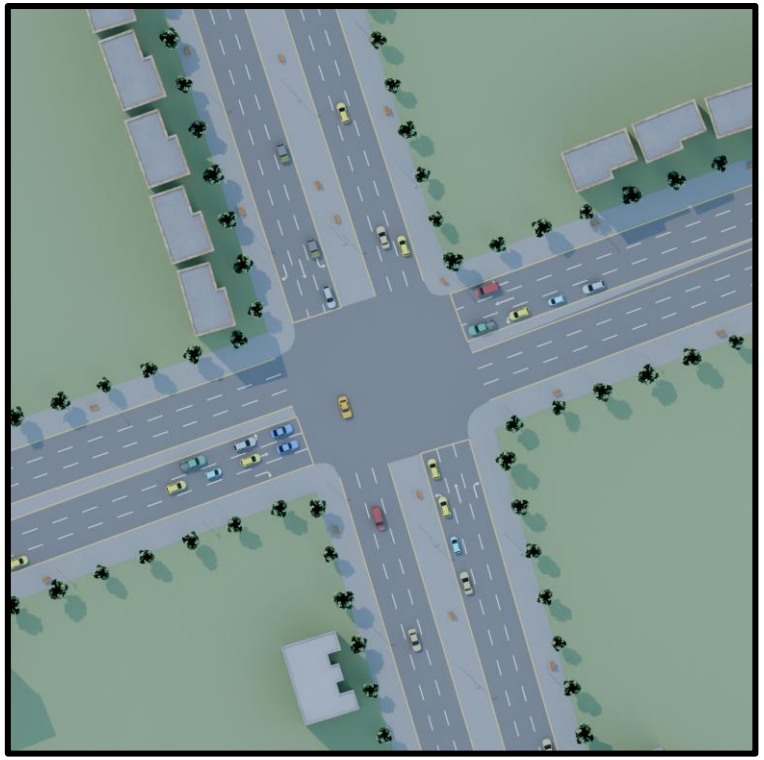
Unmanned Aerial Vehicles (UAVs) provide a **low-cost**, **bird-view**, and **high-mobility** solution for smart traffic management.





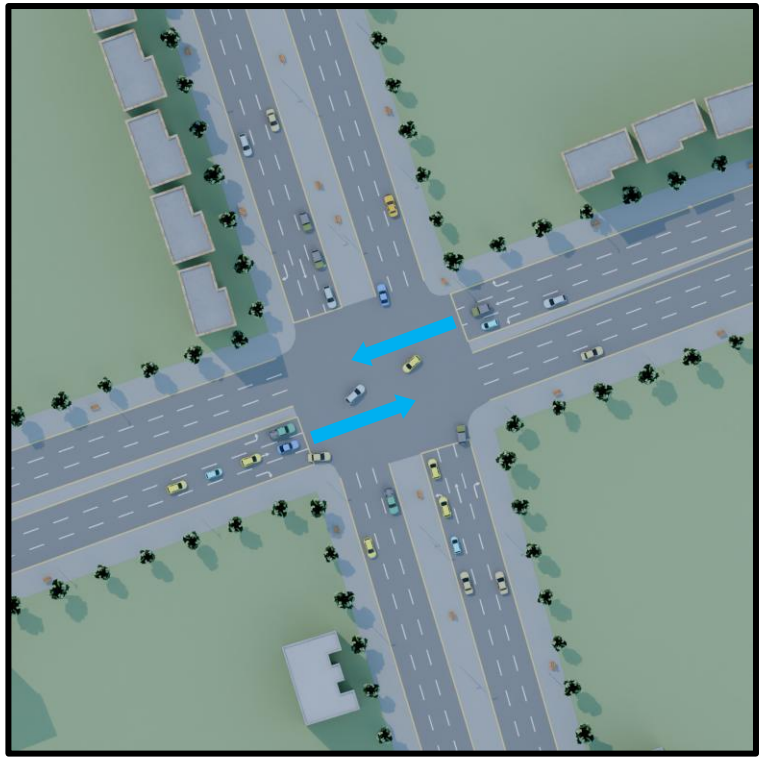


Turning aerial images into traffic intelligence



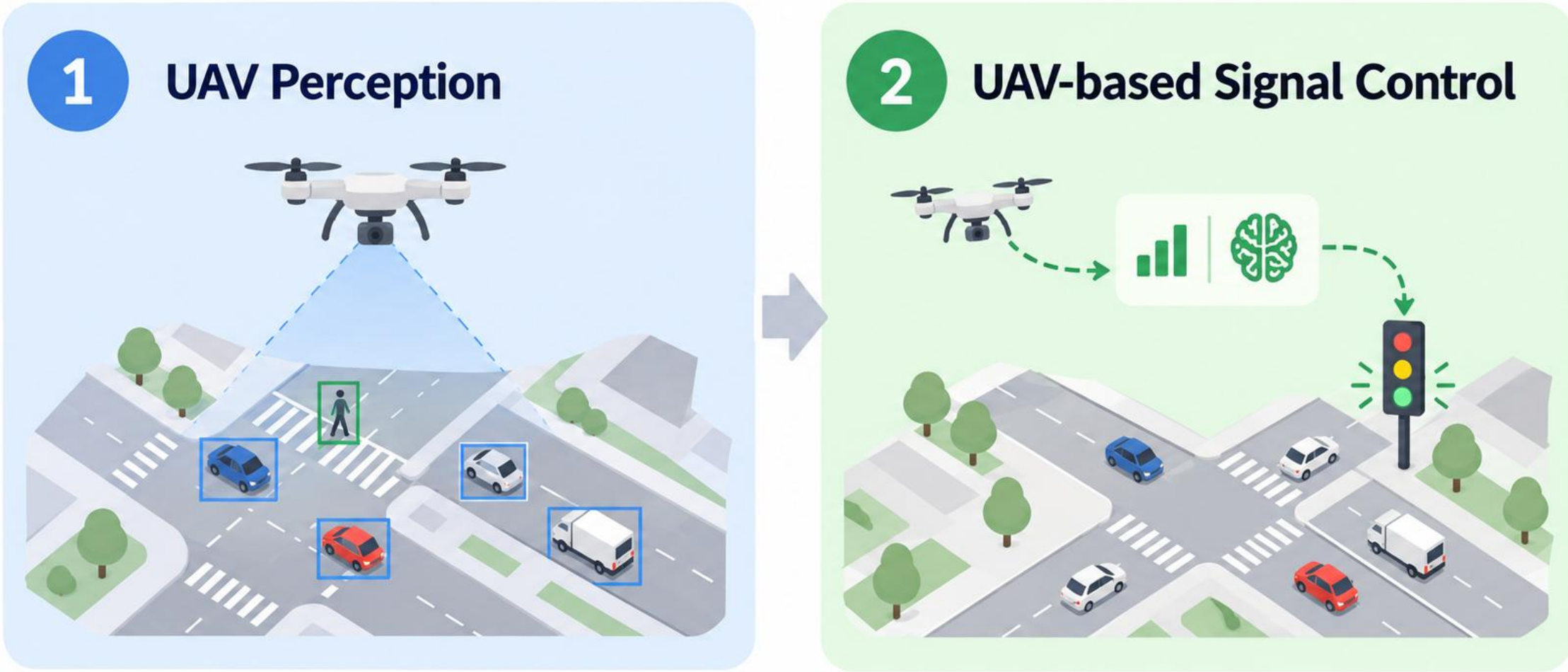


How to make decision?

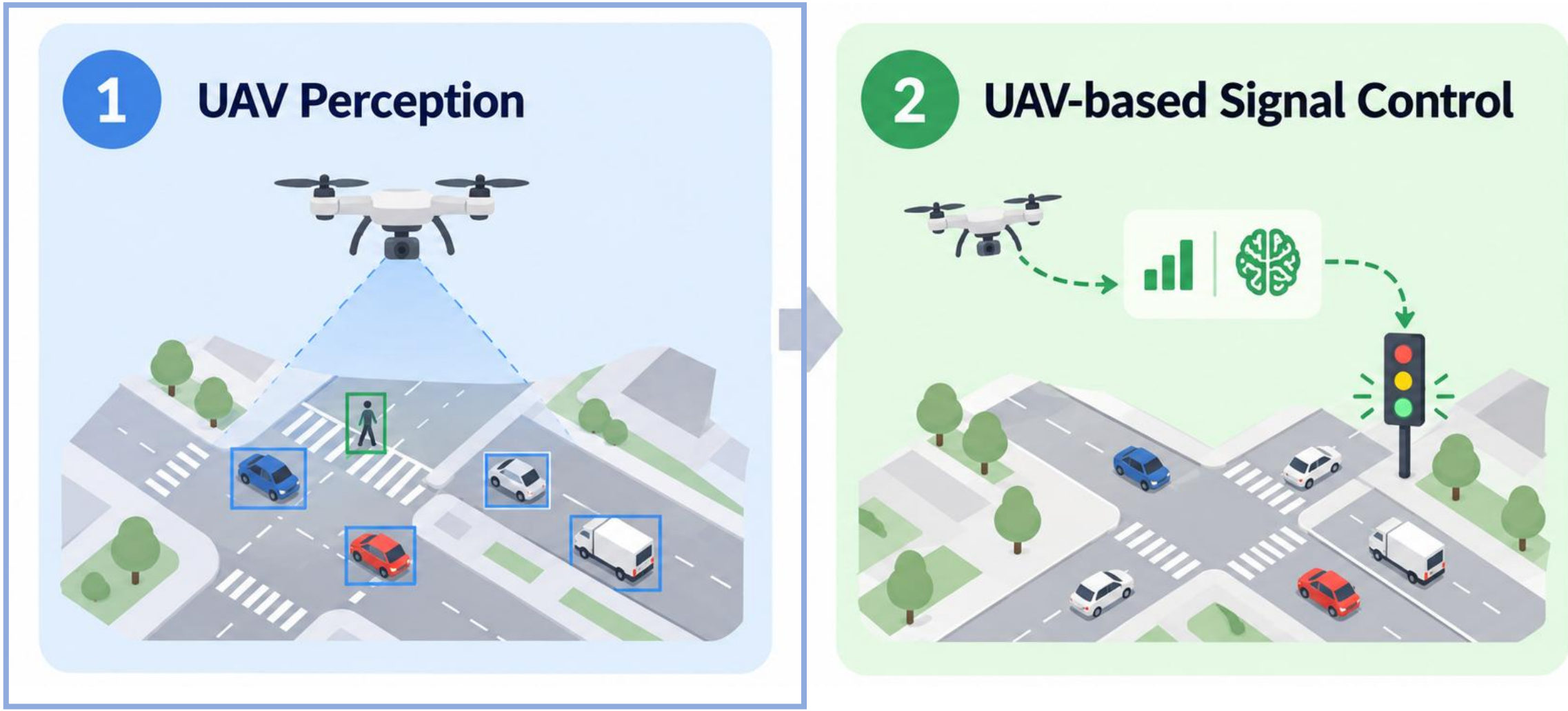


AI turns a UAV from a flying camera into an intelligent sensing platform.

Roadmap

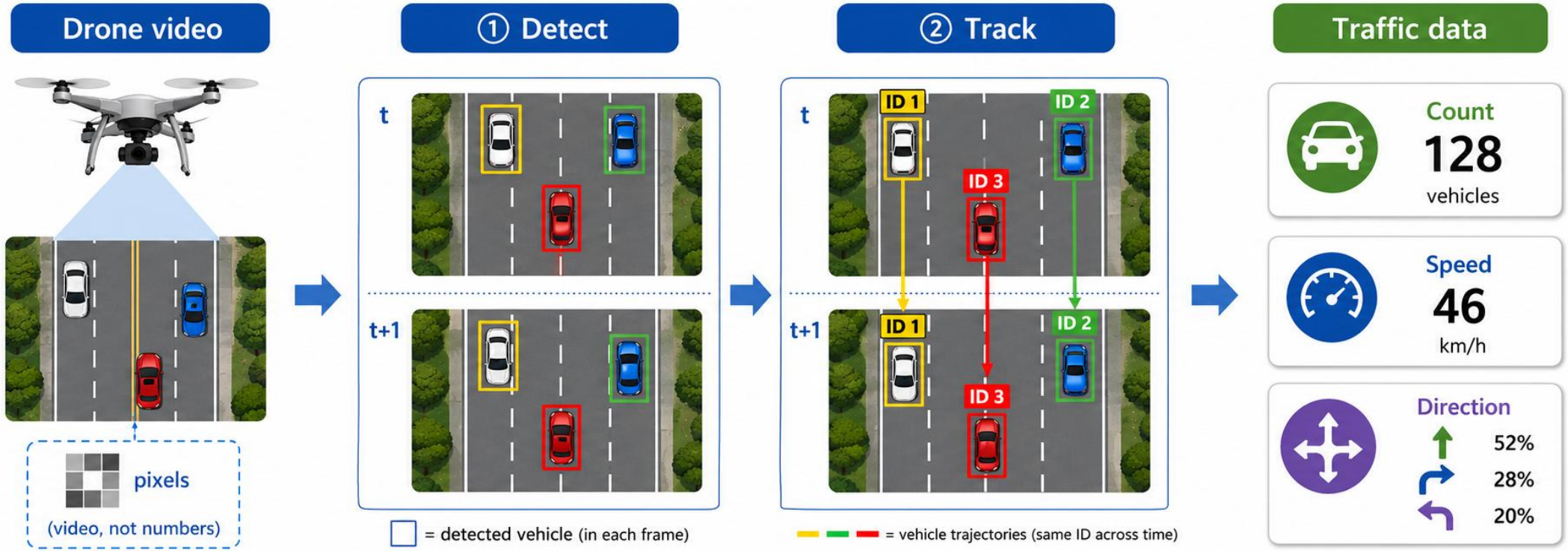


Roadmap -- Perceive



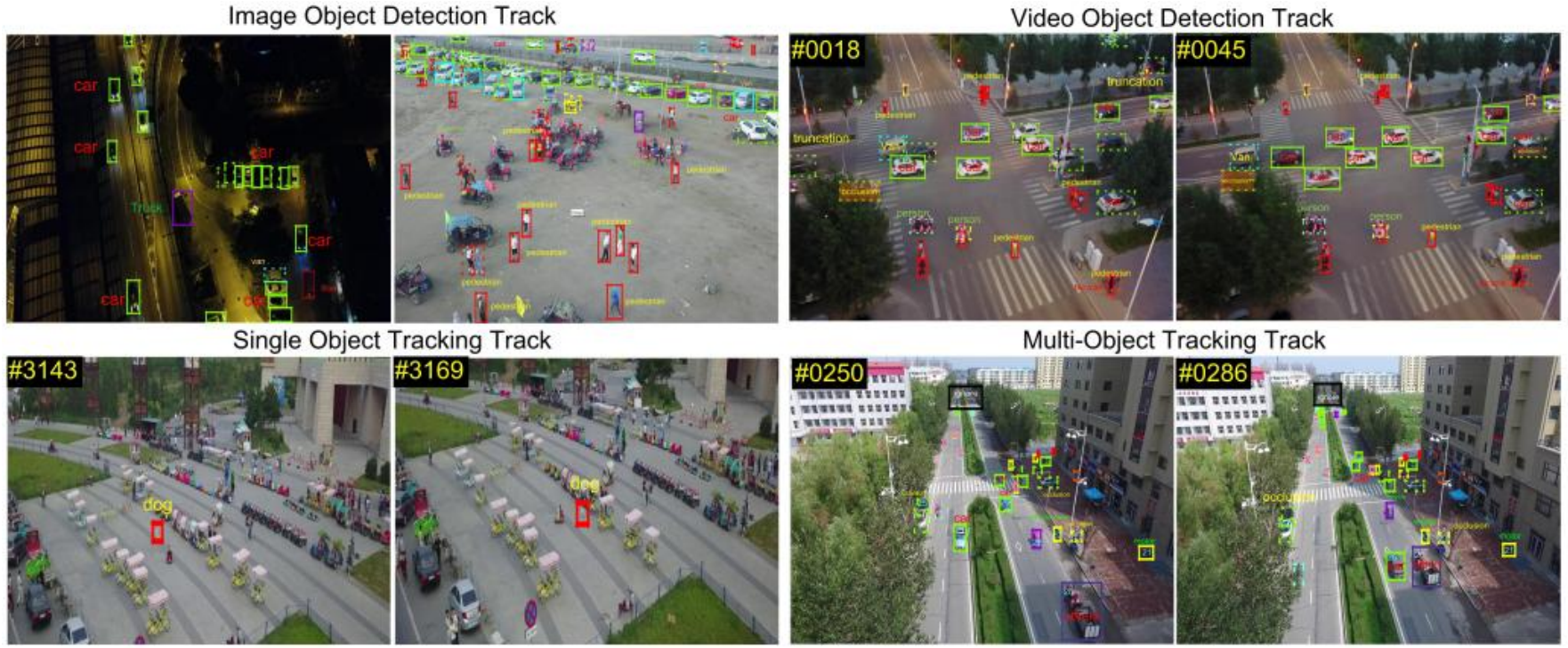
From Footage to Traffic Data

- A drone gives us video — but a video is just pixels.
- Traffic management needs numbers: how many cars, how fast, going where.
- Two AI steps turn pixels into data:
  - ① Detect every vehicle
  - ② Track them into trajectories.



Challenges of Object Detection in UAV Imagery

1. **Small objects** — vehicles and pedestrians may occupy only a few dozen pixels
2. **Dense objects** — intersections, parking lots, and commercial areas pack many overlapping targets
3. **Large images** — high-resolution frames are mostly background; targets fill only small regions
4. **Viewpoint variation** — top-down / oblique views, rotation, and night scenes change appearance



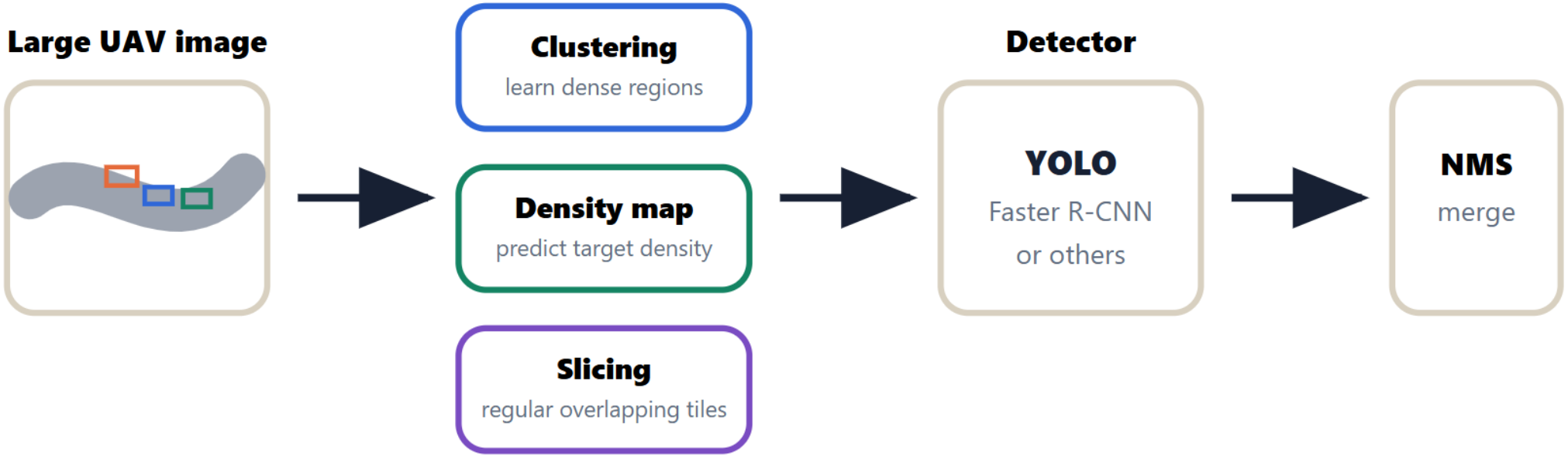
Zhu, Pengfei, Longyin Wen, Dawei Du, Xiao Bian, Heng Fan, Qinghua Hu, and Haibin Ling. "Detection and tracking meet drones challenge." *IEEE transactions on pattern analysis and machine intelligence* 44, no. 11 (2021): 7380-7399.

Three method families: make small objects larger

Cluster first, then zoom
Objects often appear in groups. Find dense cluster chips, enlarge them, and detect again.

Estimate density, then crop
A density map tells the model where objects are likely to be and where cropping is useful.

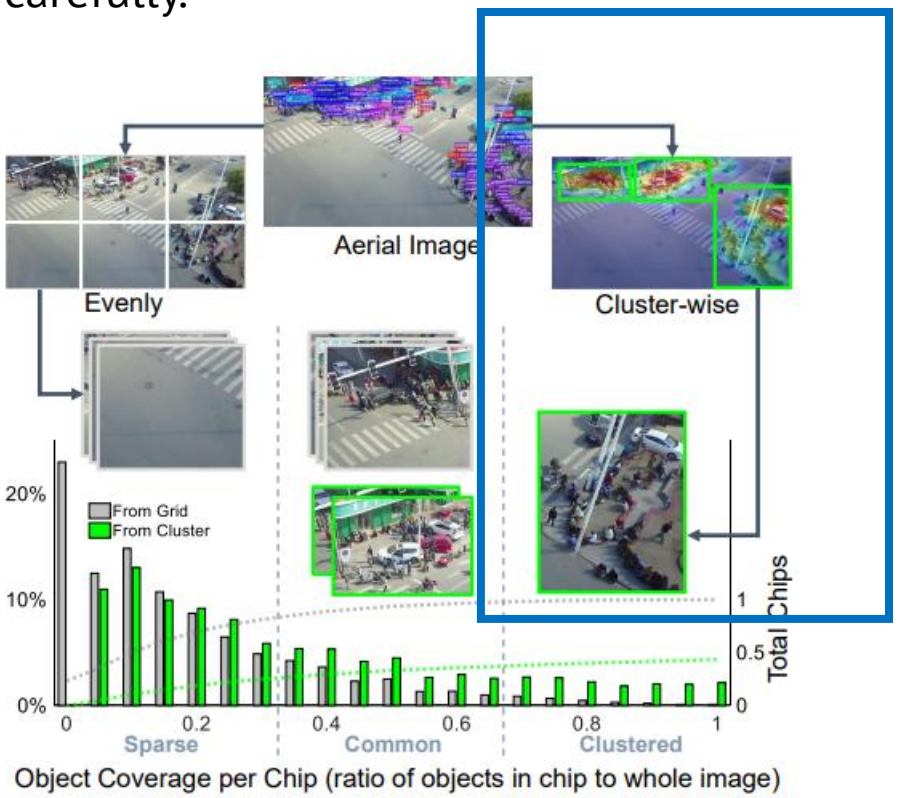
Slice first, then merge
Split the large image into overlapping tiles, run detection on each tile, and map boxes back.



ClusDet: Find Dense Regions & Zoom In

ClusDet does not tile the image uniformly.

- It first predicts regions where objects cluster, crops and enlarges those regions
- Then detects small objects more carefully.



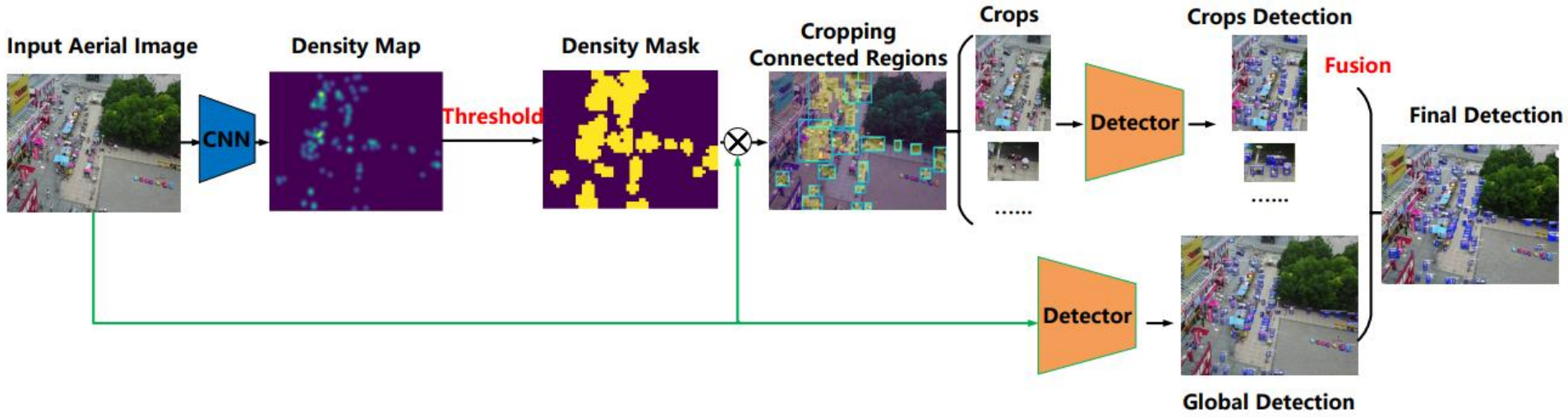
Yang, Fan, Heng Fan, Peng Chu, Erik Blasch, and Haibin Ling. "Clustered object detection in aerial images." In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 8311-8320. 2019.

DMNet: Use a Density Map to Guide Cropping

DMNet first generates a density map. Higher density indicates regions likely to contain more small objects, so the model uses that map to decide where to crop and enlarge before detection.

Relation to ClusDet

- Both solve small objects and non-uniform spatial distribution.
- ClusDet emphasizes cluster chips.
- DMNet emphasizes density intensity as a spatial guide for cropping.



Li, Changlin, Taojiannan Yang, Sijie Zhu, Chen Chen, and Shanyue Guan. "Density map guided object detection in aerial images." In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, pp. 190-191. 2020.

SAHI: Sliced Inference for Large Images

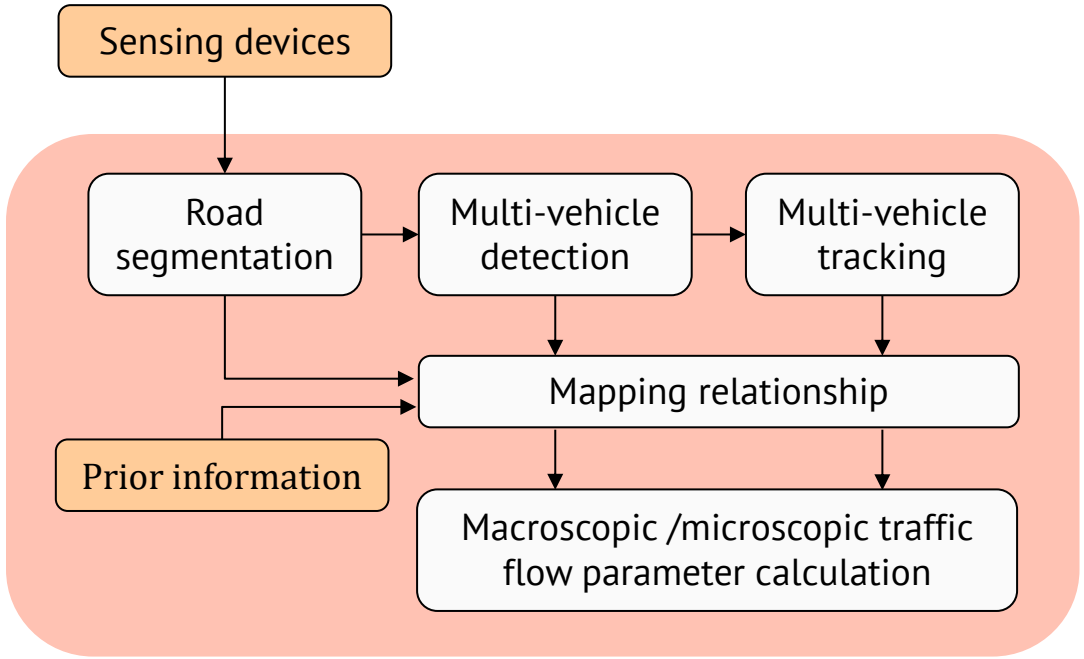
SAHI is primarily an inference and fine-tuning pipeline. It is not a new backbone. The same detector is applied repeatedly to slices.

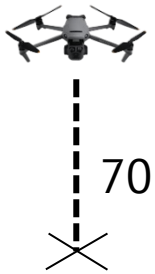
- It can wrap around common detectors such as YOLO, MMDetection, or Detectron2 models.
- Small objects become relatively larger inside each tile.
- Overlap reduces boundary truncation but creates duplicate boxes that must be merged.

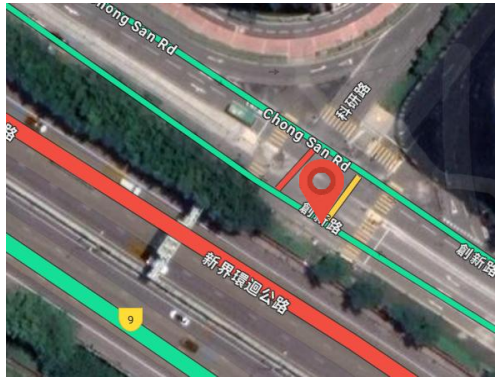


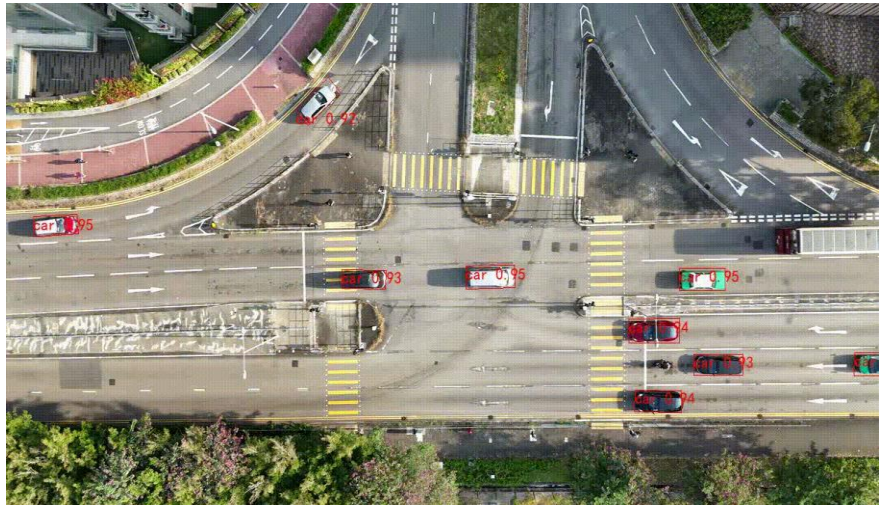
Akyon, Fatih Cagatay, Sinan Onur Altinuc, and Alptekin Temizel. "Slicing aided hyper inference and fine-tuning for small object detection." In 2022 IEEE international conference on image processing (ICIP), pp. 966-970. IEEE, 2022.

Project: Traffic Information Extraction

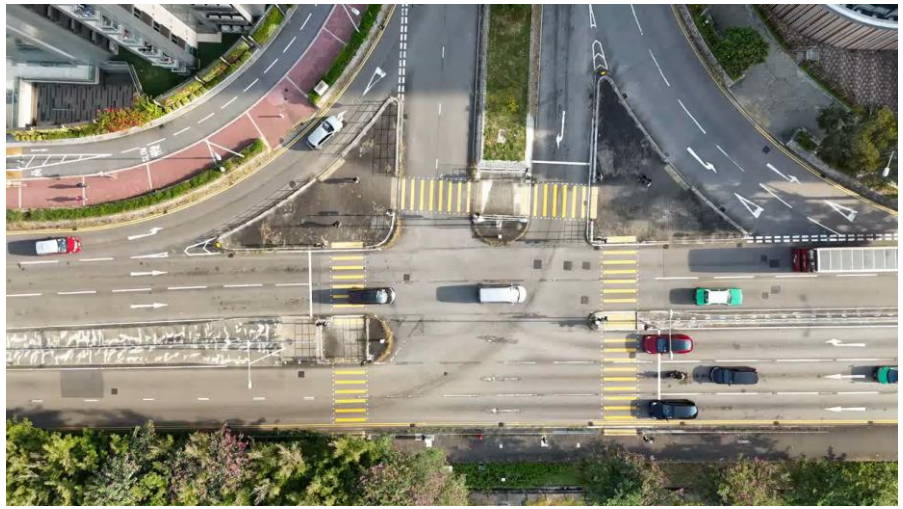








Vehicle detection



Speed estimation

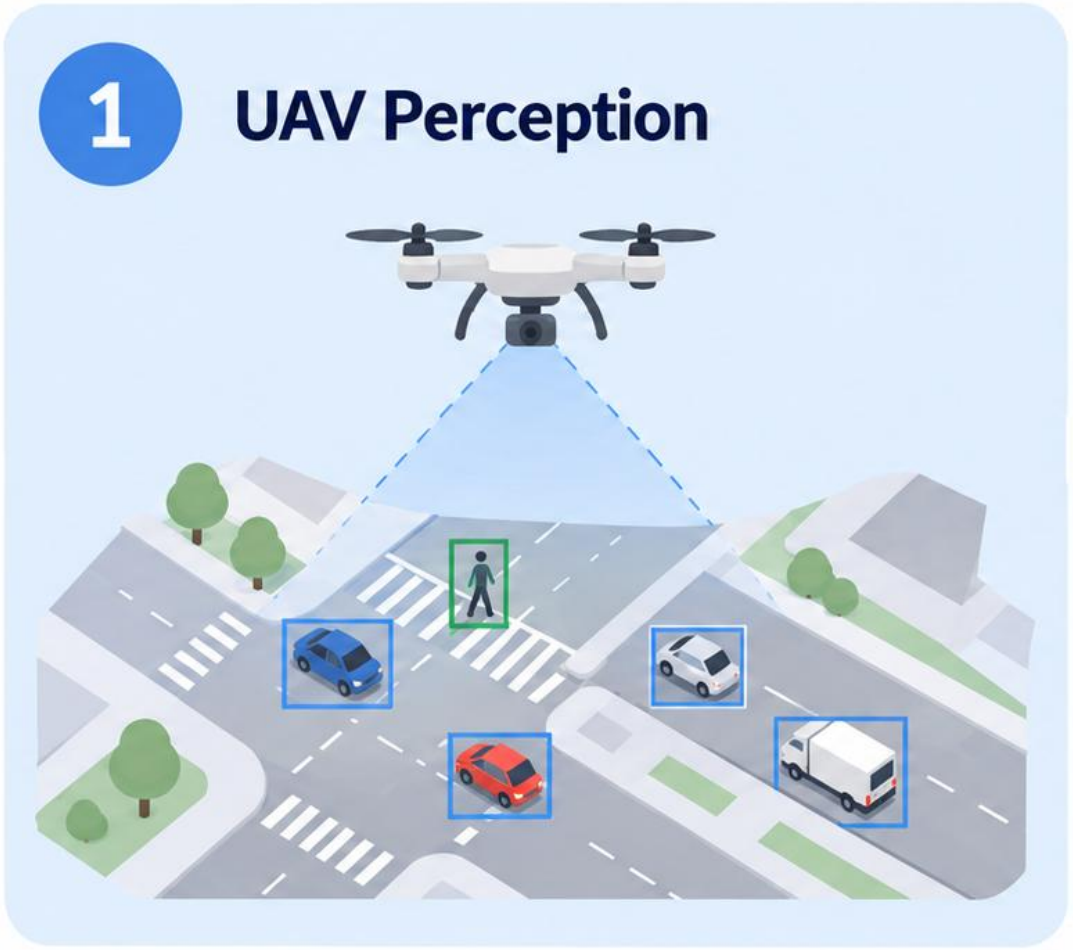
Trajectory Extraction

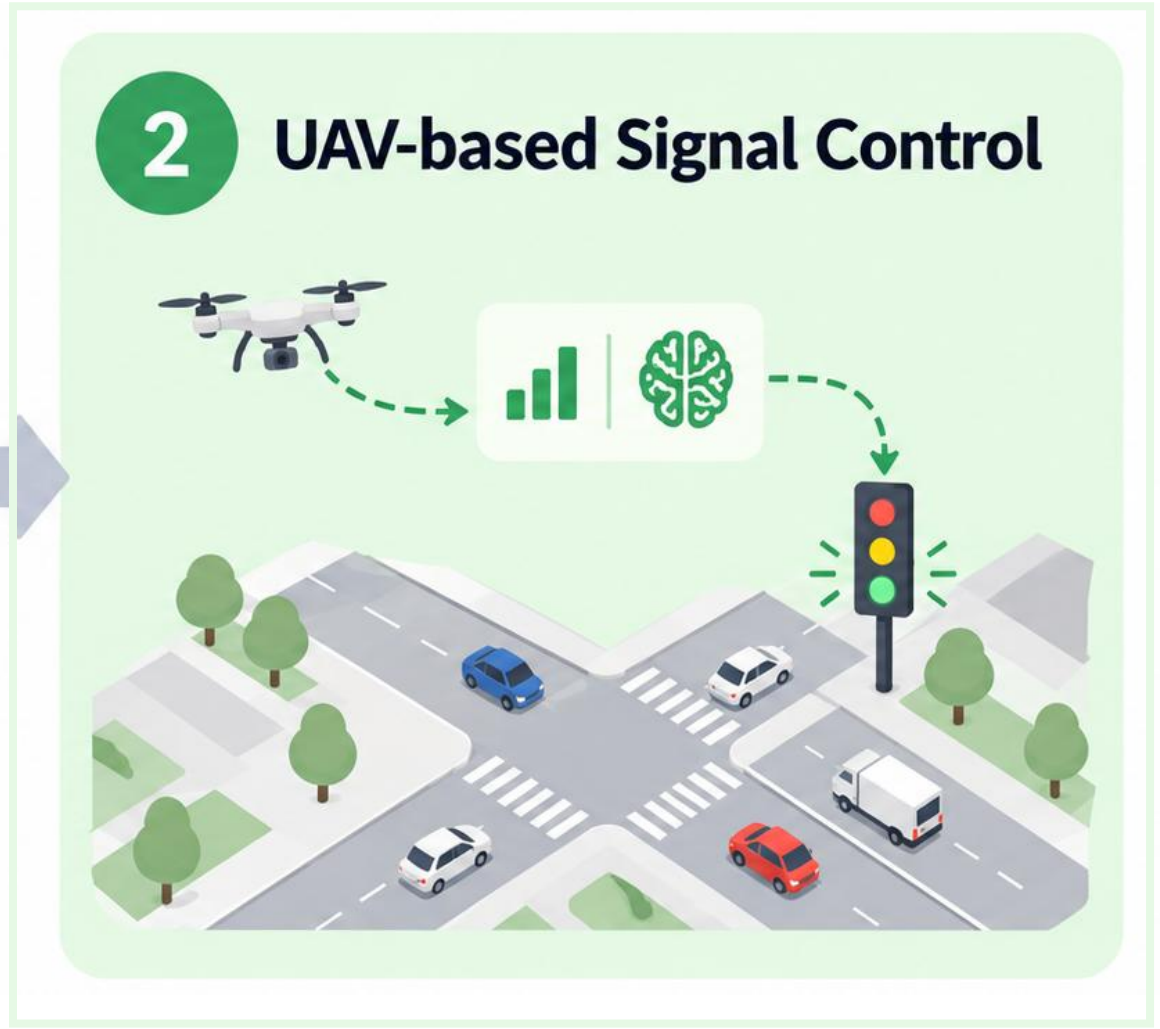


Conclusion

- UAVs give a low-cost, bird's-eye, flexible way to monitor traffic
- Object detection (YOLO) finds every vehicle — a box + a class, in a single pass
- Aerial imagery is hard (tiny, dense objects) → crop & zoom: ClusDet (learned) and SAHI (sliced)
- Detection + tracking turns raw footage into vehicle trajectories and traffic data
- Our project (STF) collects drone data and produces annotated trajectories

Roadmap





Outline

- Part I: Background & Motivation
- Part II: Preliminaries: How Traffic Signals Work
- Part III: RL-based Traffic Signal Control (TSC)
- Part IV: Generalizable RL-based TSC
- Part V: Large Language Model-based TSC
- Part VI: Vision Language Model-based TSC
- Part VII: Conclusion

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Traffic Signals as the City's Scheduler

Intersections are critical bottlenecks in urban networks

Objectives of traffic signal control:

- Reduce delay and queue length;
- Improve throughput and fairness;
- Support safety and priority vehicles;



Effective traffic signal control can significantly improve road efficiency

Traffic Signals History



1868
The first traffic light installed in London



1913
G. Morgan designed a cross-shaped traffic signal



1921
William Pott created the first three-section traffic light



1924
Traffic light towers appeared on Fifth avenue. They were twenty-five feet high and policemen operated them manually



1936
The Marshalite traffic signal used a dial and was created by Charles Marshall



Three-section traffic lights became dominant in traffic control



1990
Traffic signals acquire super-bright LEDs, which consume less power



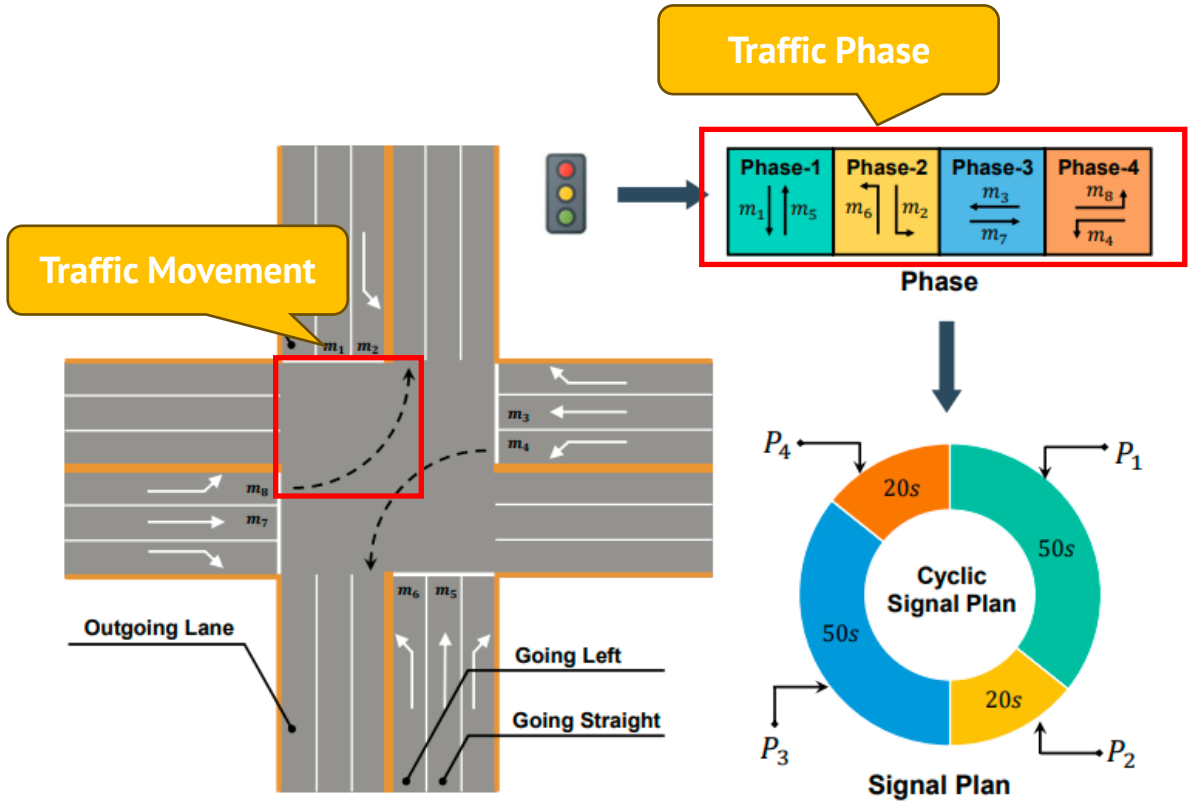
2008
Art. Lebedev Studio designed a traffic light with square sections

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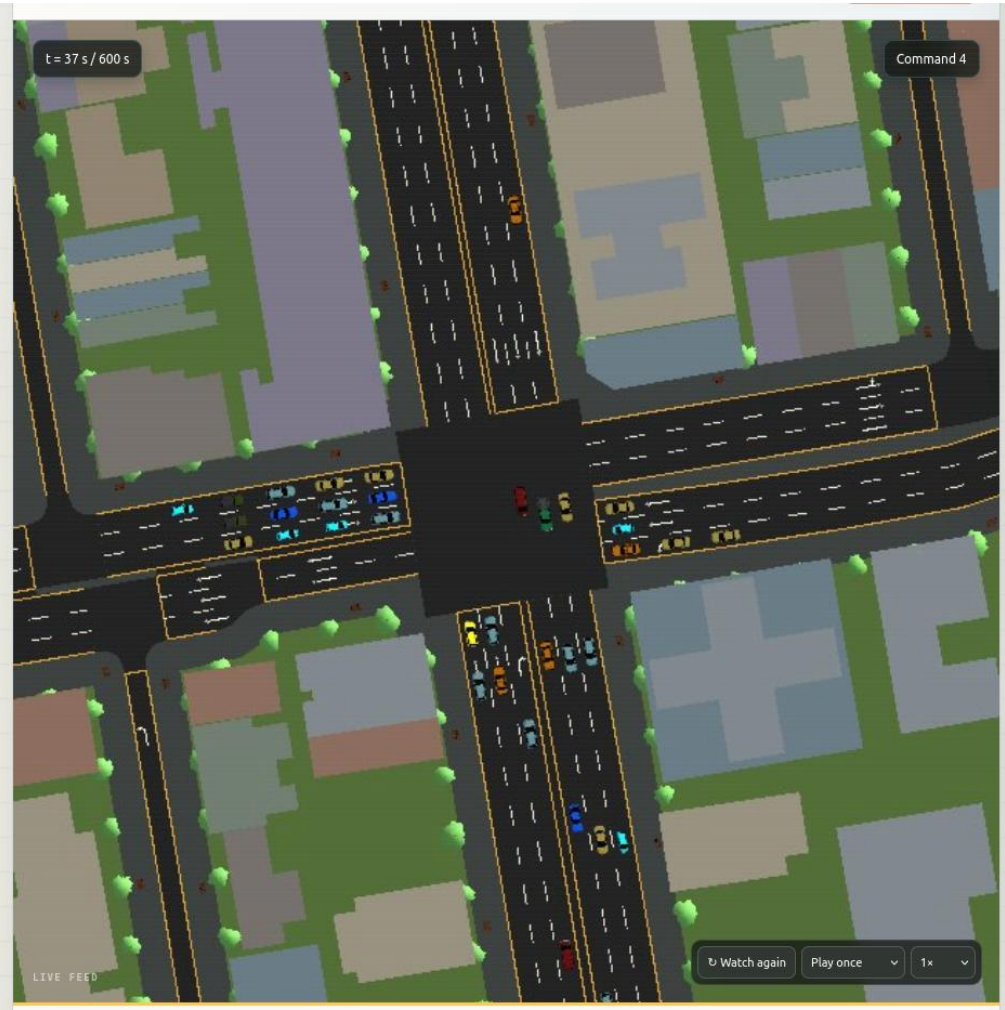
Key Concepts for TSC

- **Traffic Movement**
  - Connection from an incoming to an outgoing lane.
- **Movement Signal**
  - Green: Movement allowed.
  - Red: Movement prohibited.
- **Traffic Phase**
  - A set of movements allowed simultaneously.
- **Signal Plan**
  - Predefined sequence and timing of traffic phases.



TSC focuses on the ordering and timing of traffic phases

Key Concepts for TSC

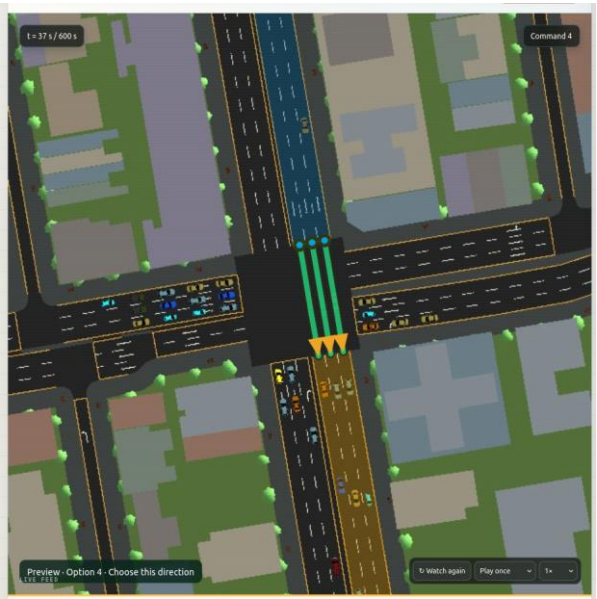




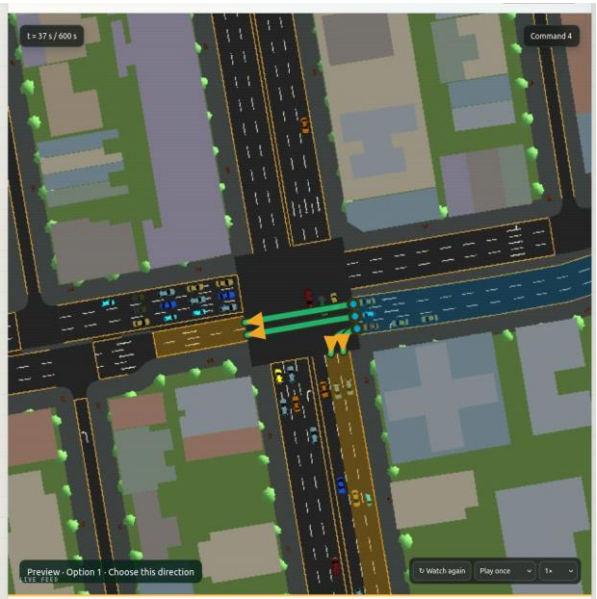
Key Concepts for TSC



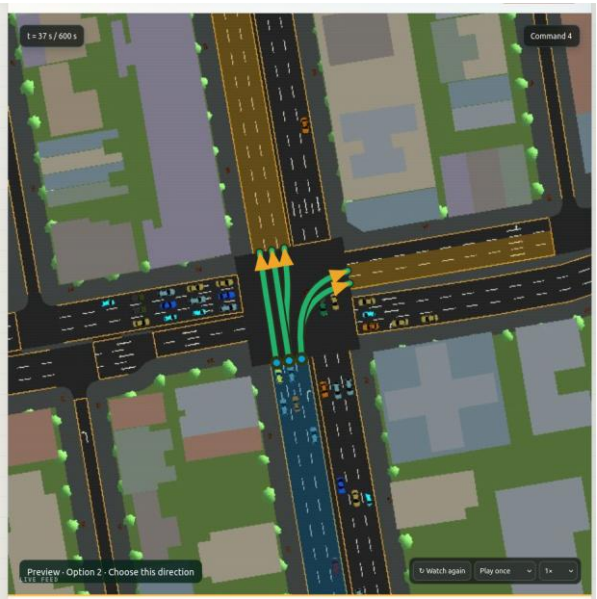
Phase-1



Phase-2

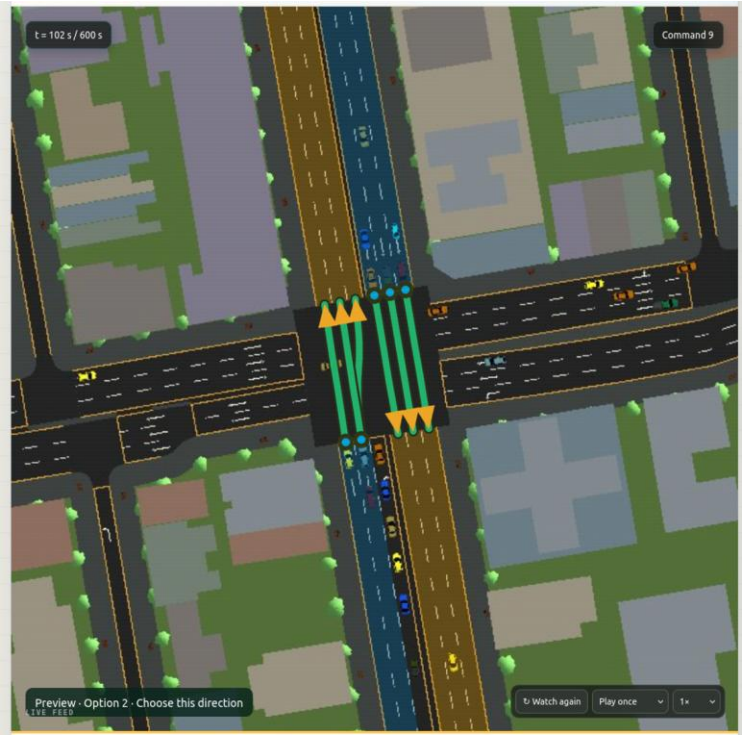


Phase-3



Phase-4

Key Concepts for TSC



Phase-1



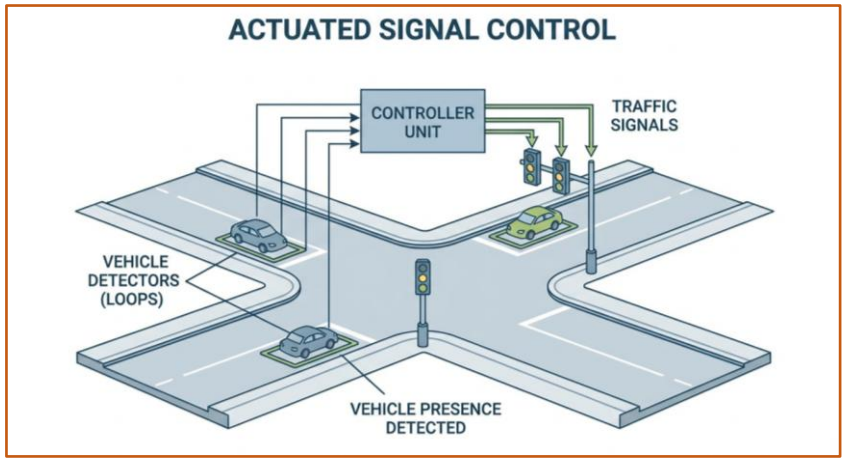
Phase-2



Phase-3

Existing Approaches to TSC

Traditional TSC Methods



1. Strengths

Simple and easy to deploy

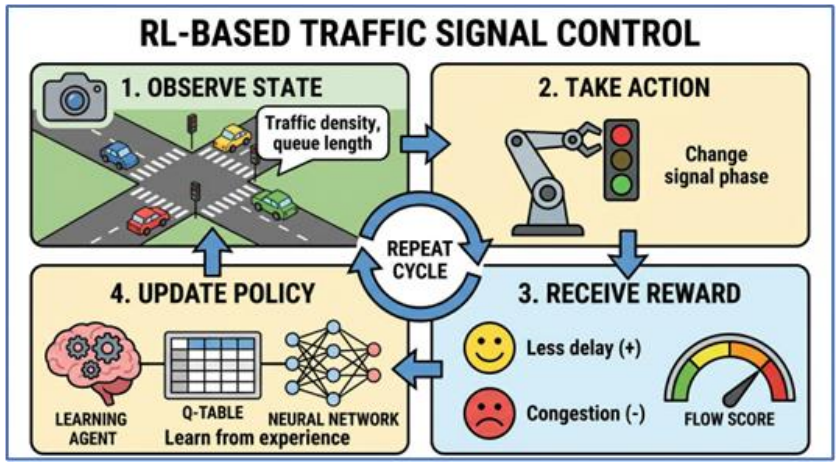
Low computational cost

2. Limitations

Limited adaptability to dynamic traffic patterns

Traffic models rely on strong assumptions

RL-based TSC Methods



1. Strengths

Ability to learn complex, nonlinear traffic dynamics

End-to-end decision making from states to actions

2. Limitations

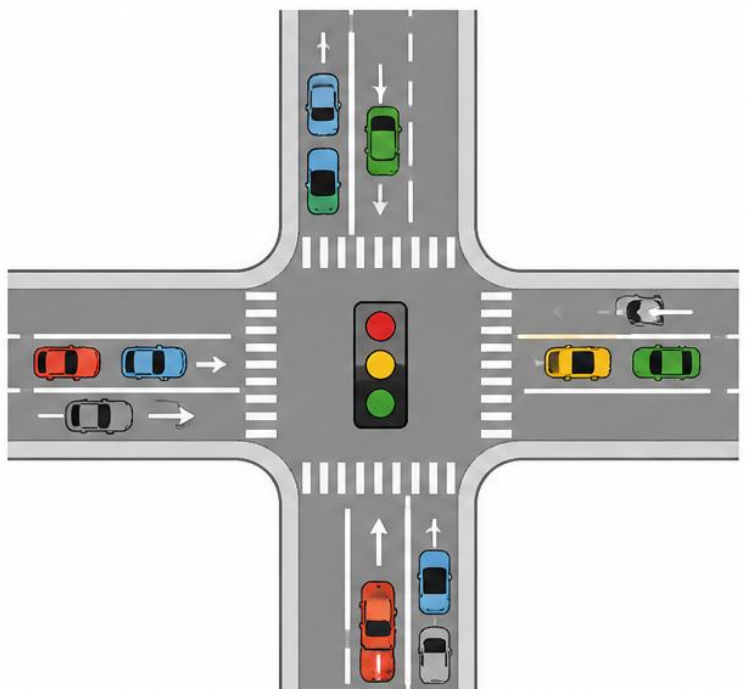
Poor generalization across intersections

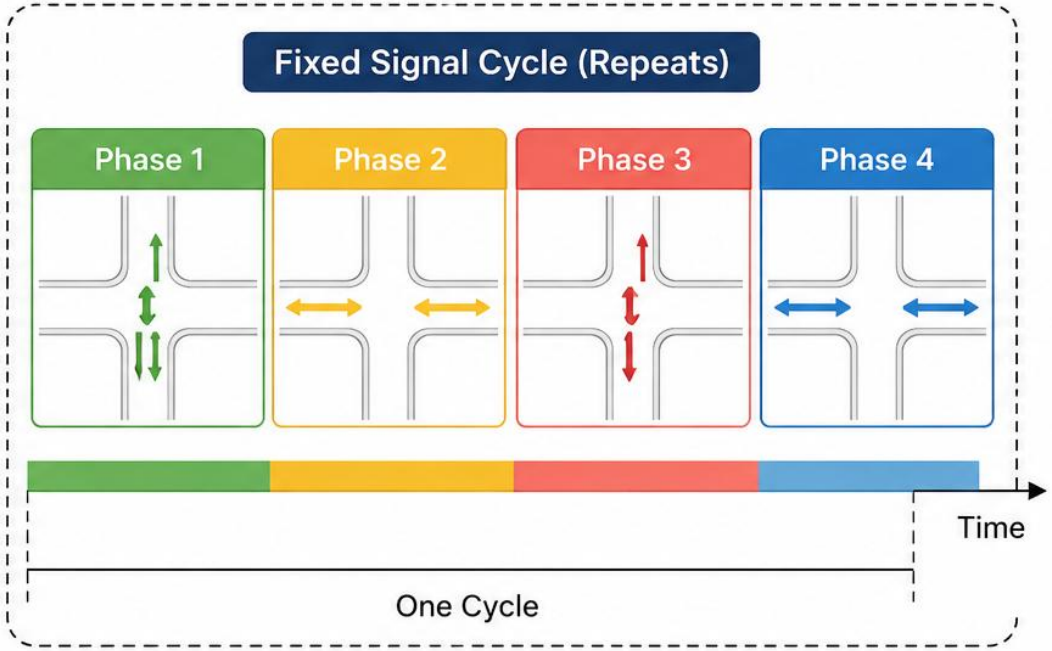
Difficulty handling rare and safety-critical scenarios

Existing Approaches to TSC – Fix-Time Control

Fix-Time Key Points:

1. The timing plans are set with fixed-time traffic signal cycles;
2. The traffic plan is not adjustment to the traffic demand





FV Webster and BM Cobbe. 1966. Traffic Signals. Road Research Technical Paper 56 (1966).

Existing Approaches to TSC – Actuated control

Actuated traffic signals come in three forms of control:

- **Semi-Actuated** is where the sensors are only on the direction of the side streets. The side streets default to red, and only turn green when a vehicle is detected by the sensor.
- **Full-Actuated** is where all directions have sensors. The traffic signal controller makes decisions based on the data supplied by the sensors.
- **Volume-Density** is similar to Full-Actuated, but it uses more information from the sensors to make its decision. For example, the traffic signal controller may choose to remain green until a variable minimum gap between vehicles is achieved

Existing Approaches to TSC – Practical systems

TRANSYT, SCATS, RHODES …





Refer:

- Stevanovic, Aleksandar. Adaptive traffic control systems: domestic and foreign state of practice. No. Project 20-5 (Topic 40-03). 2010.

🚦 Traffic Police Challenge – Game Time !!!

- **Can you beat the built-in signal controller?**
- **Select 5 students,** act as traffic police to control lights in real time.
- **<u>Your Goal</u>**
  - Observe the traffic situation
  - Decide **which direction gets green**
  - Reduce congestion and vehicle waiting time



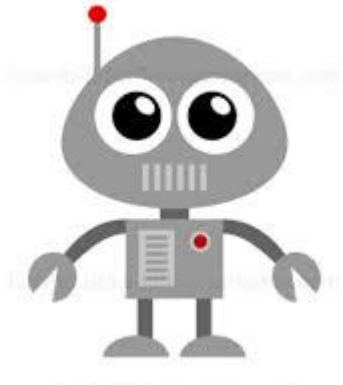
Outline

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- Part II: Preliminaries: How Traffic Signals Work
- **Part III: RL-based Traffic Signal Control (TSC)**
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- Part V: Large Language Model-based TSC
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- Part VII: Conclusion



https://www.youtube.com/watch?v=8dMFJpEGNLQ

RL-based Traffic Signal Control -- Background

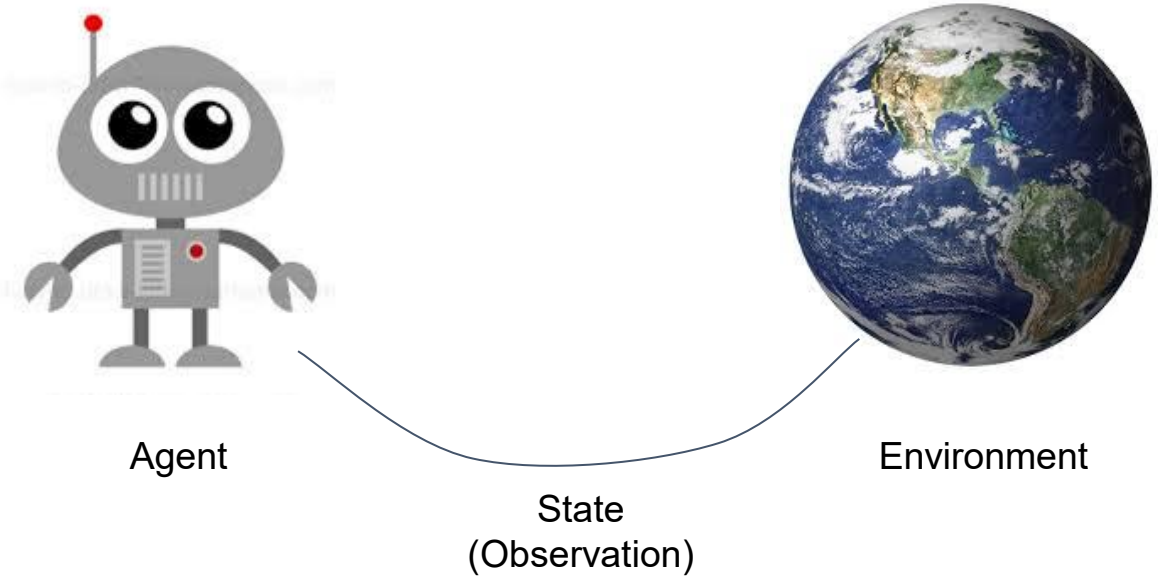


Agent

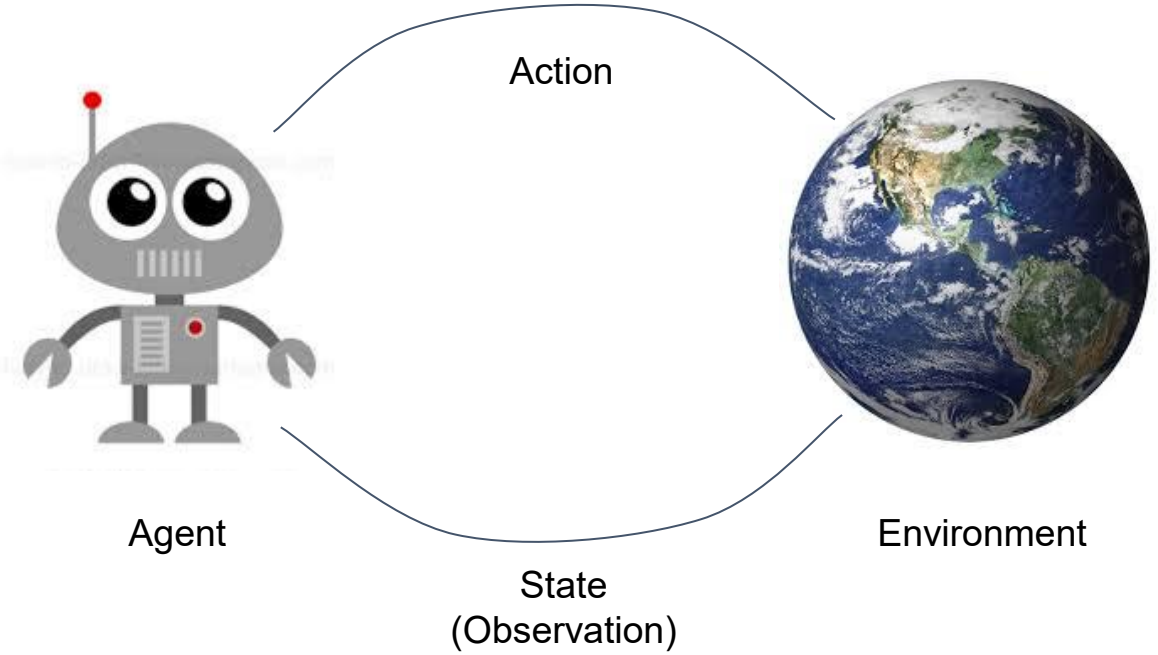


Environment

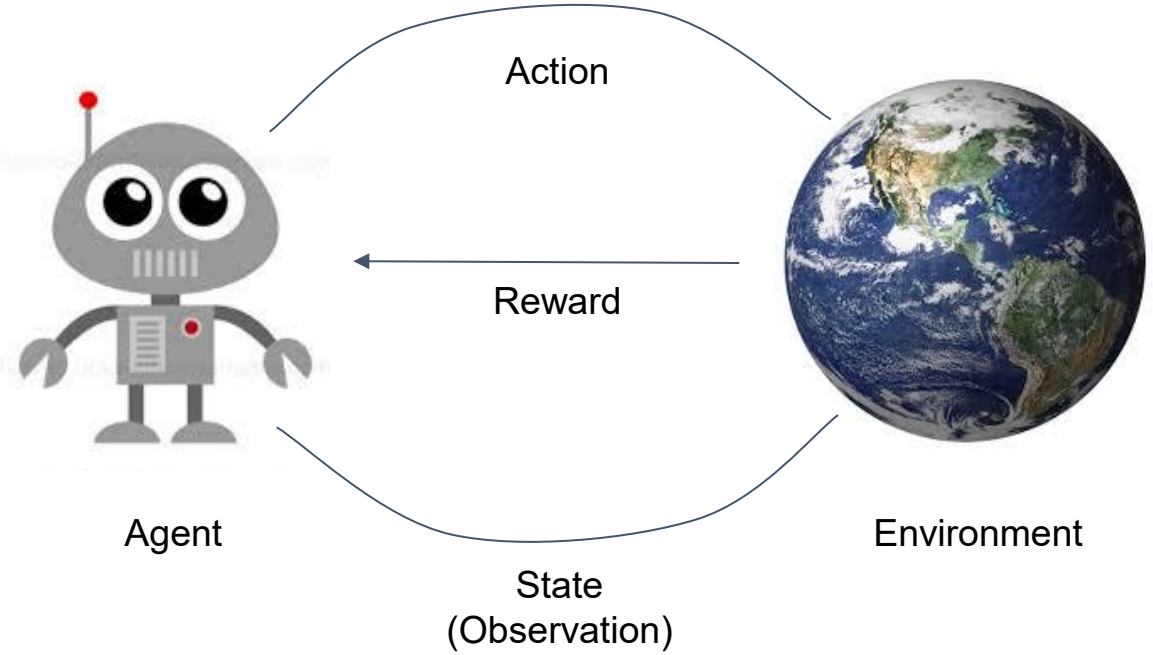
RL-based Traffic Signal Control -- Background



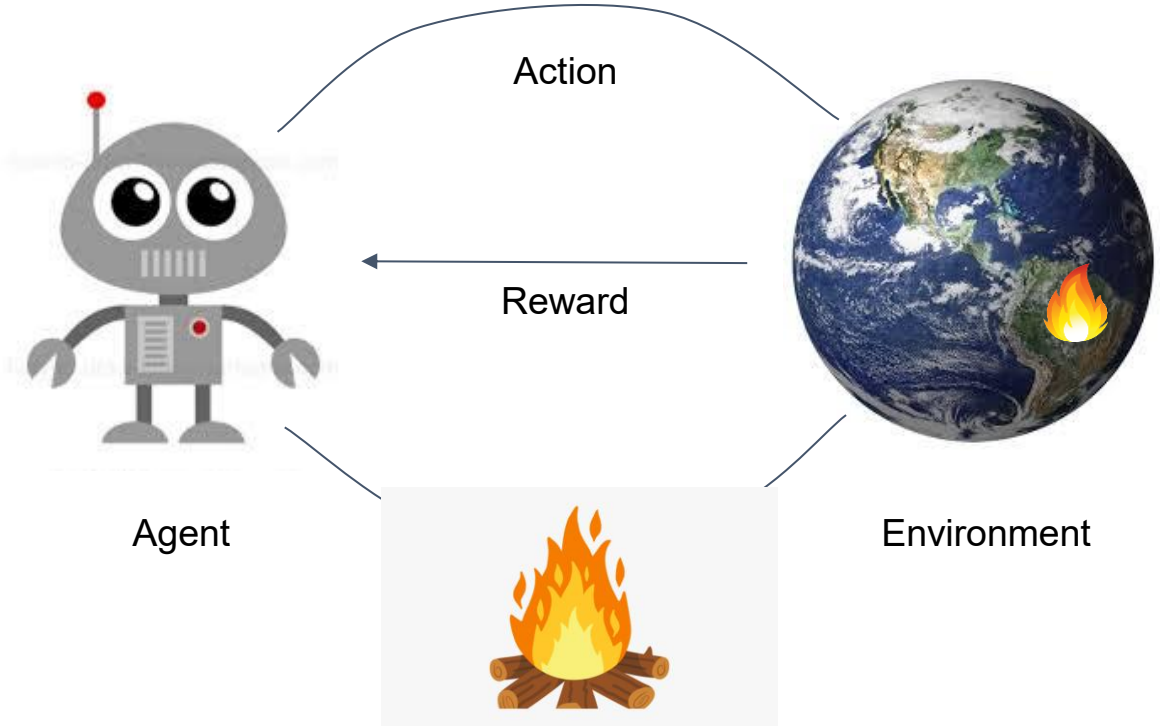
RL-based Traffic Signal Control -- Background



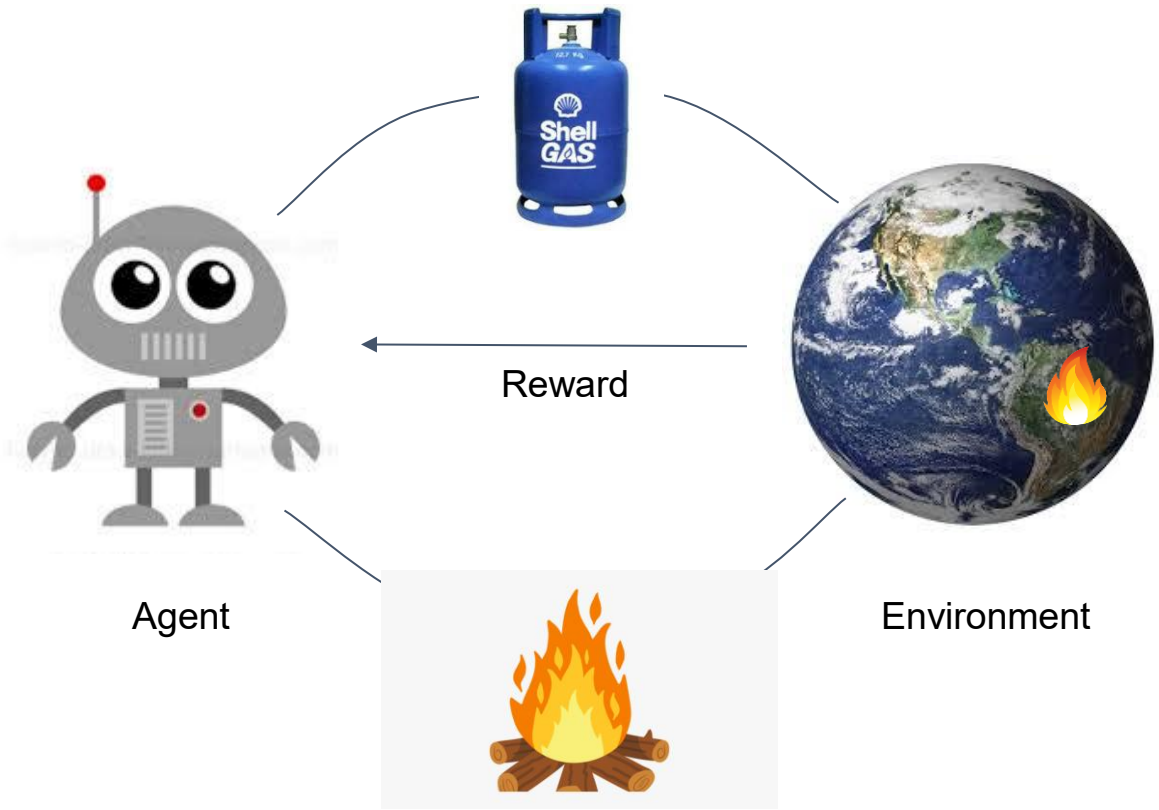
RL-based Traffic Signal Control -- Background



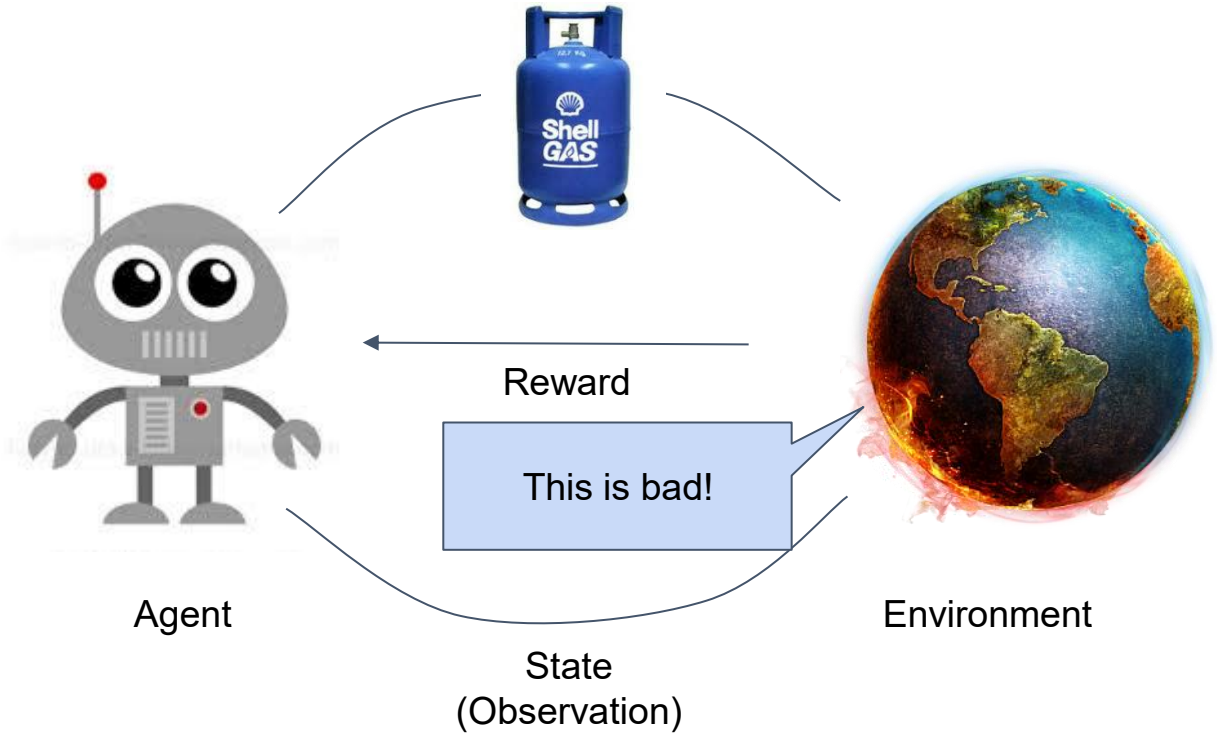
RL-based Traffic Signal Control -- Background



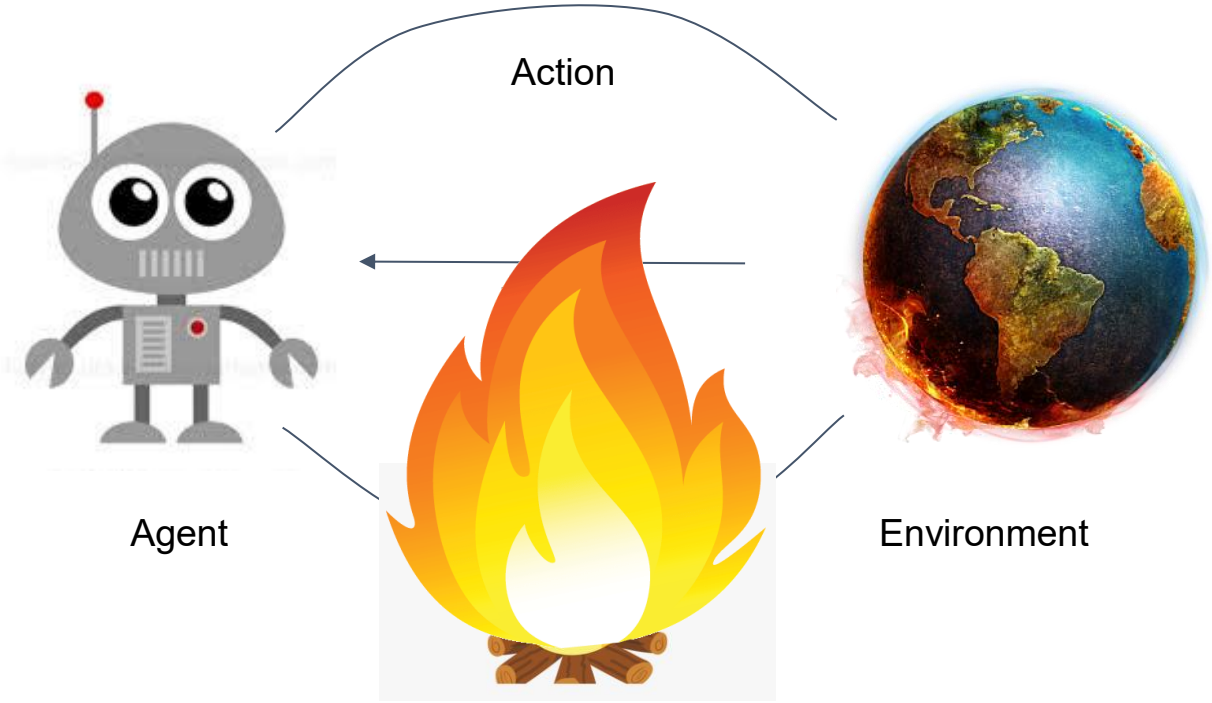
RL-based Traffic Signal Control -- Background



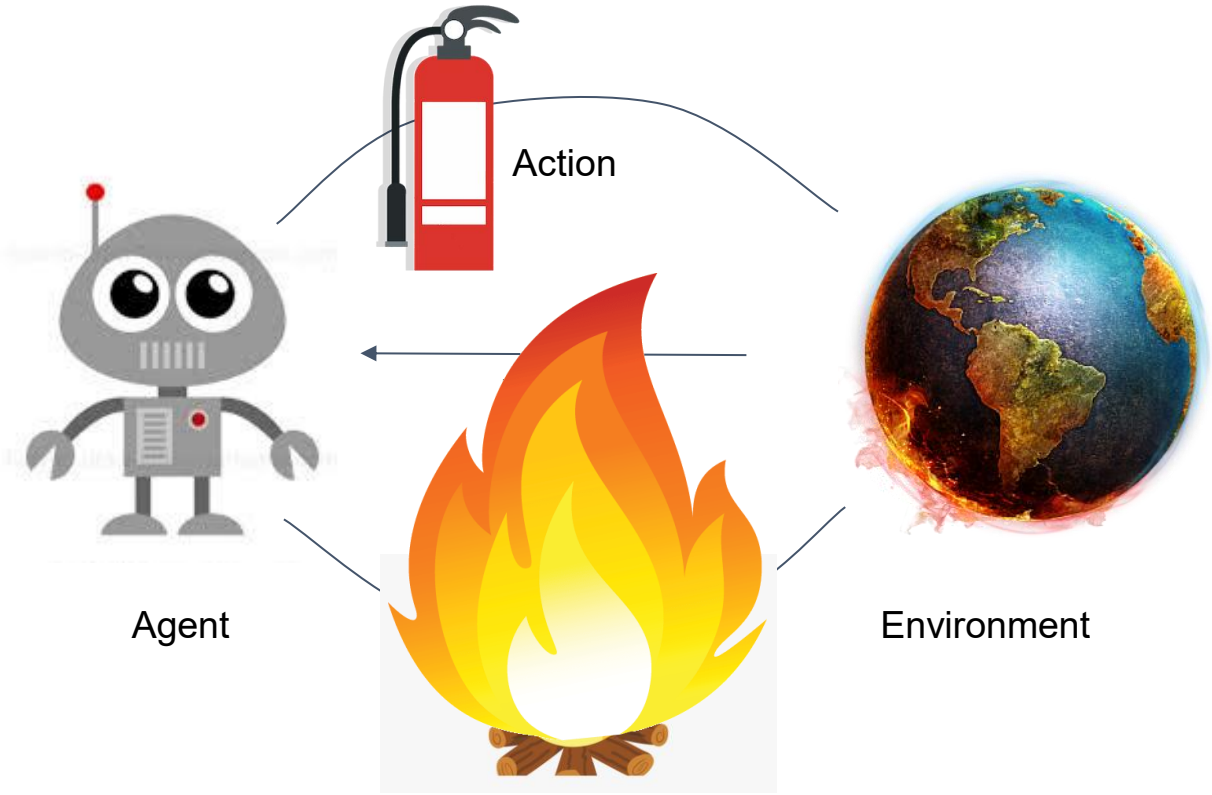
RL-based Traffic Signal Control -- Background



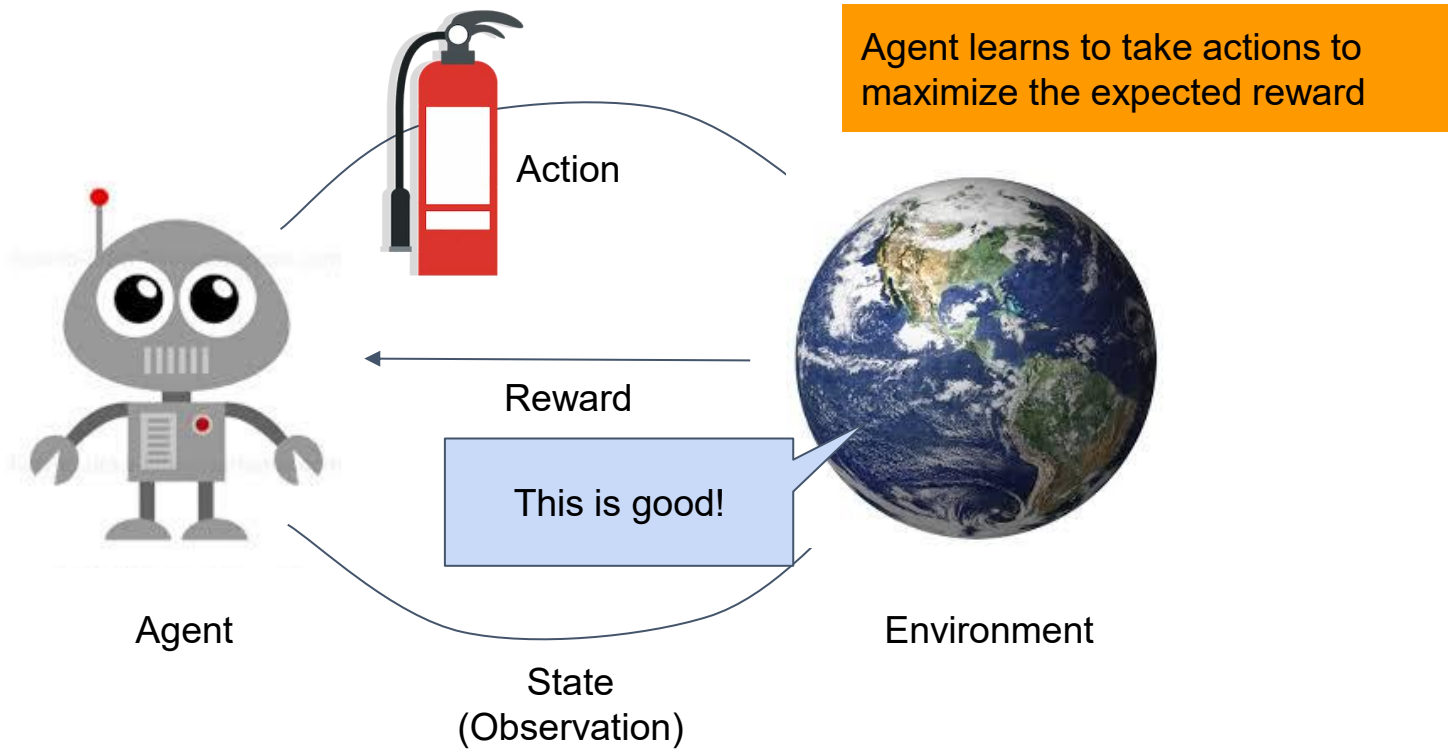
RL-based Traffic Signal Control -- Background



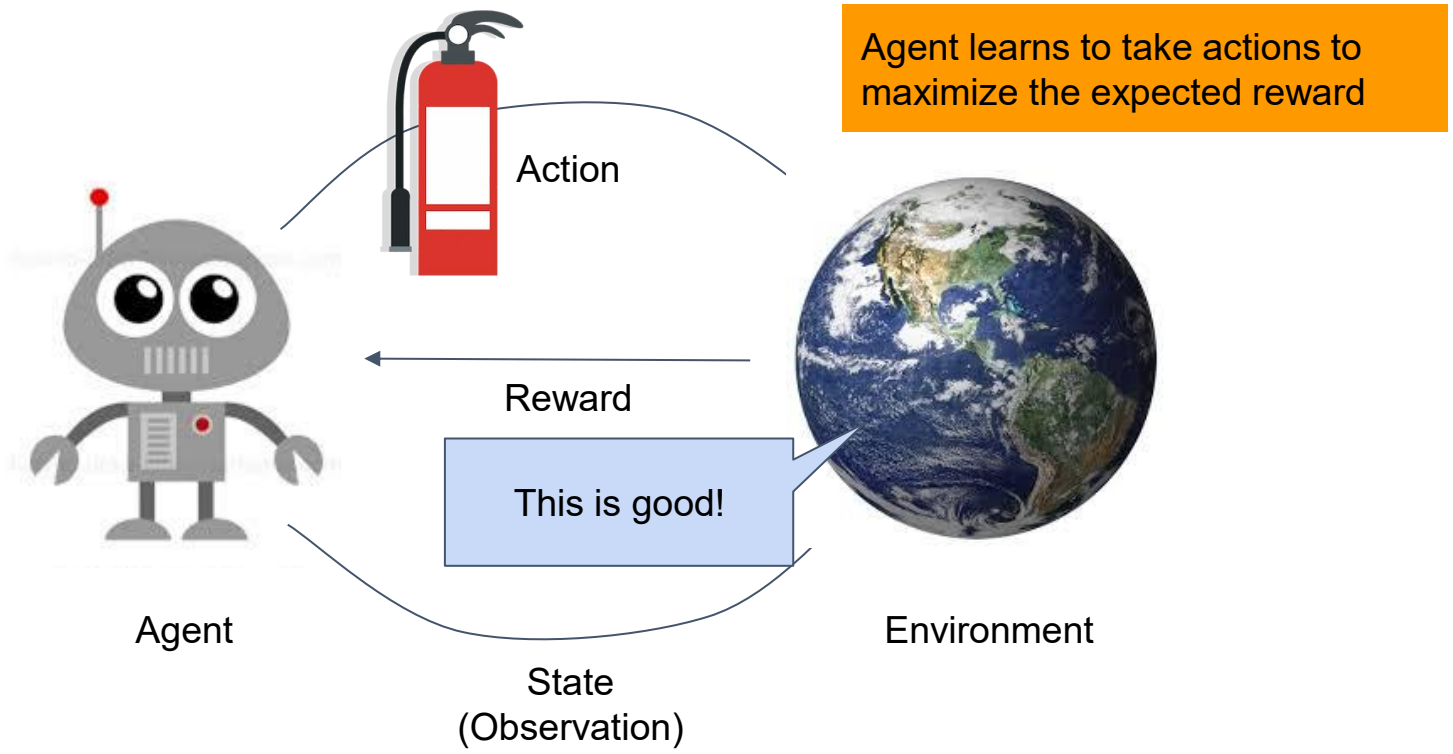
RL-based Traffic Signal Control -- Background



RL-based Traffic Signal Control -- Background

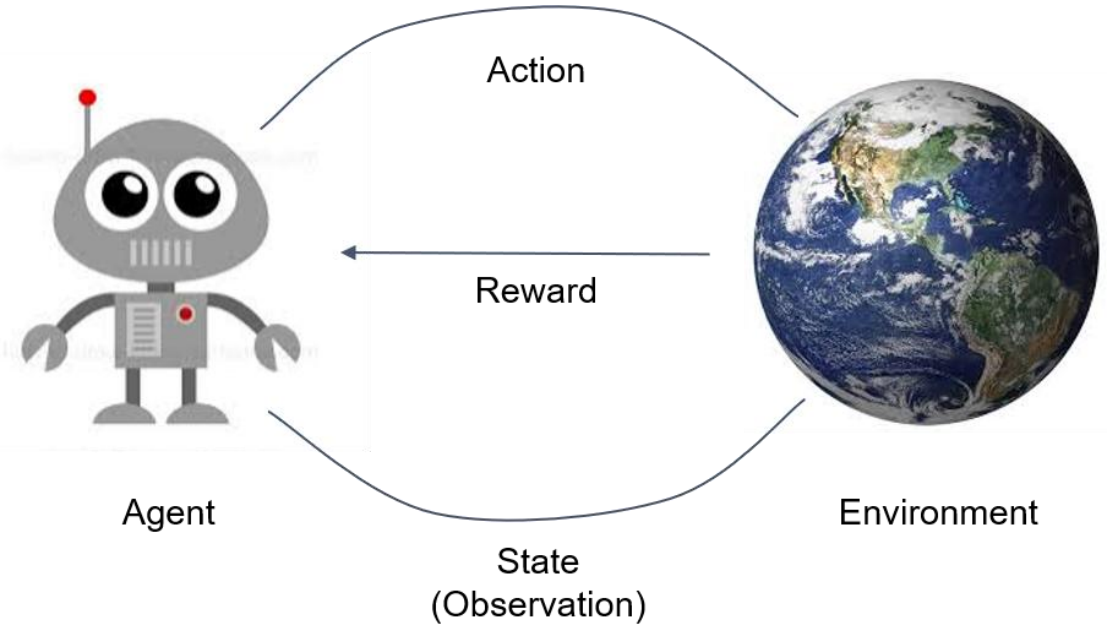


RL-based Traffic Signal Control -- Background

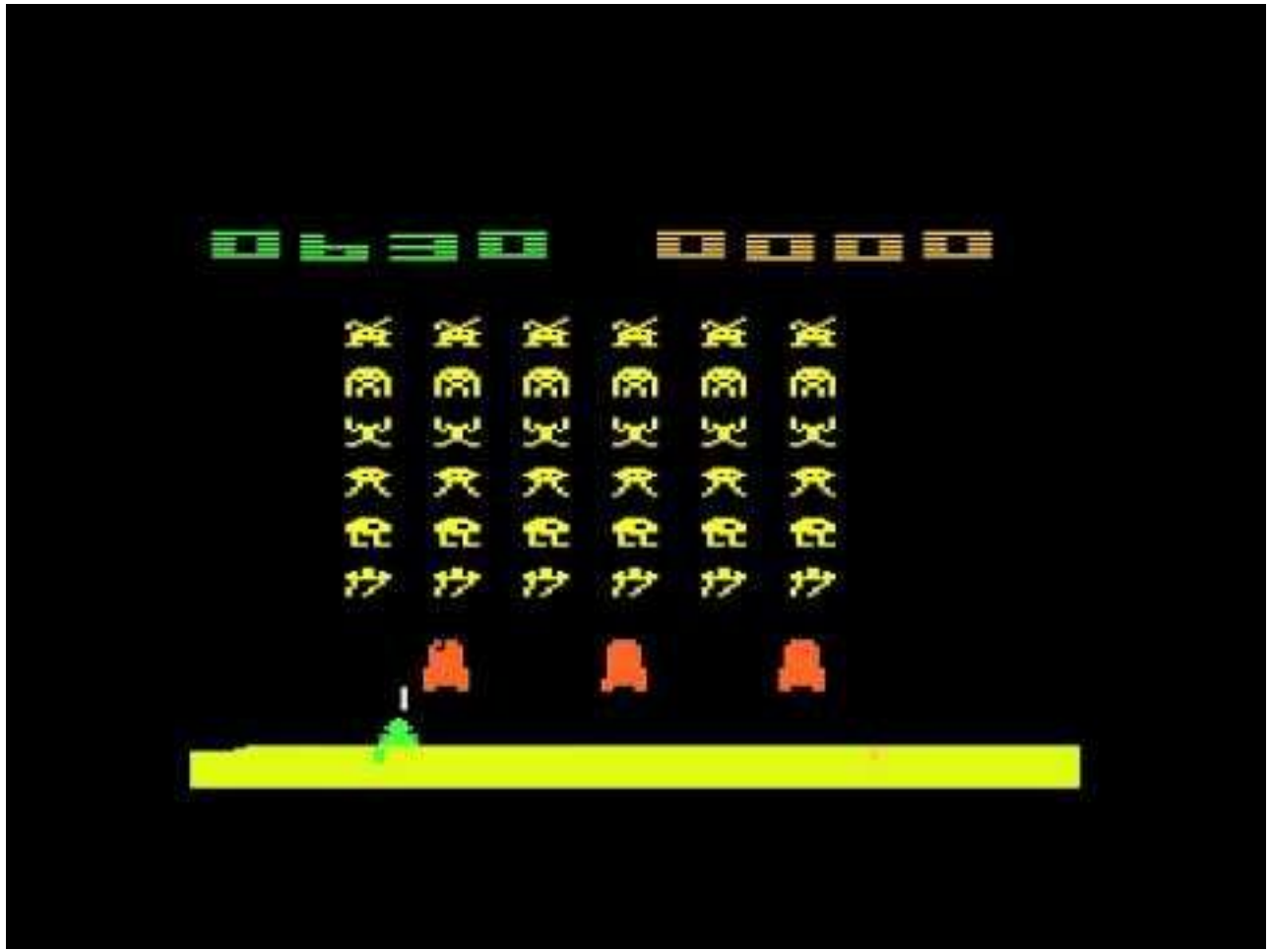


Key Components of RL Agents

- **State**
  - Feature representation of the observation on the env
- **Action**
  - Actions may have long-term consequences in sequential decision making process
- **Reward**
  - A scalar feedback signal
  - Reward maybe delayed



Example: Learning to Play Video Game

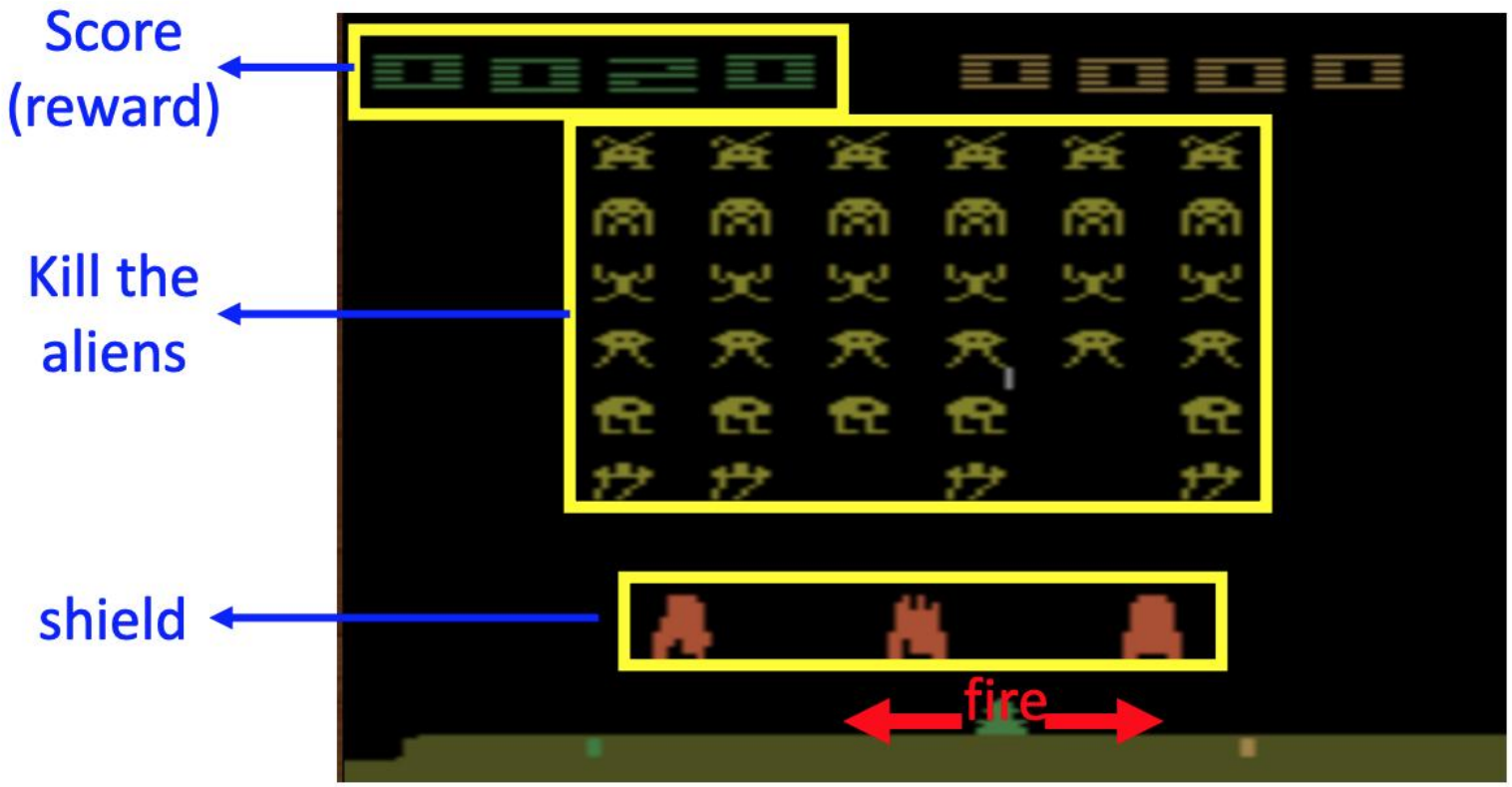


Space Invader

Example: Learning to Play Video Game

- Space invader

Termination: all the aliens are killed, or your spaceship is destroyed.



Example: Learning to Play Video Game

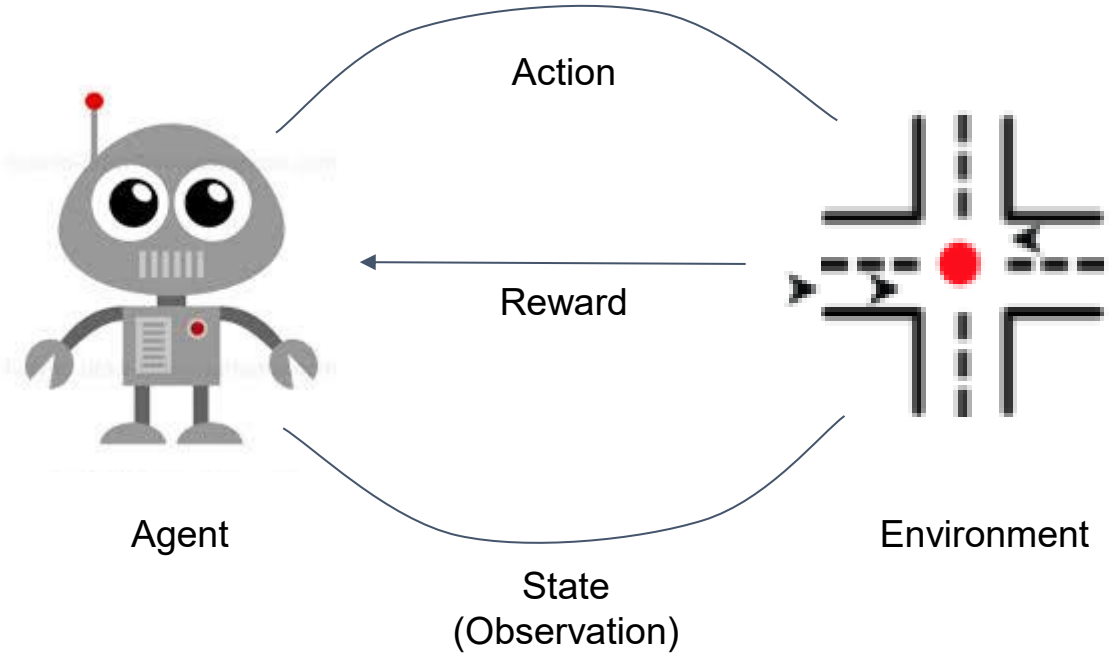


https://www.youtube.com/watch?v=TmPfTpjtdgg

Example: Learning to Control Traffic Signals

What's your state, action and reward?

Goal: take actions to minimize the travel time of all vehicles.



Example: Learning to Control Traffic Signals

- **State Choices**

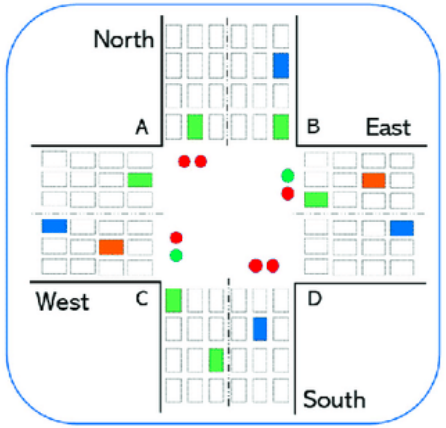
1. **Queue length**
2. Waiting time
3. Volume
4. Delay
5. Speed
6. **Positions of vehicles**
7. Phase duration, Congestion, Phase

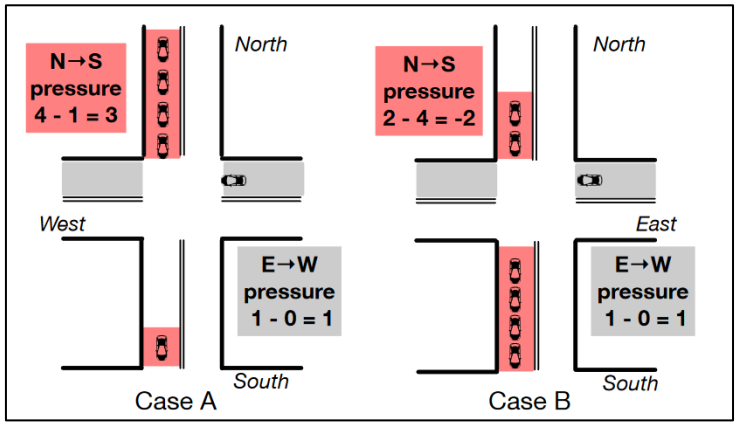
- **Reward Choices**

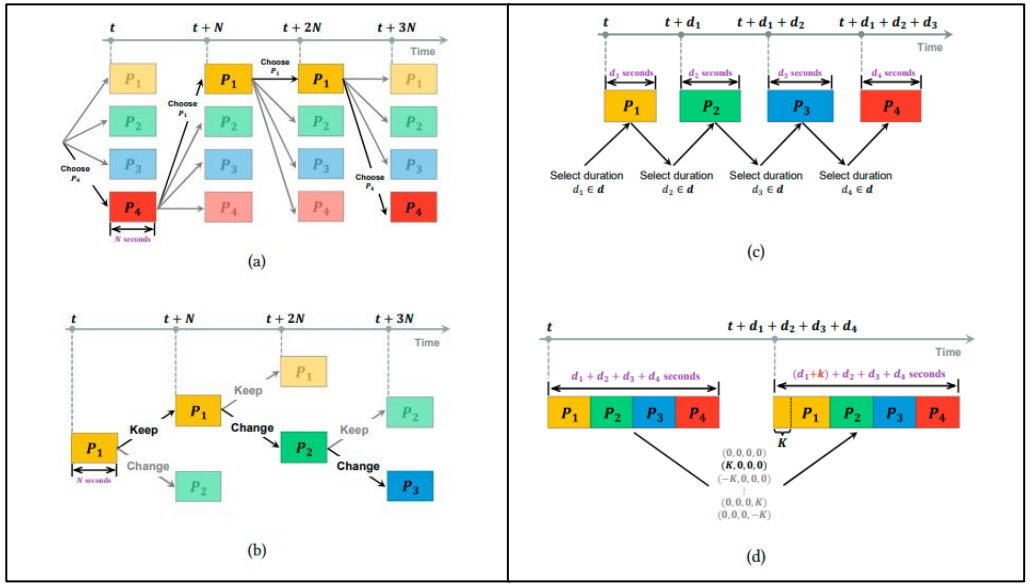
1. Queue length
2. Waiting time
3. Speed
4. Throughput
5. **Frequency of change**
6. Accident avoidance
7. **Pressure**

- **Action Choices**

1. Choose Next Phases
2. Next or Not
3. Set Phase Duration
4. Adjust Single Phase Duration







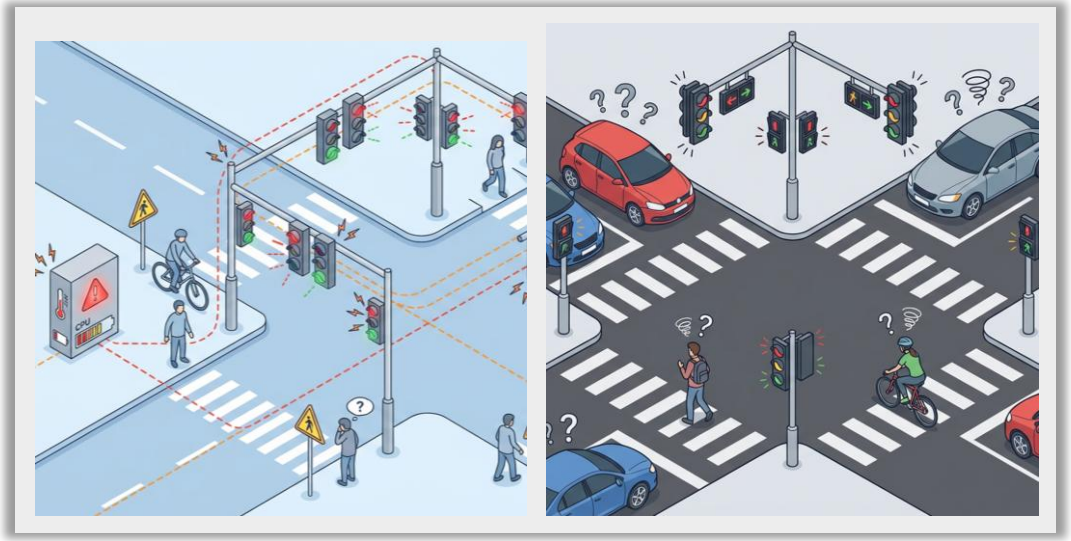
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Generalizable TSC under Constraints



Challenge 1: Generalization for RL-based TSC

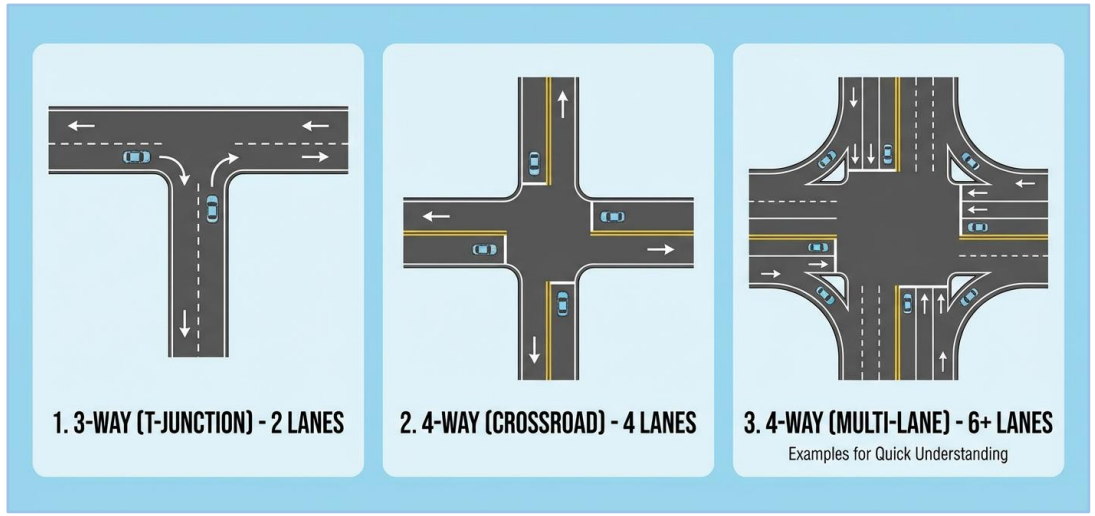


Challenge 2: TSC under Limited Computation Resources

Practical RL-based TSC requires both strong generalization to unseen scenarios and efficient decision-making under limited computation.

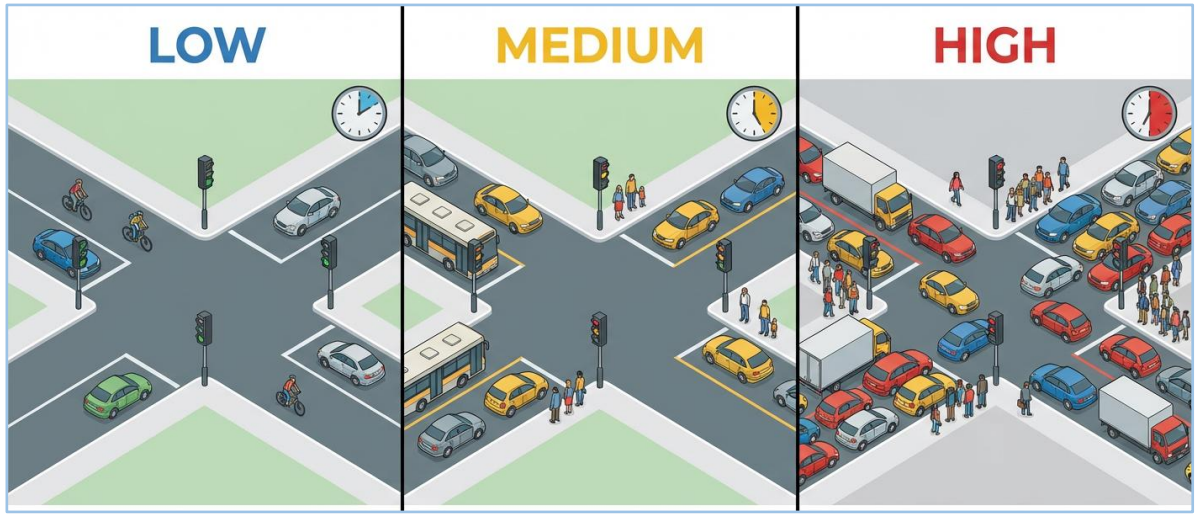
Generalizable TSC under Constraints

Challenge 1: Generalization for RL-based TSC



Different junction layouts

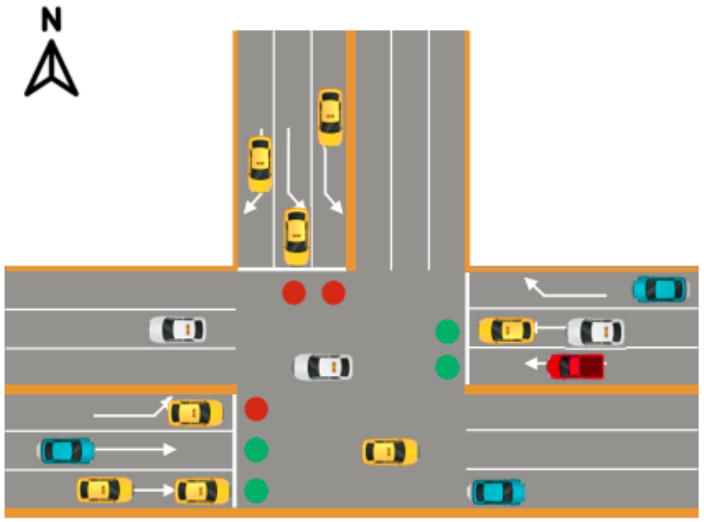
(3-way vs. 4-way, varying lane numbers)

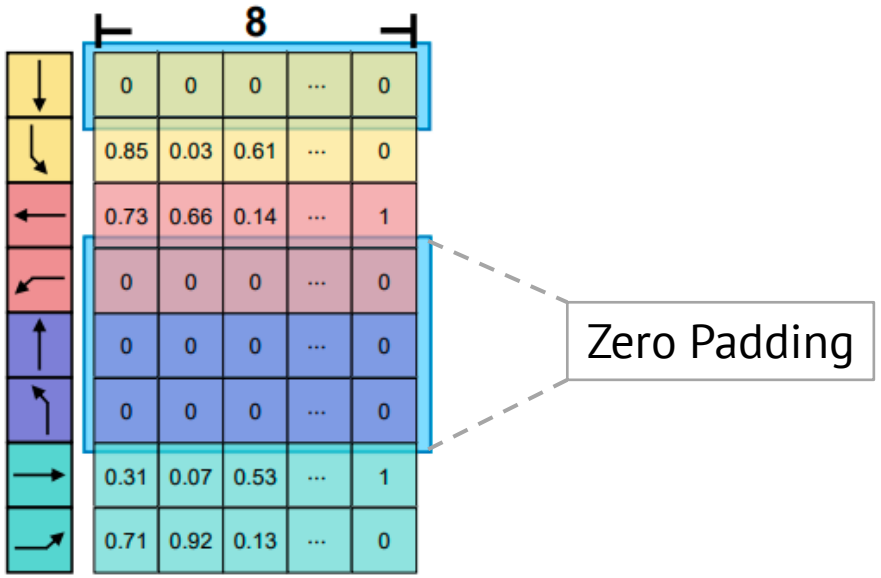


Different traffic flow patterns

Generalizable TSC under Constraints

A Unified Representation for Heterogeneous Intersections

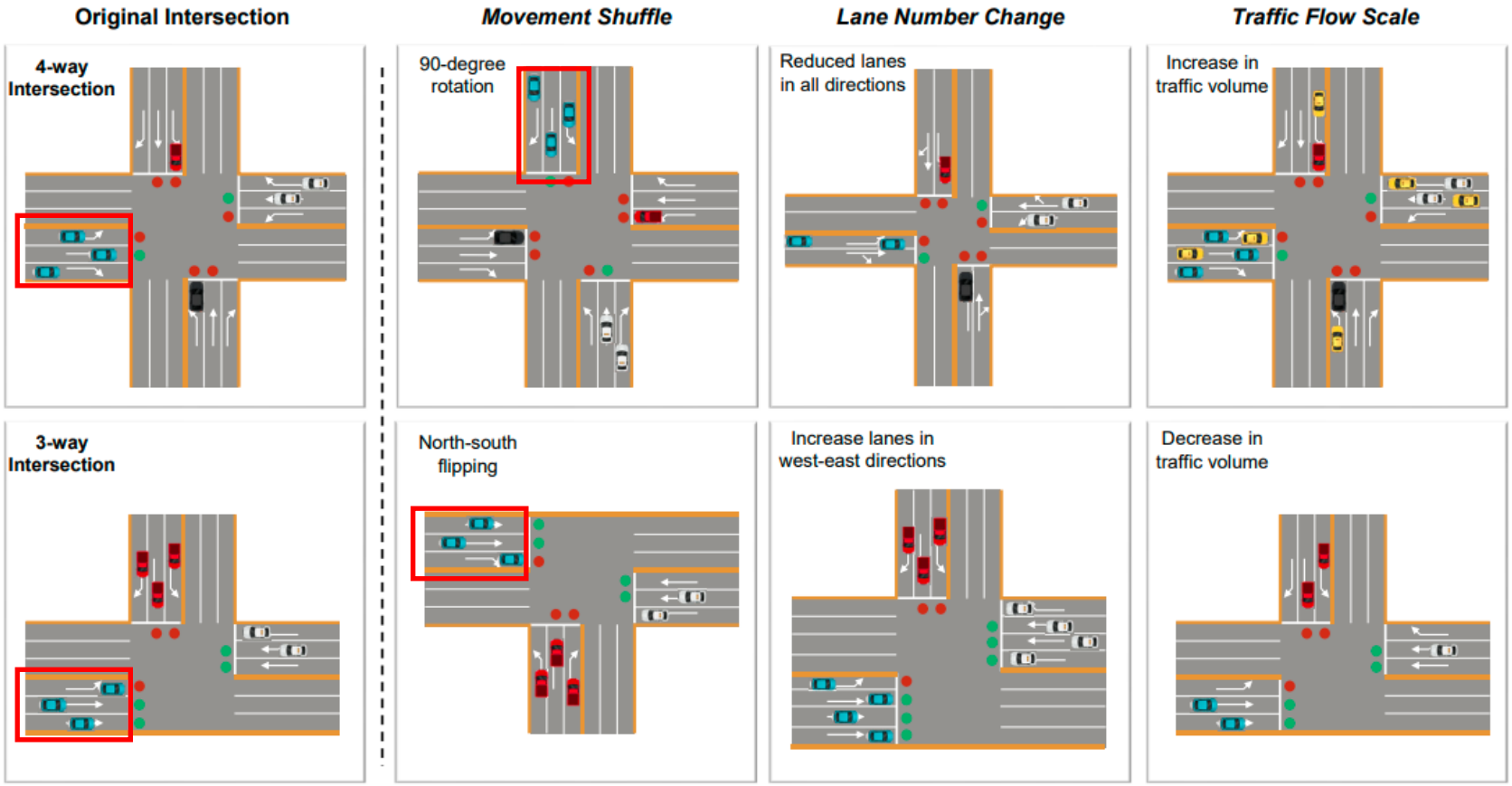


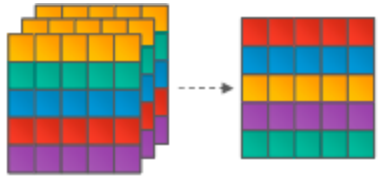


- Extract traffic information from 8 movements (Fixed movement representation, independent of lane count);
- Supports both 3-way and 4-way intersections via zero padding;
- Include traffic state and signal state per movement, such as *queue length, lane number, and signal status*;

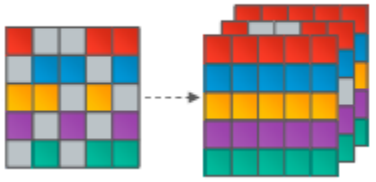
Generalizable TSC under Constraints

Improve Generalization to Unseen Traffic Scenarios





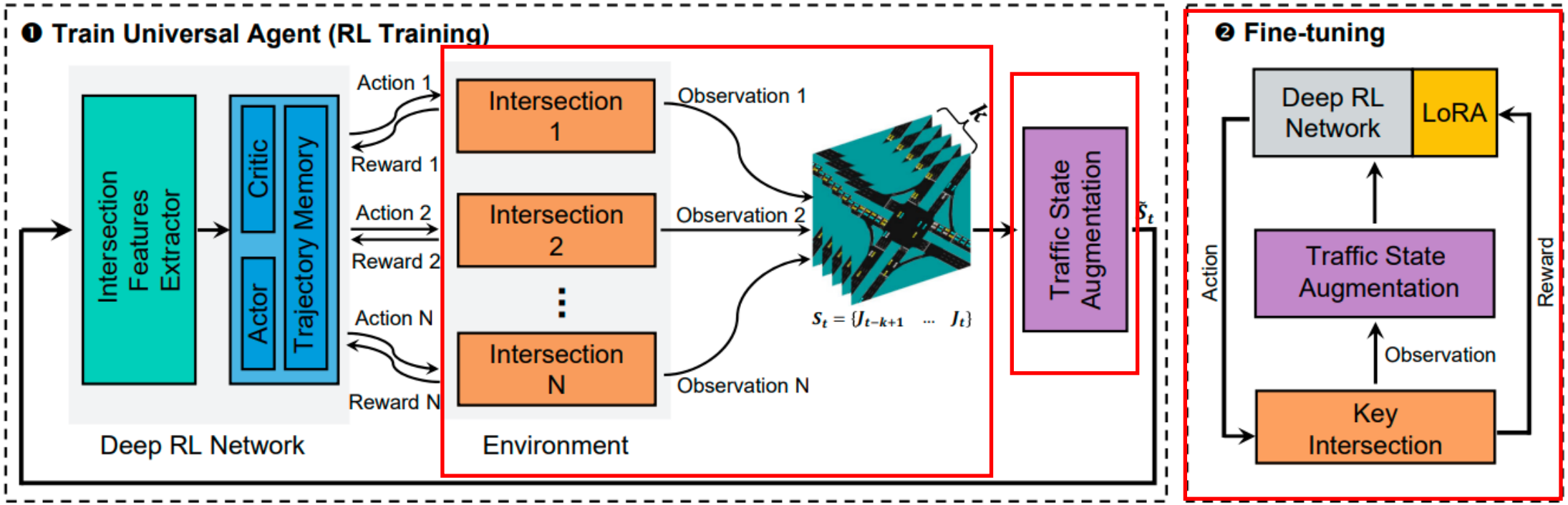
Row Shuffle



Matrix Mask

Rotating or mirroring an intersection **does not change the optimal control decision**

Generalizable TSC under Constraints



Training Phase

Unified junction matrix representation

Structural traffic state augmentation

Adaptation Phase

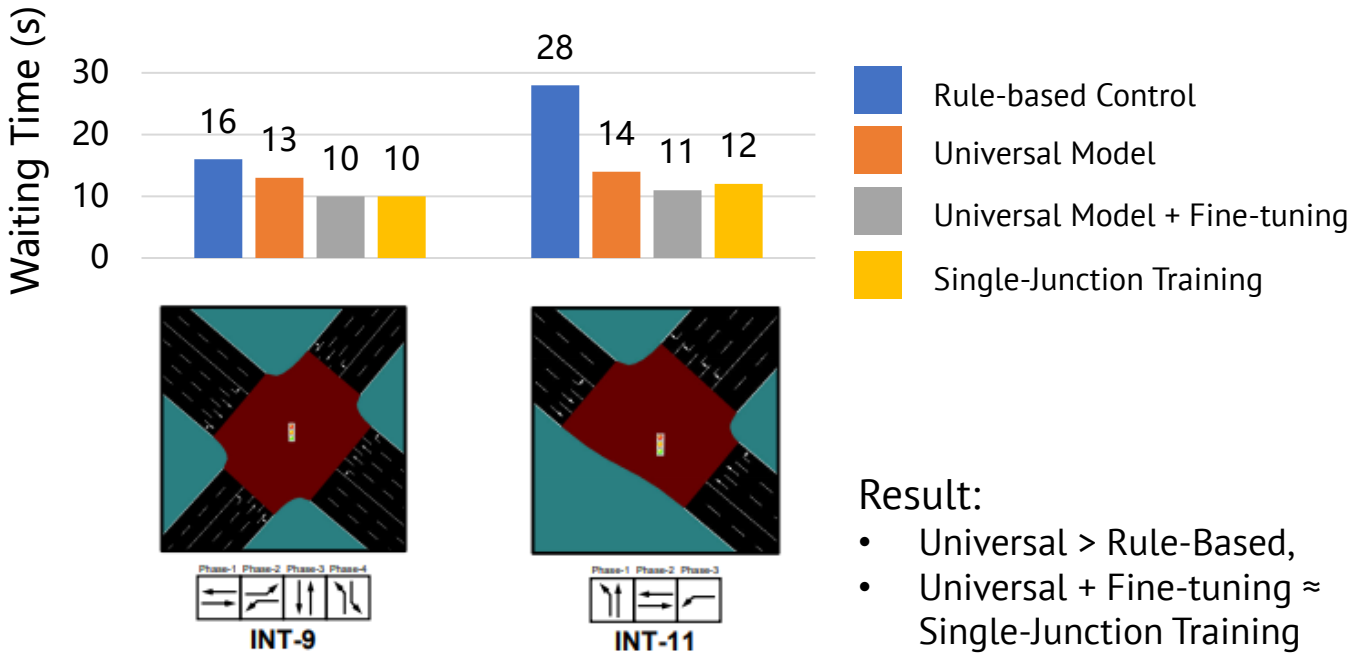
Based on pretrained general policy

LoRA-based fine-tuning

Generalizable TSC under Constraints

Converge Speed

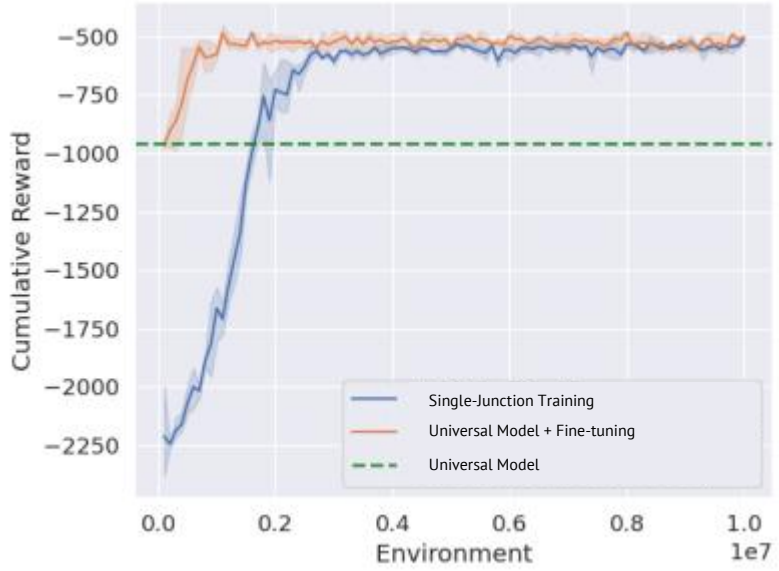
Performance on Unseen Junctions



Result:

- Universal > Rule-Based,
- Universal + Fine-tuning ≈ Single-Junction Training

Training Efficiency



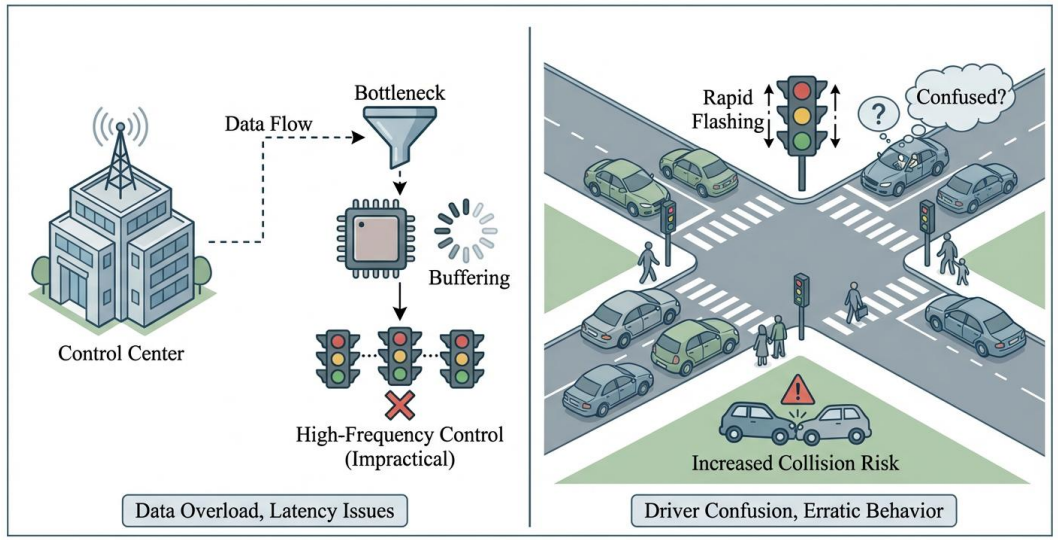
The universal model enables strong generalization to unseen junctions with significantly reduced training cost, while fine-tuning provides efficient specialization.

Generalizable TSC under Constraints

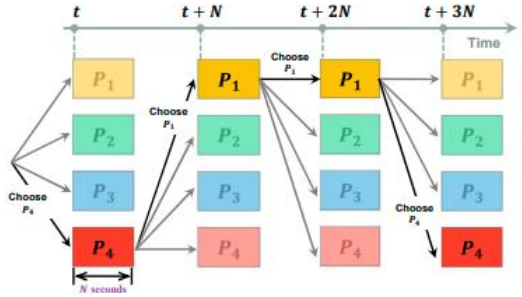
Challenge 2: TSC under Limited Computation Resources

High-frequency decision making is impractical in real-world systems.

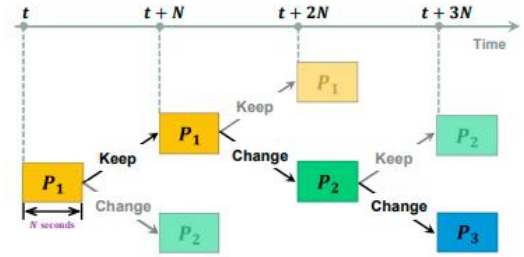
- Limited computational and communication resources;
- Safety concerns with frequent signal change;



High-Frequency Control in Existing Methods



Choose Next Phase



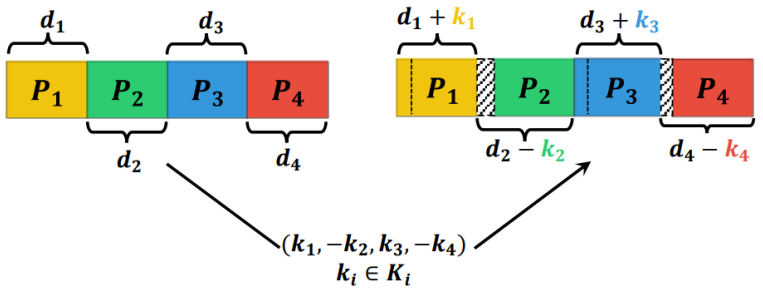
Keep or Change

Decisions are made every **5–10 seconds**

There is a mismatch between real-world constraints and existing high-frequency control methods.

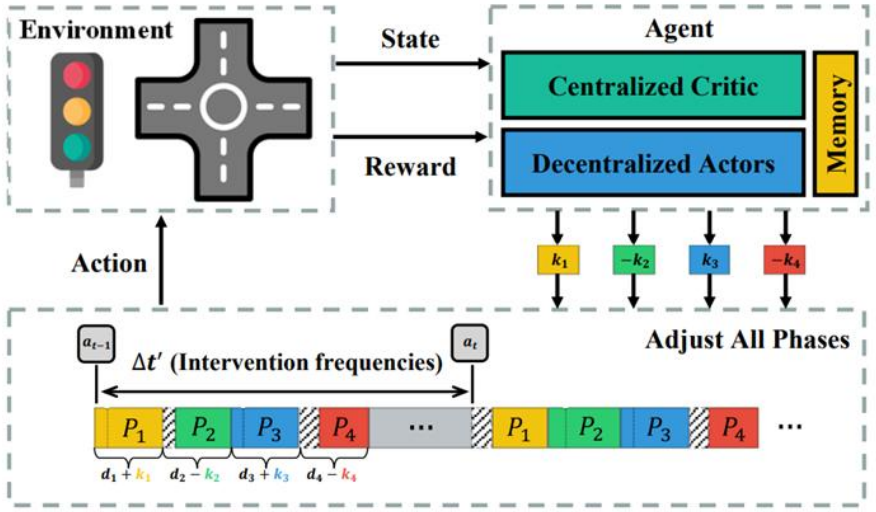
Generalizable TSC under Constraints

Cycle-Level Traffic Signal Control



- Make control decisions once per traffic cycle;
- Adjust the duration of all traffic phases;
- Each phase duration is modified by ±k;

Centralized–Decentralized Control Framework



Centralized Critic

Observes the full state and coordinates the decision across all phases

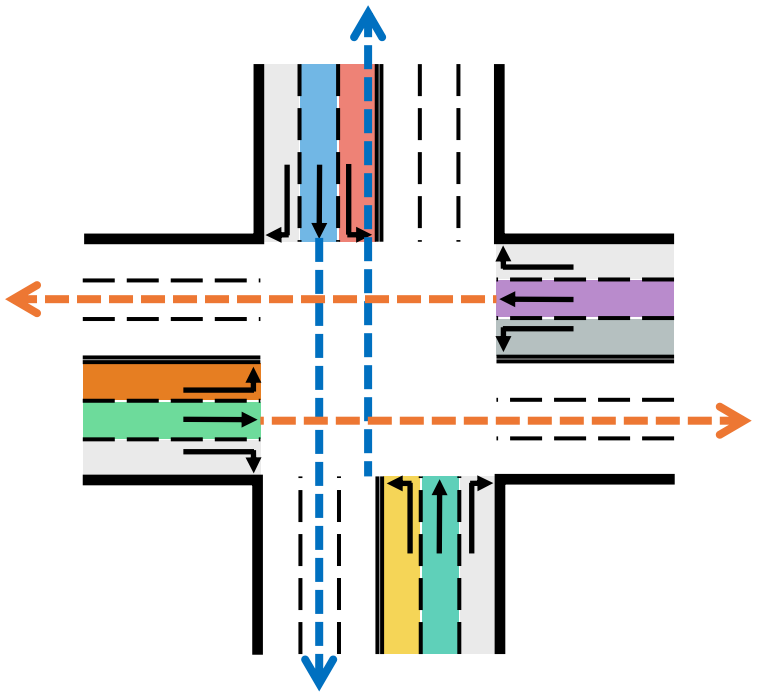
Decentralized Actors

One actor per traffic phase, independently adjusting phase duration

This design enables low-frequency, stable, and coordinated traffic signal control under practical constraints.

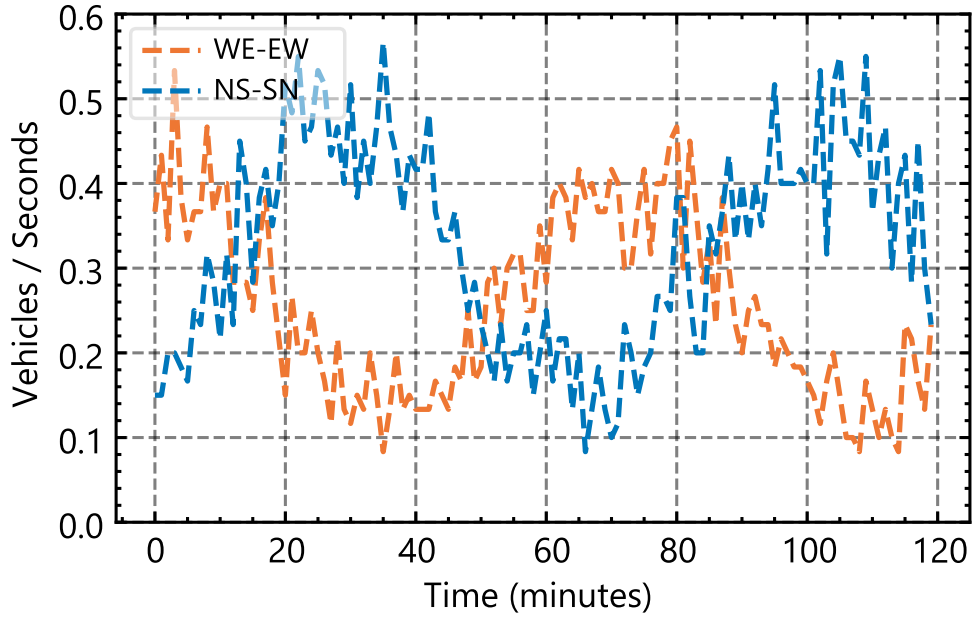
Generalizable TSC under Constraints

Experimental Road Network



- Four-way intersection scenario;
- Focus on NS and EW through movement;

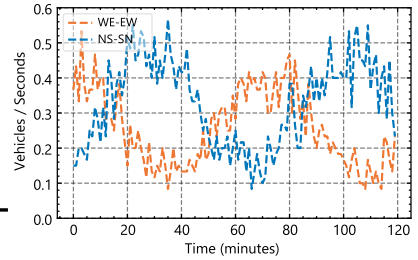
Traffic Flow Dynamics



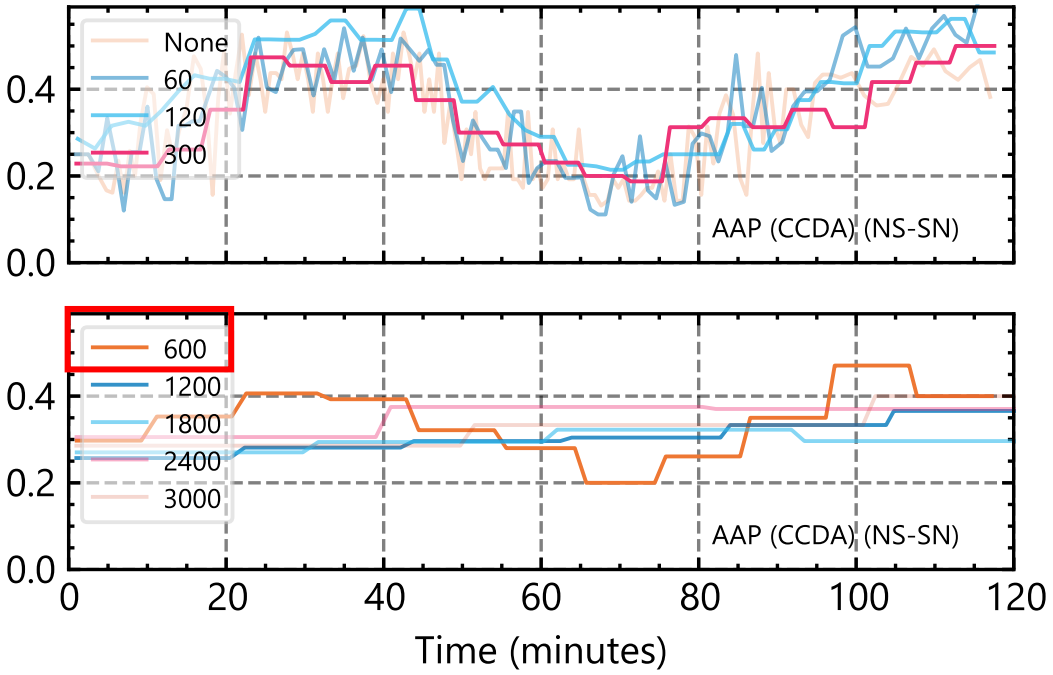
- Traffic flow varies over time;
- Different demand patterns across two movements;

Generalizable TSC under Constraints

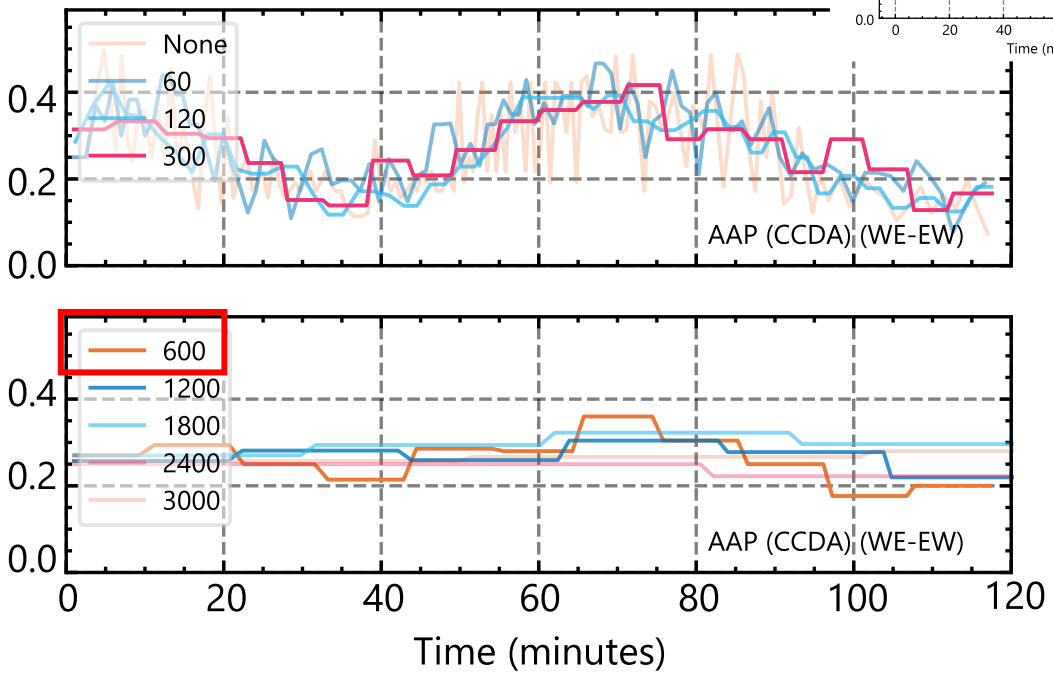
Traffic Flow



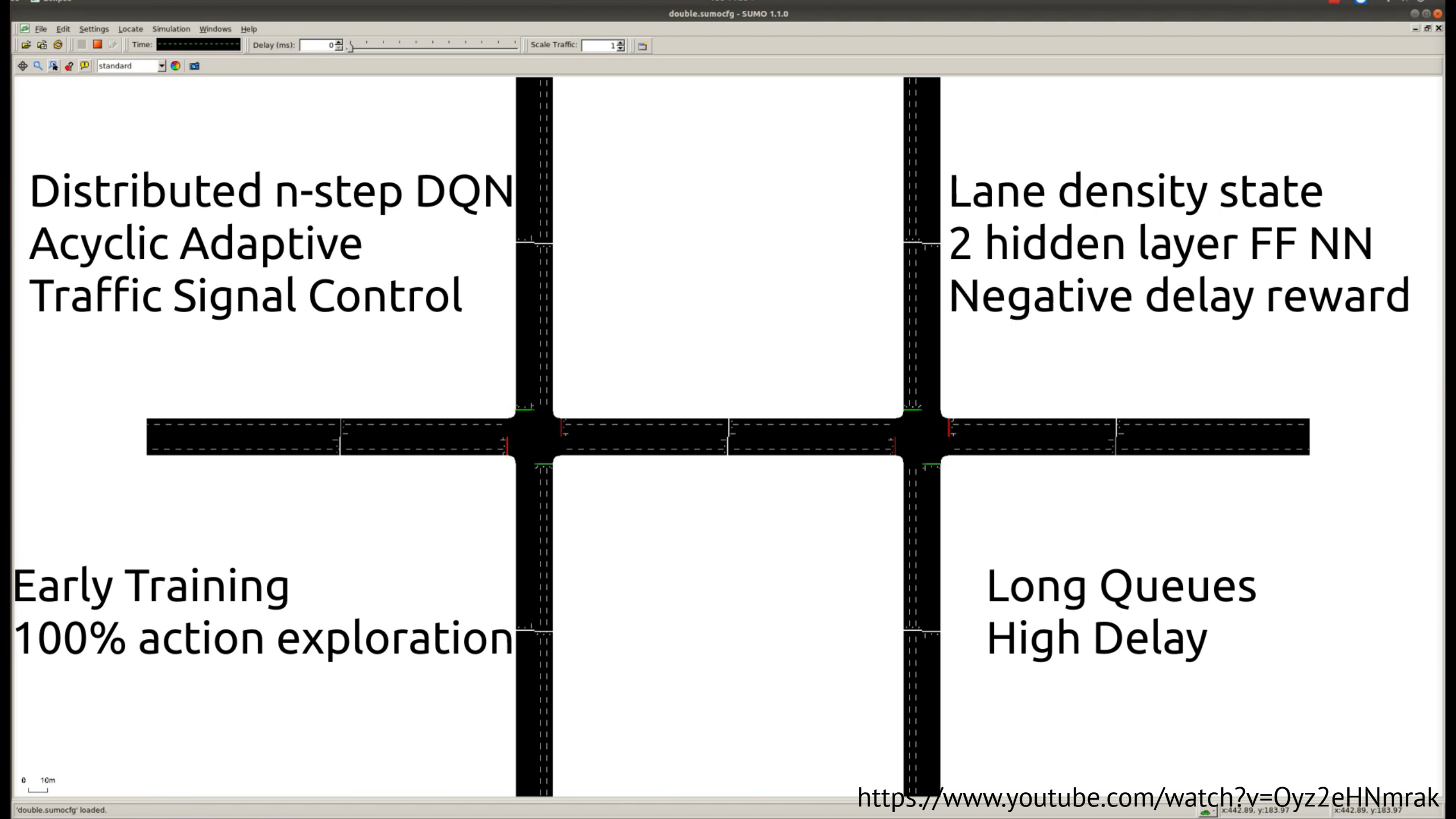
NS-SN direction



EW-WE direction



Frequent phase-level control tends to overfit traffic noise, while cycle-level control captures traffic trends.



https://www.youtube.com/watch?v=Oyz2eHNmrak

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LLM-Guided Agent Control for TSC

Limitations of Current Traffic Signal Control

- Most TSC methods optimize average waiting time;
- Traffic states are primarily based on queue length;
- Designed for normal traffic conditions;



Failure in Rare and Emergency Scenarios

- Sudden events such as accidents and emergency vehicles;
- Fixed objectives become inappropriate;
- The existing methods cannot response effectively without retraining;



Existing traffic signal control methods lack flexibility in rare and safety-critical scenarios

LLM-Guided Agent Control for TSC

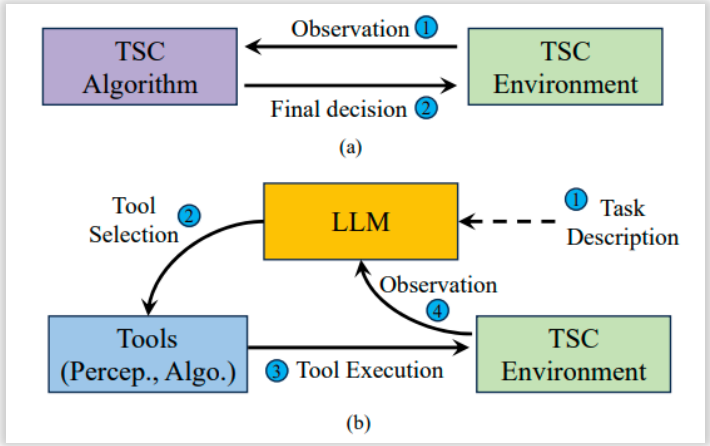
Human-Inspired Traffic Signal Control

- Decision based on context, not fixed objectives;
- Strategies adapted to both normal and special scenarios;
- In emergencies, police assess context first;



From Convention Control to Agent-based Decision Making

- Conventional TSC maps states to actions directly;
- Lacks flexibility in rare and dynamic scenarios;
- Shift to agent-based approach: reasoning before acting;

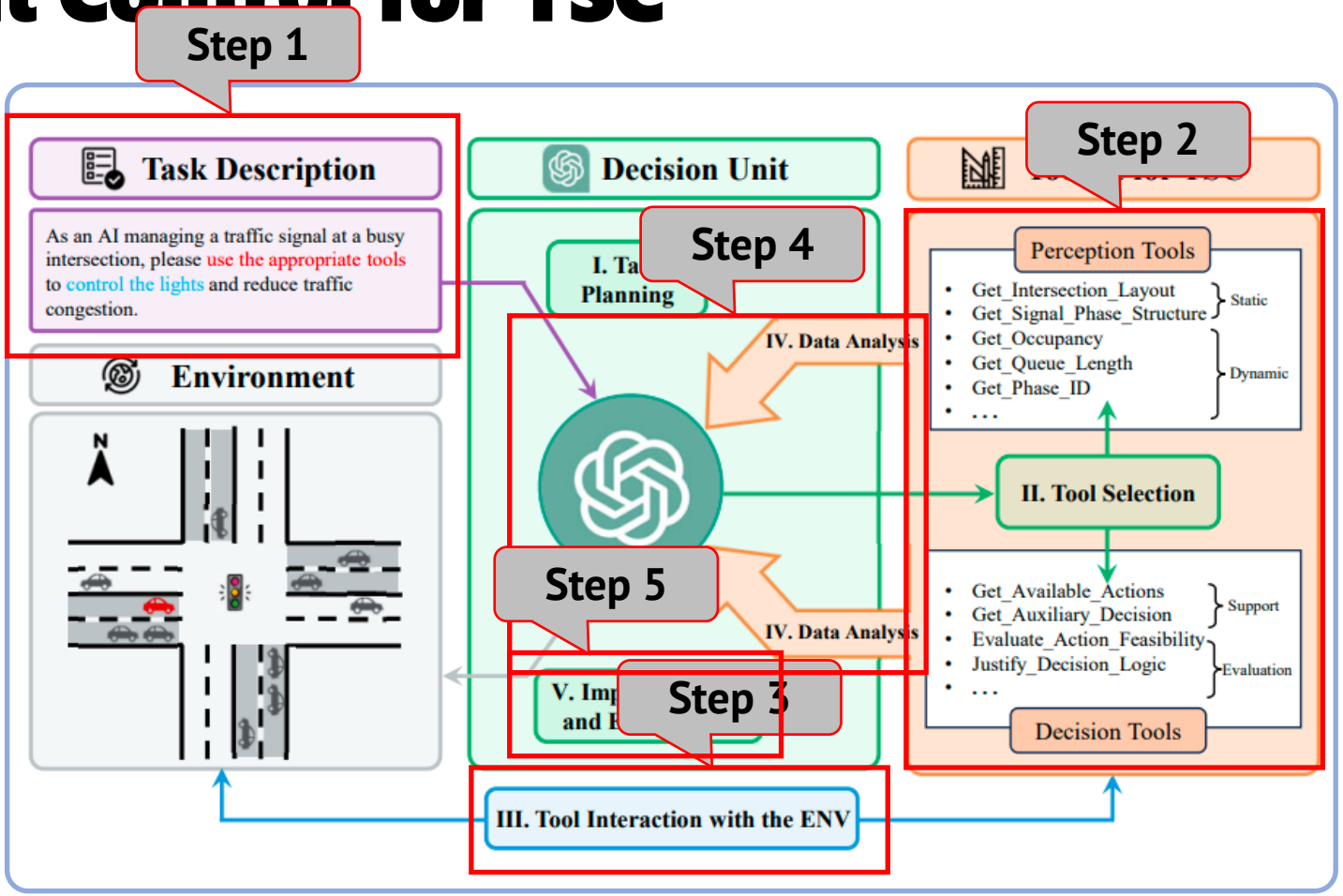


LLM-Guided Agent Control for TSC

LLM-Guided TSC Agent Pipeline

- ✓ Step 1, Task Planning
- ✓ Step 2, Tool Selection
- ✓ Step 3, Data Gathering
- ✓ Step 4, Data Analysis
- ✓ Step 5, Decision Making





The LLM-guided agent separates reasoning from execution, enabling flexible and interpretable traffic signal control

LLM-Guided Agent Control for TSC

Example Tool: Get Intersection Layout

Description

Retrieve the layout of the target intersection.

Input: Junction ID (string)

Output

A dictionary where each key represents a traffic movement id, the corresponding value represents its information.

```
{
  "E1": {"direction": "l", "number_of_lanes": 2}, ...
}
```

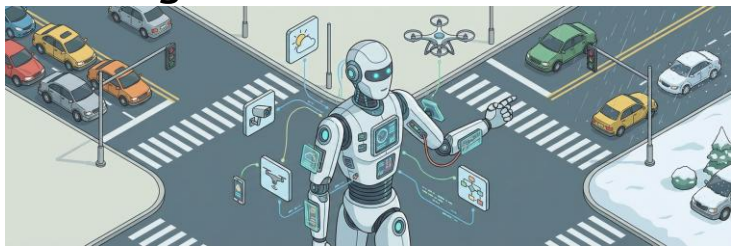
Customization

Tool Extensibility

Researchers can define new tools with standardized interfaces;

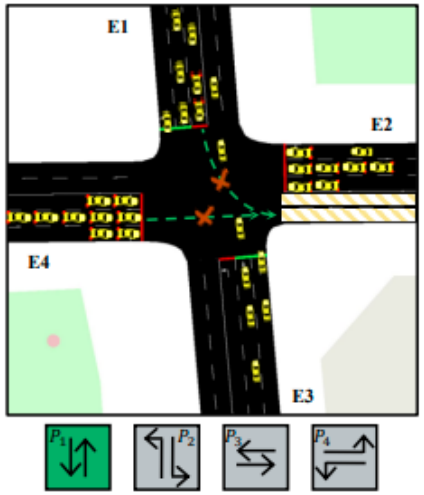
Environment Adaptability

The agent system can adapt to different environments without retraining the models.



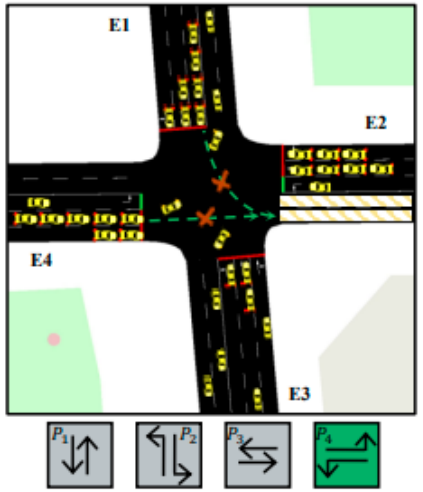
Standardized tool interfaces enable flexible, extensible, and environment-adaptive LLM-guided traffic signal control

LLM-Guided Agent Control for TSC



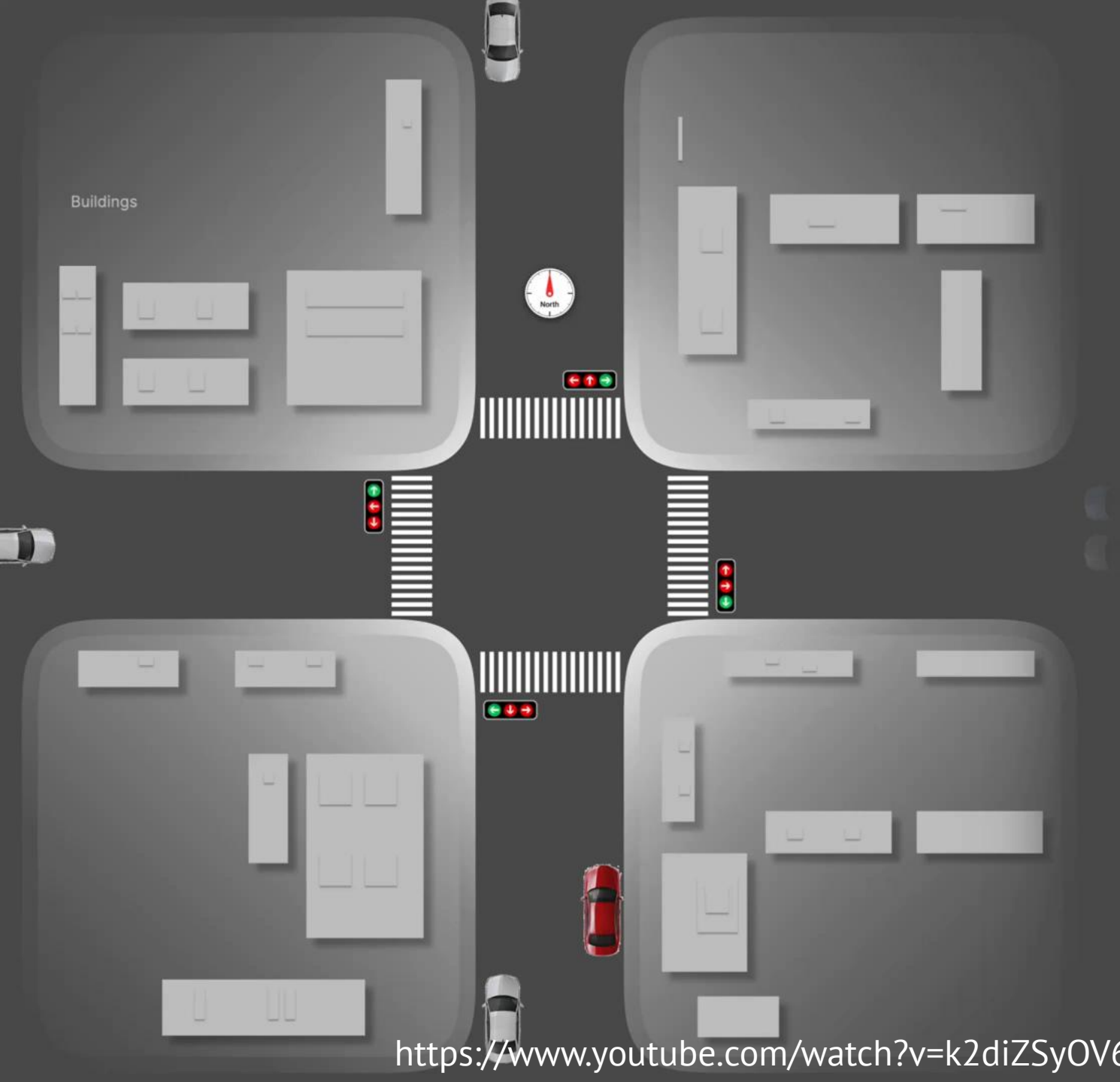
Round 1: Detect and Localize the Problem

- **Thought**: A roadblock may exist; inspect network;
- **Action**: Invoke perception tools to observe intersection.
- **Observation**: An accident blocks the eastbound exit (from west to east);



Round 2: Traffic Phase Reasoning and Decision

- **Thought**: Avoid traffic phases with blocked movements;
- **Action**: Map affected movements to phases, evaluate the options.
- **Observation**: Phase 2 and Phase 4 feasible; Phase 2 selected;



Traffic-R1

https://www.youtube.com/watch?v=k2diZSyOV6U

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THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

香 港 大 學
THE UNIVERSITY OF HONG KONG

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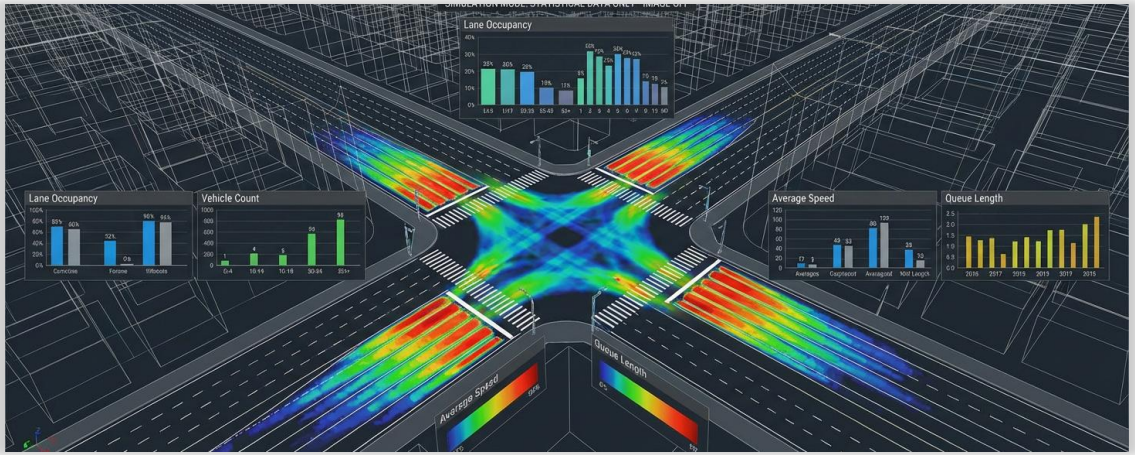
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VLM for Safety-Critical TSC

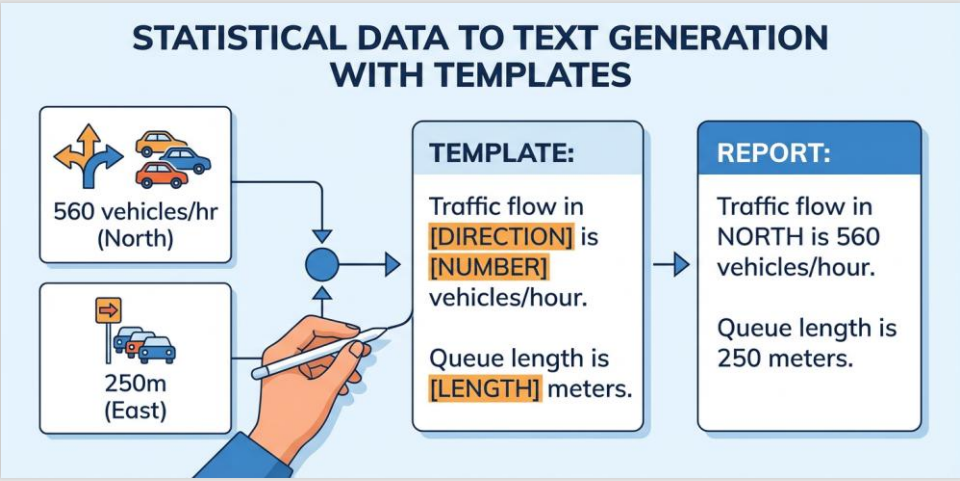
Statistic-Based Traffic Simulators

- Aggregated traffic statistics;
- Queue length, speed, traffic flow;
- Limited abstraction of traffic state;



Statistic-to-Text Prompting

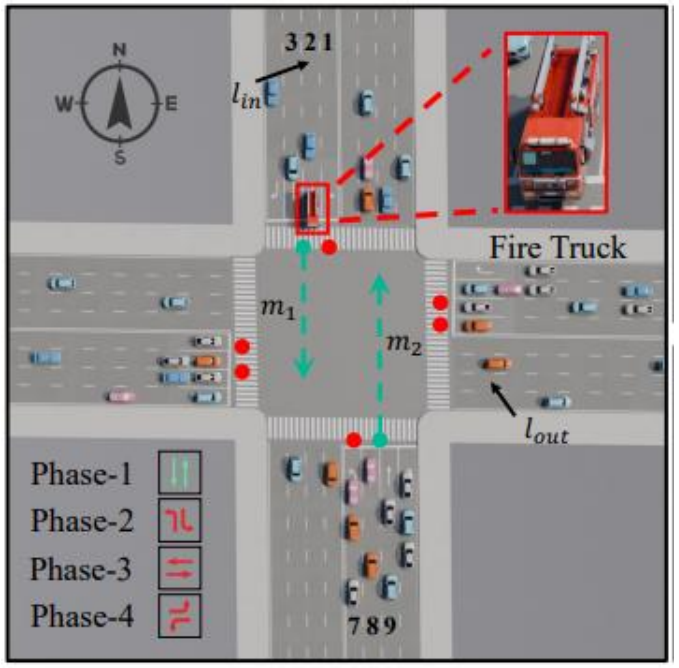
- Statistics converted to text templates;
- Spatial and visual cues removed;
- Safety-critical details missing;

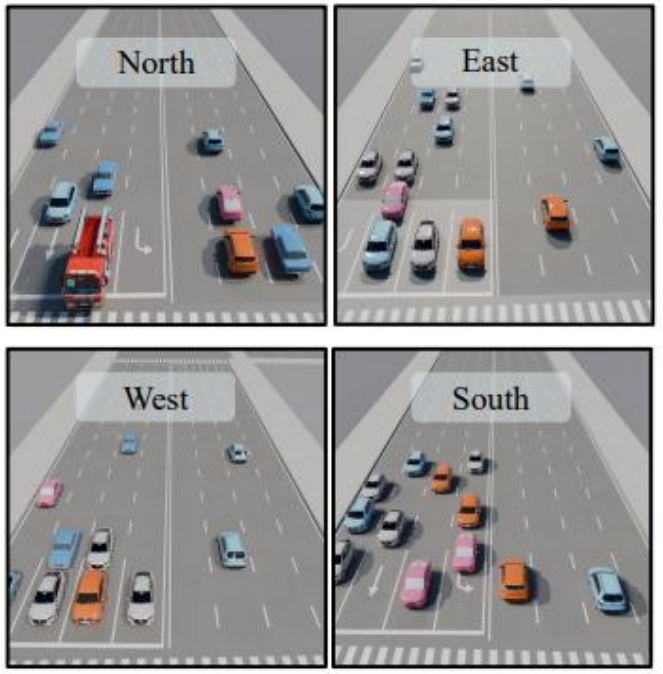


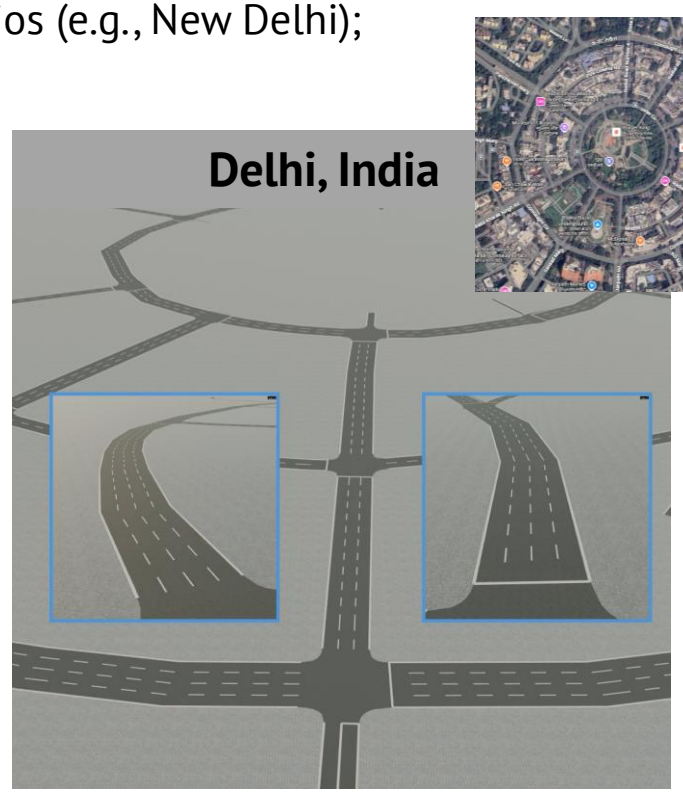
Converting statistical traffic states into text prompts based on template removes critical visual and spatial information.

VLM for Safety-Critical TSC

- **Multi-Modal Simulator**: Develop a multi-modal traffic simulator with visual rendering;
- **Multi-View & Special Scenarios**: Support multi-view observations and diverse traffic participants (e.g. emergency vehicles);
- **Multi-Intersections**: Enable both single-intersection and multi-intersection scenarios (e.g., New Delhi);





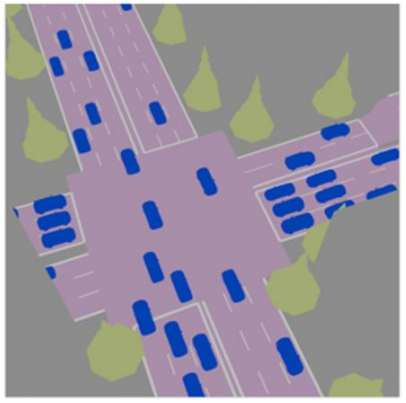
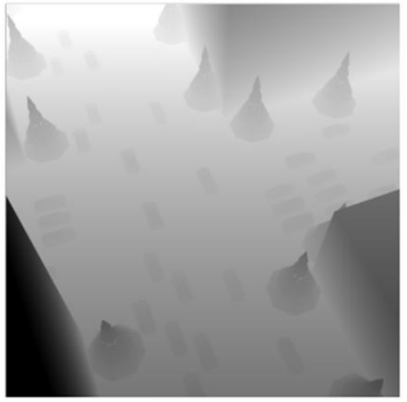


VLM for Safety-Critical TSC





Different Views

Multi-modal Rendering Outputs





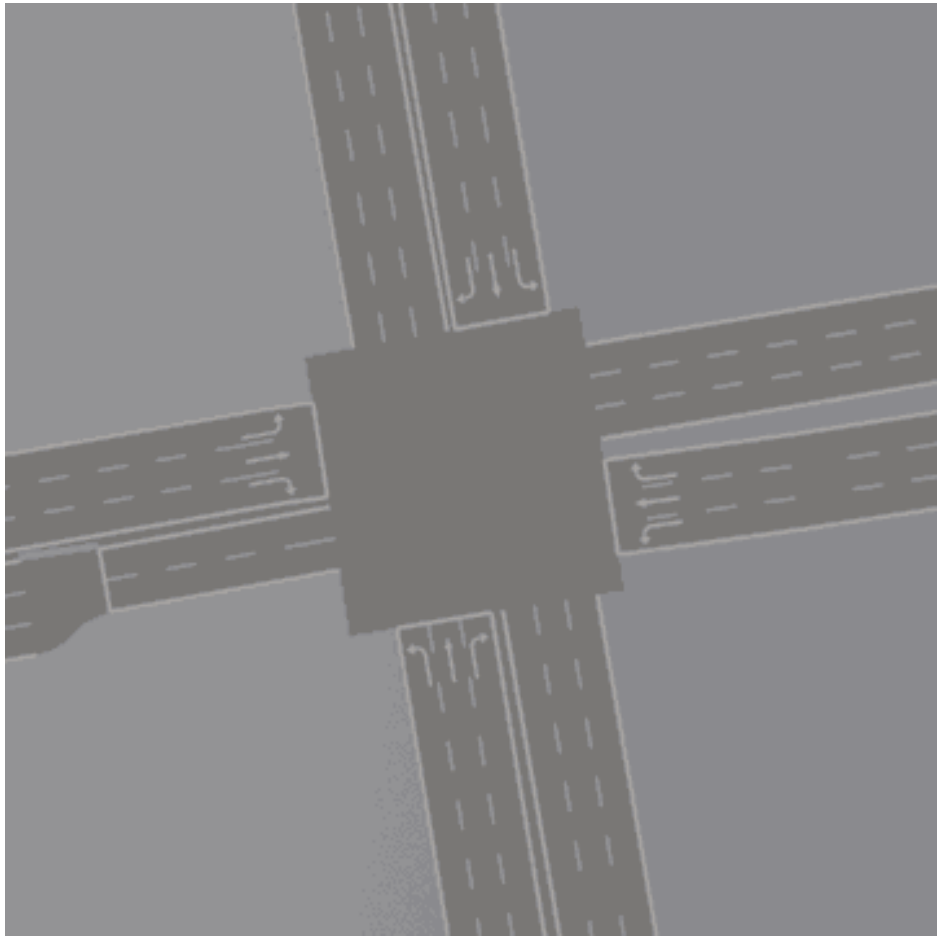
Different Incidents



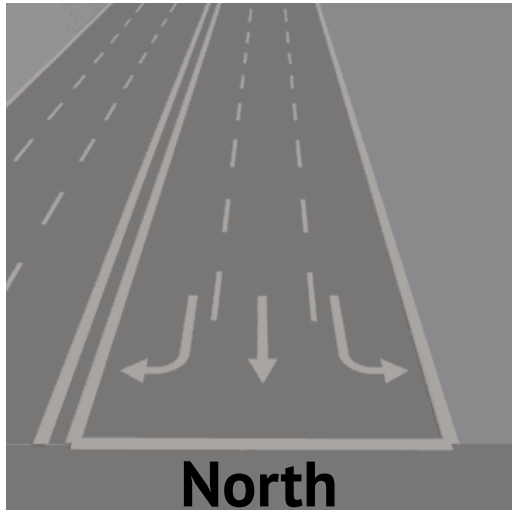


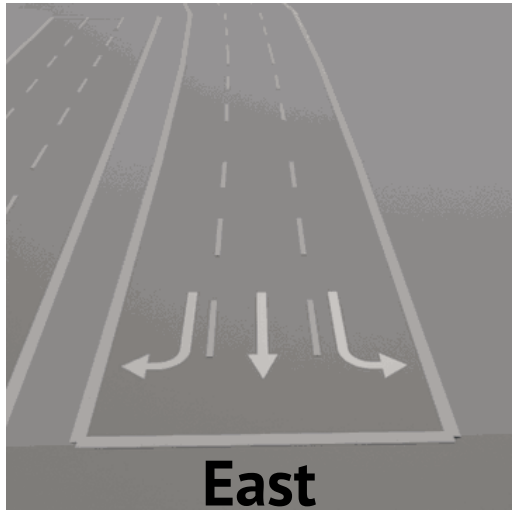
Different Weather and Lighting Conditions

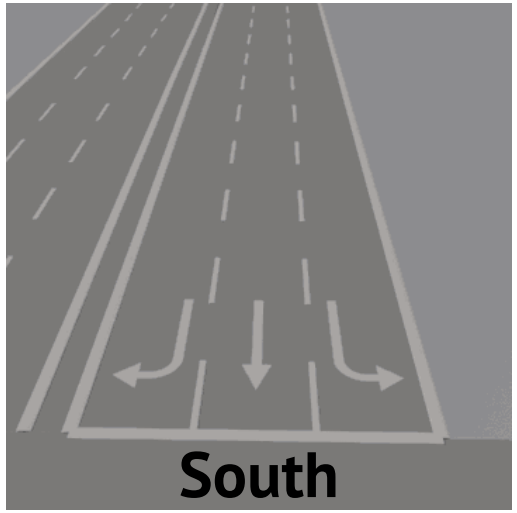
VLM for Safety-Critical TSC

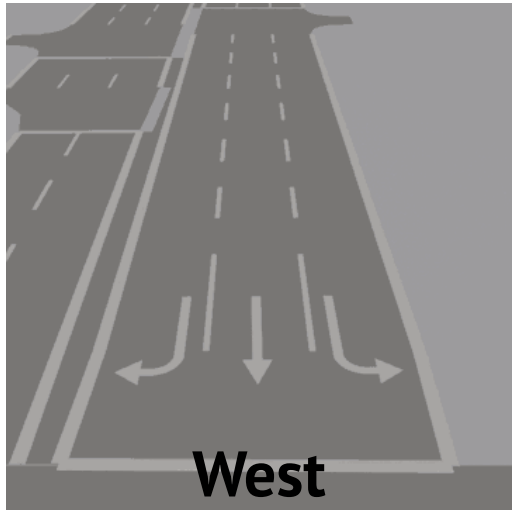


BEV





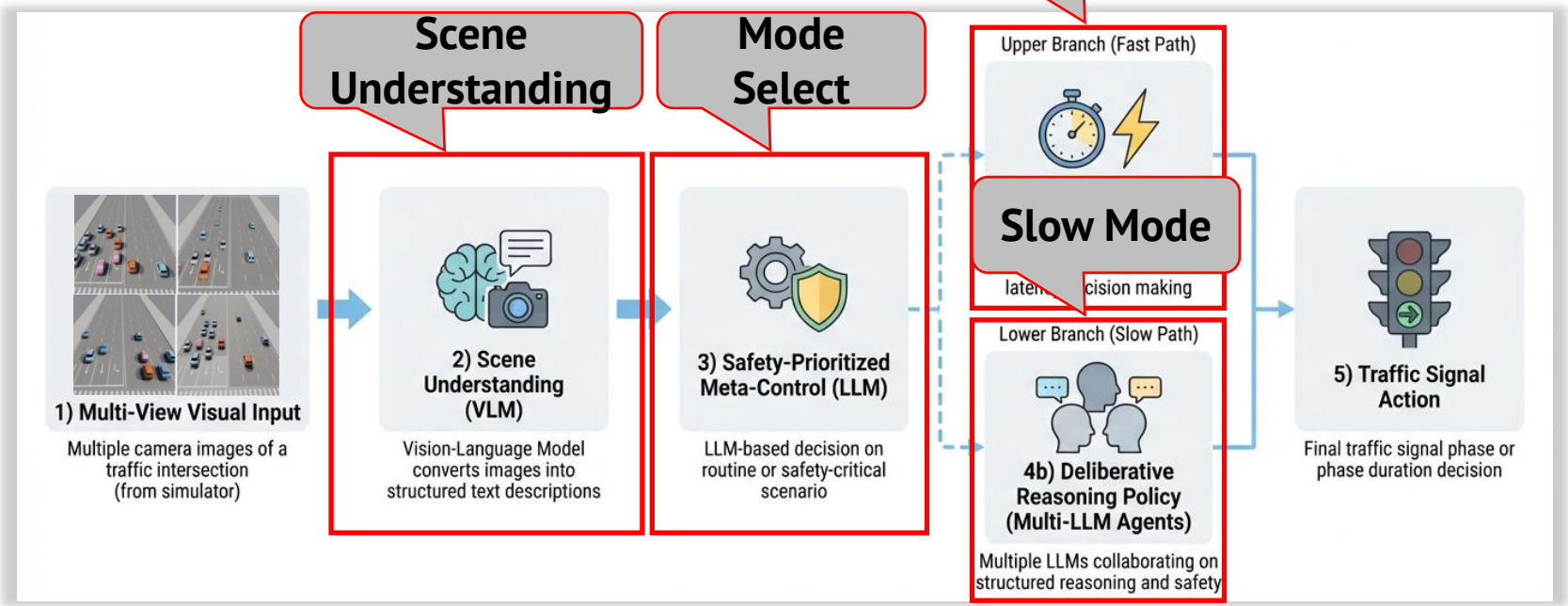


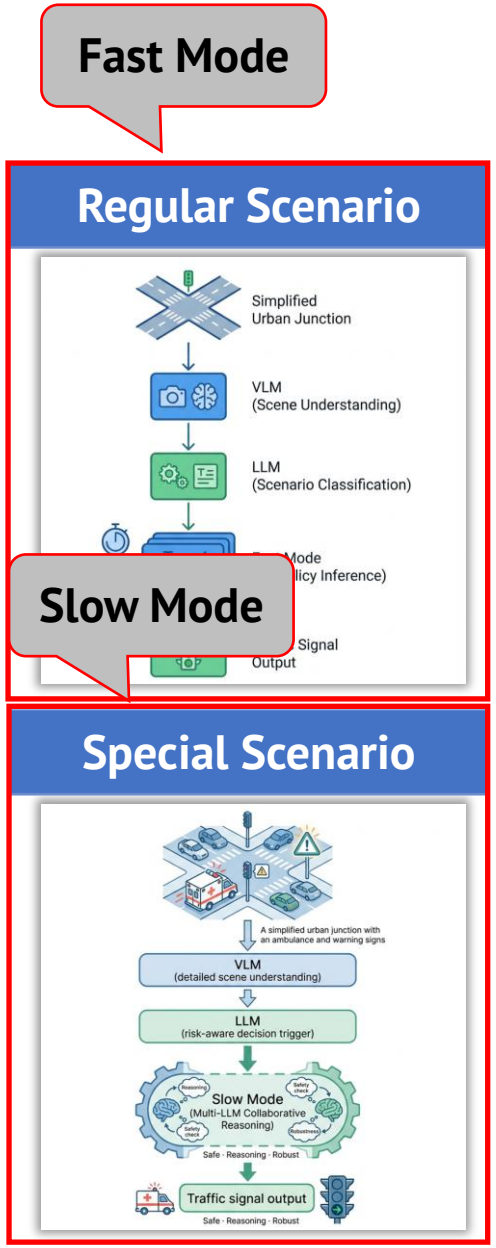


VLM for Safety-Critical TSC

The Overall Architecture of VLMLight

1. **Scene Understanding**: Converts multi-view images into directional text descriptions;
2. **Safety-Prioritized Meta-Control**: Selects control mode: fast policy or structured LLM reasoning;
3. **Routine Control Policy**: Support existing RL-based TSC for routine traffic;
4. **Reasoning Policy**: Multiple LLM agents collaborate to asse… …d give the actions;

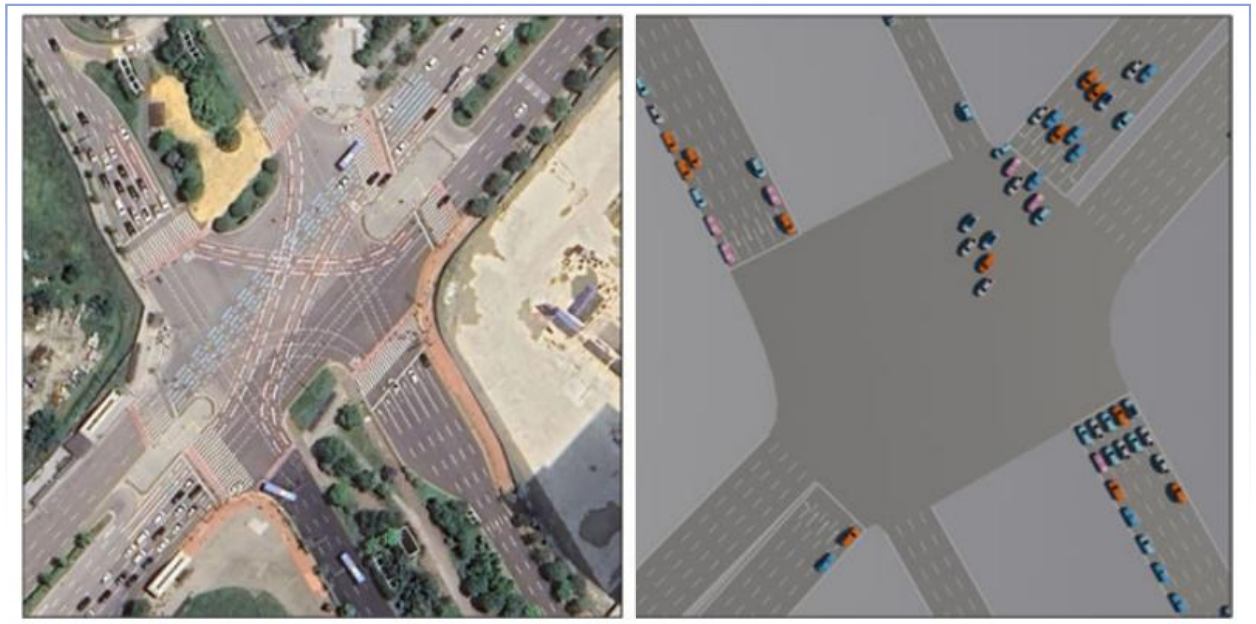


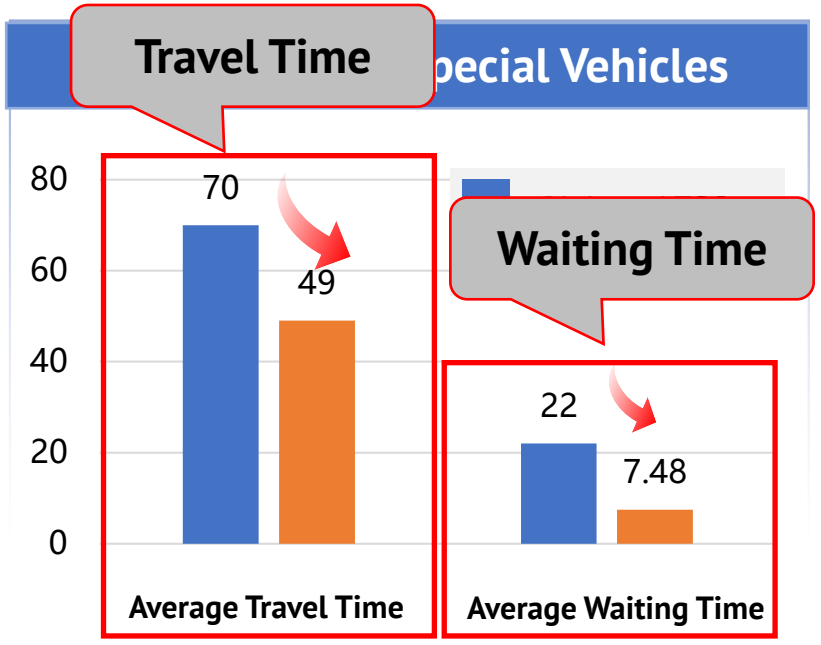


VLM for Safety-Critical TSC

Experimental Results of VLMLight

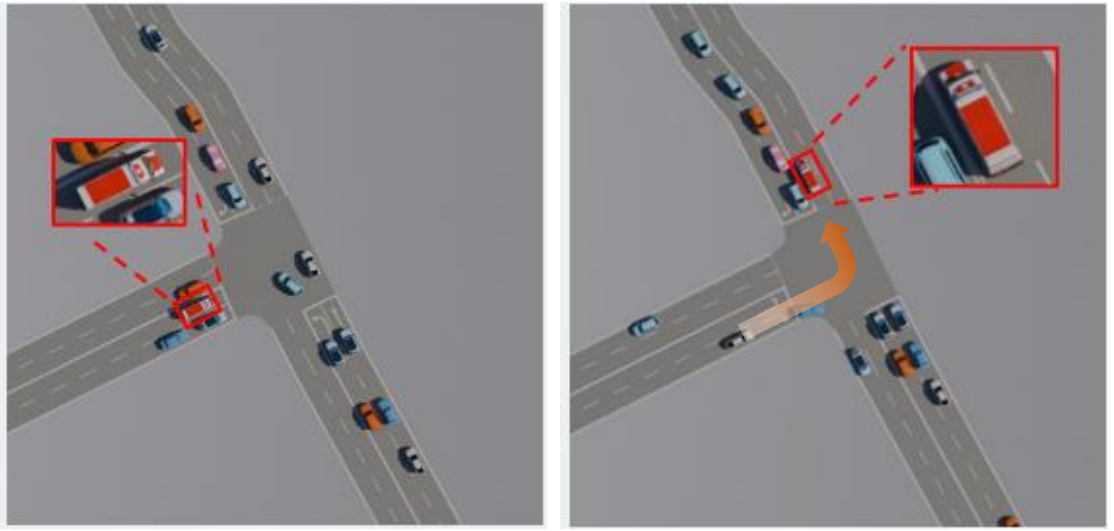
1. **Experimental Setup:** Conduct in Songdo, South Korea, a dense urban area with heavy traffic flow;
2. **Key Results:** Cuts waiting time to 7 s and travel time to 49 s for safety-critical vehicles.





VLM for Safety-Critical TSC

Safety-Critical Scene (Before -> After)



Emergency Detection: Ambulance approaching the intersection;

Safety-Critical Decision: Recognized the priority event;

Final Decision: A signal enabling the left-turn movement;

Safety-Critical Scene (Before -> After)



Emergency Detection: Fire engine entering the intersection;

Safety-Critical Decision: Recognized the priority event;

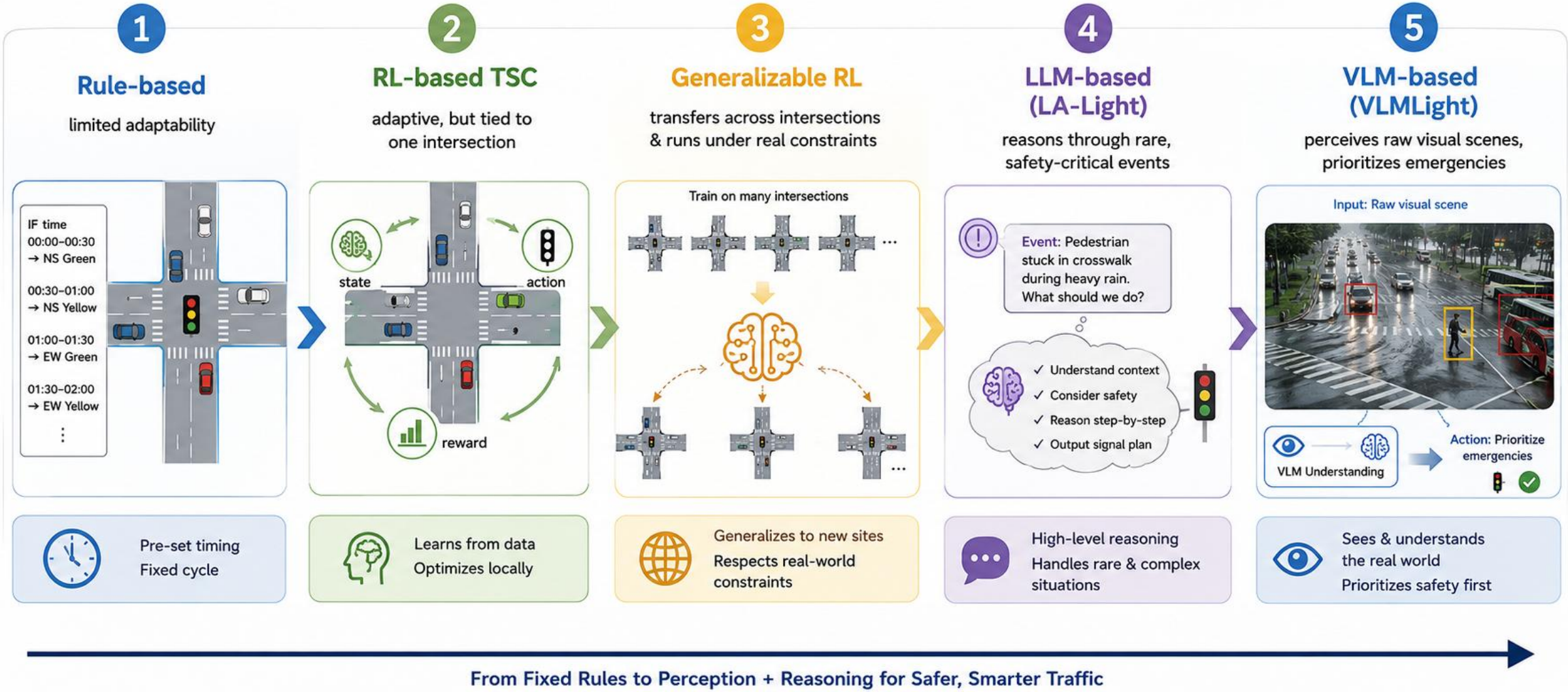
Final Decision: A signal enabling the straight-through movement;

VLMLight identifies safety-critical situations from visual input and prioritizes emergency vehicles through LLM Agents' decision

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The Journey: From Timers to Large Models



From Fixed Rules to Perception + Reasoning for Safer, Smarter Traffic

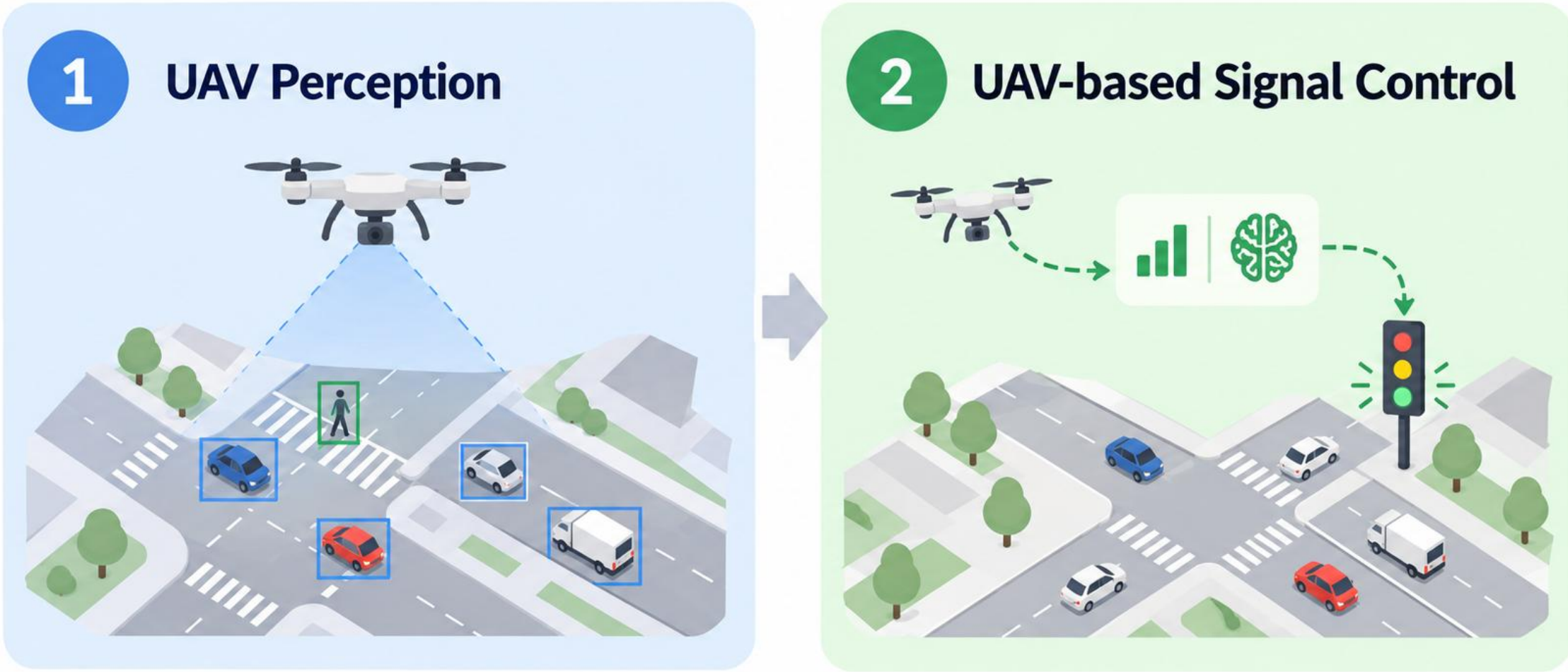
Key Takeaways

- **From hand-crafted rules to learned control**
  - RL lets signals adapt to real-time demand instead of fixed plans.
- **Making learning-based TSC deployable**
  - A unified junction representation + state augmentation + LoRA give cross-intersection generalization;
  - Cycle-level, low-frequency control fits real compute/safety constraints.
- **Foundation models close the long-tail gap**
  - LLM agents add interpretable reasoning for emergencies;
  - VLMs add direct visual perception, moving toward human-like, safety-aware control.

Open Challenges & Future Directions

- **Sim-to-real**: closing the gap between simulators and live intersections
- **Latency & cost**: real-time decisions with large models (the fast/slow idea points the way)
- **Safety & trust**: verifiable, reliable control for safety-critical events
- **City-scale coordination**: from single intersections to networked, multi-intersection control
- **A unified perceive–reason–act loop**: integrating VLM perception, LLM reasoning, and RL control

Conclusion -- The City Traffic Brain



Future Work

- **Sim-to-real** — close the gap between simulators and live roads
- **Real-time & low cost** — fast decisions, even with large models
- **Safety, reliability & trust** — especially for rare, safety-critical events
- **City-scale coordination** — from single intersections to whole networks
- **Toward integration** — one unified *perceive – reason – act* loop

Questions and Discussion

Thank you for Listening!